

Principle of Financial Computing

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1. Introduction

— Slides pp. 2–3 —

Lecture 1
September 10,
2026

You must go into finance, Amory.
— F. Scott Fitzgerald (1896–1940),
This Side of Paradise (1920)

*The two most dangerous words in Wall Street
vocabulary are “financial engineering.”*
— Wilbur Ross (2007)

The two epigraphs above already frame the tension this course lives in. Finance is where quantitative modelling pays off most visibly, and it is also where a misplaced faith in a model does the most damage. Our job is to learn the models *and* to keep track of what they assume.

§1.1. Class Information

— Slides pp. 4–6 —

The course follows the instructor’s own textbook, and everything administrative lives on the course web page rather than in these notes.

- **Textbook.** Yuh-Dauh Lyuu, *Financial Engineering & Computation: Principles, Mathematics, Algorithms*. Cambridge University Press, 2002.
- **Official web page.**

www.csie.ntu.edu.tw/~lyuu/finance1.html

- Lecture notes will be uploaded *before* class.
- Homeworks and teaching assistants will also be announced there.
- Do not mistake last year’s homeworks for this year’s!

- **Software.** Some of the software used in the course is collected at

www.csie.ntu.edu.tw/~lyuu/capitals.html

- **Questions.** Please ask many questions in class. In a large class this is the best way for the instructor to remember you.¹

§1.2. Grading

— Slides p. 7 —

Assessment is by **programming assignments** plus **one test**. Every homework is to be treated as an examination, so the academic-integrity rules are strict:

- You are expected to write your own codes and to turn in your source code.
- Do not copy or collaborate with fellow students.

¹“[A] science concentrator [...] said that in his eighth semester of [Harvard] college, there was not a single science professor who could identify him by name.” (*New York Times*, September 3, 2003.)

- Never ask your friends to write programs for you.
- Do not ask AI to code for you.
- Never give your codes to other students or publish your codes.

§1.3. Scope of the Course

— Slides pp. 8–9 —

§1.3.i. What This Course Is About

The course runs from financial theory down to running code, in roughly this order:

- Financial theories in pricing.
- Mathematical backgrounds.
- Derivative securities.
- Pricing models.
- Efficient algorithms in pricing financial instruments.
- Research problems.
- Help in finding your thesis directions.

The last two items are worth emphasising: pricing problems that are merely “computable” are usually not good enough, and the gap between a correct algorithm and a *fast* one is exactly where the open research questions sit.

§1.3.ii. What This Course Is *Not* About

Equally important is what will *not* be covered.

- **How to program.**² Programming ability is assumed, not taught — but it matters. A software bug cost Knight Capital Group, Inc. US\$457.6 million on August 1, 2012.³
- **Basic calculus, probability, combinatorics, and algebra.**
- **Details of the financial markets.**
- **How to be rich.**
- **How the markets will perform tomorrow.**
- **Professional behavior.**

Pricing theory tells you what a security is worth *given* a model of the market. It does not tell you what the market will do next.

§1.4. Useful Journals

— Slides pp. 10–13 —

The literature of the field is spread across finance, mathematics, operations research and computer science journals. The following list is the one given in class.

²See <http://www.csie.ntu.edu.tw/train/>.

³Kirilenko & Lo (2013).

- *Applied Mathematical Finance.*
- *Applied Mathematics and Computations.*
- *Communications on Pure and Applied Mathematics.*
- *Computers & Mathematics with Applications.*
- *European Journal of Finance.*
- *European Journal of Operational Research.*
- *Finance and Stochastics.*
- *Finance Research Letters.*
- *Financial Analysts Journal.*
- *Financial Management.*
- *Insurance: Mathematics and Economics.*
- *International Journal of Finance & Economics.*
- *International Journal of Theoretical and Applied Finance.*
- *Journal of Banking & Finance.*
- *Journal of Computational Finance.*
- *Journal of Derivatives.*
- *Journal of Economic Dynamics & Control.*
- *Journal of Empirical Finance.*
- *Journal of Finance.*
- *Journal of Financial Economics.*
- *Journal of Financial Markets.*
- *Journal of Fixed Income.*
- *Journal of Futures Markets.*
- *Journal of Financial and Quantitative Analysis.*
- *Journal of Portfolio Management.*
- *Journal of Real Estate Finance and Economics.*
- *Journal of Risk and Uncertainty.*
- *Management Science.*
- *Mathematical Finance.*
- *Quantitative Finance.*
- *Review of Financial Studies.*
- *Review of Derivatives Research.*
- *Review of Finance.*
- *Risk Magazine.*
- *SIAM Journal on Financial Mathematics.*
- *Stochastics and Stochastics Reports.*

§1.5. The Modelers' Hippocratic Oath

— Slides p. 14 —

Before writing down a single model, it is worth reading the oath proposed by Emanuel Derman and Paul Wilmott on January 7, 2009, in the wake of the financial crisis.

- I will remember that I didn't make the world, and it doesn't satisfy my equations.
- Though I will use models boldly to estimate value, I will not be overly impressed by mathematics.
- I will never sacrifice reality for elegance without explaining why I have done so.
- Nor will I give the people who use my model false comfort about its accuracy. Instead, I will make explicit its assumptions and oversights.
- I understand that my work may have enormous effects on society and the economy, many of them beyond my comprehension.

Intuition . A model is a deliberate simplification. The discipline is not in believing it, but in knowing precisely *where* it is wrong and *how much* that costs.

§1.6. A First Look at the Markets and the Data

— Slides p. 15 —

Two facts motivate everything that follows. First, the markets we want to price in are enormous, so even small pricing errors translate into large amounts of money. Second, the

price series they produce have a very definite statistical shape, and any pricing model has to be compatible with it.

§1.6.i. Outstanding U.S. Debts

The fixed-income market alone is measured in tens of trillions of dollars, and over the period below it roughly quadrupled. Note in particular how the **mortgage-related** and **asset-backed** columns grow — these are precisely the securities whose pricing requires the algorithmic machinery of this course.

Year 年份	Municipal 市政債	Treasury 美國公債	Mortgage- related 房貸相關	U.S. corporate 公司債	Fed agencies 聯邦機構債	Money market 貨幣市場	Asset- backed 資產擔保	Total 總計
85	859.5	1,437.7	372.1	776.5	293.9	847.0	0.9	4,587.6
86	920.4	1,619.0	534.4	959.6	307.4	877.0	7.2	5,225.0
87	1,010.4	1,724.7	672.1	1,074.9	341.4	979.8	12.9	5,816.2
88	1,082.3	1,821.3	772.4	1,195.7	381.5	1,108.5	29.3	6,391.0
89	1,135.2	1,945.4	971.5	1,292.5	411.8	1,192.3	51.3	7,000.0
90	1,184.4	2,195.8	1,333.4	1,350.4	434.7	1,156.8	89.9	7,745.4
91	1,272.2	2,471.6	1,636.9	1,454.7	442.8	1,054.3	129.9	8,462.4
92	1,302.8	2,754.1	1,937.0	1,557.0	484.0	994.2	163.7	9,192.8
93	1,377.5	2,989.5	2,144.7	1,674.7	570.7	971.8	199.9	9,928.8
94	1,341.7	3,126.0	2,251.6	1,755.6	738.9	1,034.7	257.3	10,505.8
95	1,293.5	3,307.2	2,352.1	1,937.5	844.6	1,177.3	316.3	11,228.5
96	1,296.0	3,459.7	2,486.1	2,122.2	925.8	1,393.9	404.4	12,088.1
97	1,367.5	3,456.8	2,680.2	2,346.3	1,022.6	1,692.8	535.8	13,102.0
98	1,464.3	3,355.5	2,955.2	2,666.2	1,296.5	1,978.0	731.5	14,447.2
99	1,532.5	3,281.0	3,334.2	3,022.9	1,616.5	2,338.2	900.8	16,026.4
00	1,567.8	2,966.9	3,564.7	3,372.0	1,851.9	2,661.0	1,071.8	17,056.1
01	1,688.4	2,967.5	4,125.5	3,817.4	2,143.0	2,542.4	1,281.1	18,565.3
02	1,783.8	3,204.9	4,704.9	3,997.2	2,358.5	2,577.5	1,543.3	20,170.1

Table 1.1.: Outstanding U.S. debts, in billions of U.S. dollars.

§1.6.ii. The Global OTC Derivatives Market

— Slides p. 16 —

Over-the-counter derivatives grew even faster, by close to an order of magnitude between 2000 and the 2008 crisis, after which the notional outstanding plateaued and then declined as trades were compressed and cleared centrally.

Remark 1.6.1. **衍生性商品** (derivative) 是價值取決於某個標的（利率、匯率、股價、信用等）的合約。**場外交易** (over-the-counter, OTC) 指買賣雙方私下議定、不經交易所撮合的交易，與合約標準化、由結算所集中結算的**交易所交易**相對。OTC 合約可客製化（金額、到期日、標的皆可議），代價是雙方直接承擔**對手方風險**；市場由少數大型銀行（dealer）造市，未平倉部位以利率交換（interest rate swap）為大宗。

Figure 1.1 的縱軸為**名目本金** (notional amount) ——僅是計算利息與收益的參考金額，並非真正易手的資金，故市場實際市值遠小於此。2008 年金融危機後，美國 Dodd-Frank 法與歐盟 EMIR 要求標準化的 OTC 商品（尤其信用違約交換 CDS 與利率交換）改採**中央結算** (central clearing)，並藉**壓縮** (compression) 沖銷互相抵銷的部位，名目餘額因而在 2013 年前後見頂並回落。

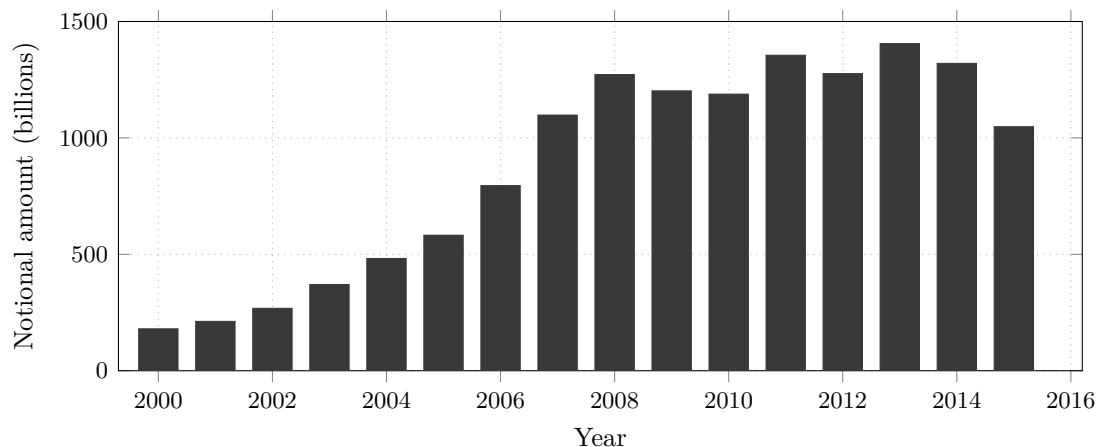


Figure 1.1.: The global OTC derivatives market. Bar heights are read off the chart shown in class.

§1.6.iii. The S&P 500 Index and Its Returns

— Slides pp. 17–19 —

Robert Shiller⁴ maintains a monthly series of the S&P 500 index reaching back to 1871. Plotted in levels (Figure 1.2), it is dominated by the post-1980 growth; the 1929 crash is barely visible on this scale.

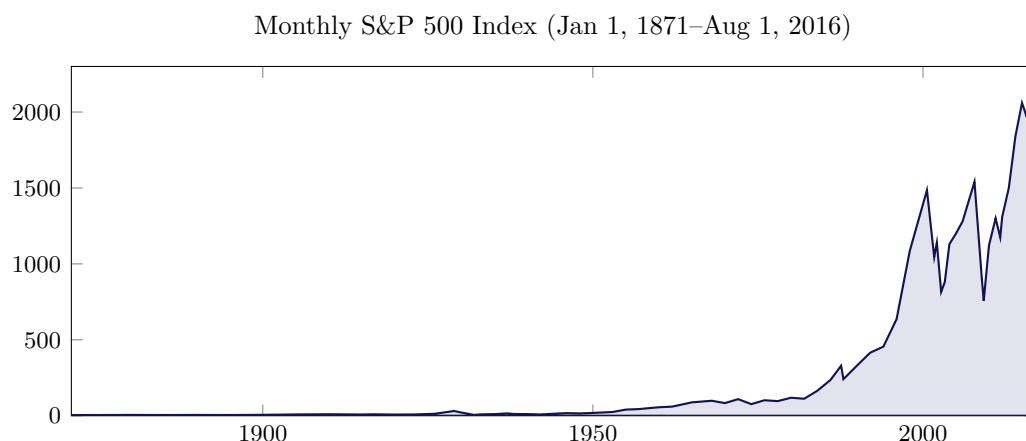


Figure 1.2.: The S&P 500 index in levels, reconstructed from the Shiller series.

Levels are not what we model, however. Prices are non-stationary and their scale is arbitrary, so we work with the **log return** over one period,

$$r_t = \ln \frac{S_t}{S_{t-1}},$$

where S_t is the index level at time t . Figure 1.3 plots the monthly log returns of the same series.

Remark 1.6.2. 為什麼不直接對價位建模？價位（levels）是**非定態**的——統計性質隨時間漂移，1900 年與 2000 年的「正常水準」天差地遠，無法用一組固定參數描述；而且指數從幾點起編是人為選擇，尺度本身沒有經濟意義。改用**對數報酬** $r_t = \ln(S_t/S_{t-1})$ 一次補齊三件事：

⁴Co-winner of the 2013 Nobel Prize in Economic Sciences.

- S_t/S_{t-1} 是本期的成長倍數（漲 2% 即 1.02）。變化小時 $\ln(1+x) \approx x$ ，所以 r_t 近似百分比變化，且與絕對點數無關。
- 可加性：跨期報酬 $\ln(S_t/S_{t-k}) = r_t + r_{t-1} + \dots + r_{t-k+1}$ ，相加即可，不必連乘。
- 報酬率序列大致定態——繞著 0 波動、沒有趨勢，正適合建模。

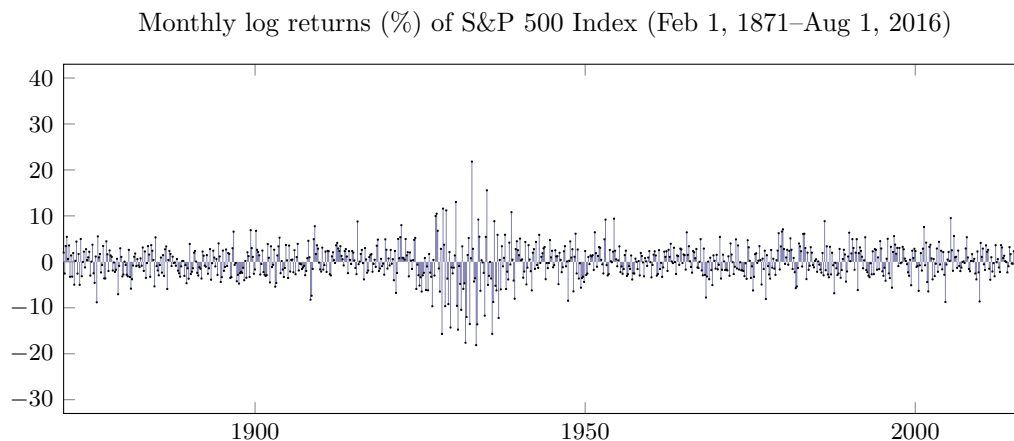


Figure 1.3.: Monthly log returns of the S&P 500 index. The series is a schematic reproduction of the plot shown in class: it reproduces the mean, the typical amplitude, the fat tails and the volatility cluster of the early 1930s, not the individual monthly observations.

Three features stand out, and they will constrain every model we build:

- The returns hover around zero and show no obvious trend, in sharp contrast with the levels.
- Their amplitude is *not* constant: quiet stretches alternate with violent ones (**volatility clustering**), the early 1930s being the extreme case.
- Large moves are far more frequent than a normal distribution would allow.

The last point is easiest to see from the empirical density (Figure 1.4): the body is a sharp peak near zero, but the tails stay noticeably above zero out to $\pm 30\%$.

Remark 1.6.3. 兩個關鍵術語：**波動聚集** (volatility clustering) 指抖動的幅度會成群出現——大波動旁邊通常還是大波動，平靜期也連成一片（1930 年代初最極端），也就是說波動率本身不是常數。**厚尾** (fat tails) 指極端漲跌遠比常態分佈預測的頻繁。看 Figure 1.4：曲線畫的是每月報酬率的出現頻率，中央為 0 附近又高又尖的峰（多數月份幾乎不動），但兩側尾巴延伸到 $\pm 30\%$ 仍明顯高於零，而常態分佈在那裡的機率早已可忽略。這種「尖峰厚尾」的形狀，正是接下來那句結論的依據。

Monthly log returns (%) of S&P 500 Index (Feb 1, 1871–Aug 1, 2016)

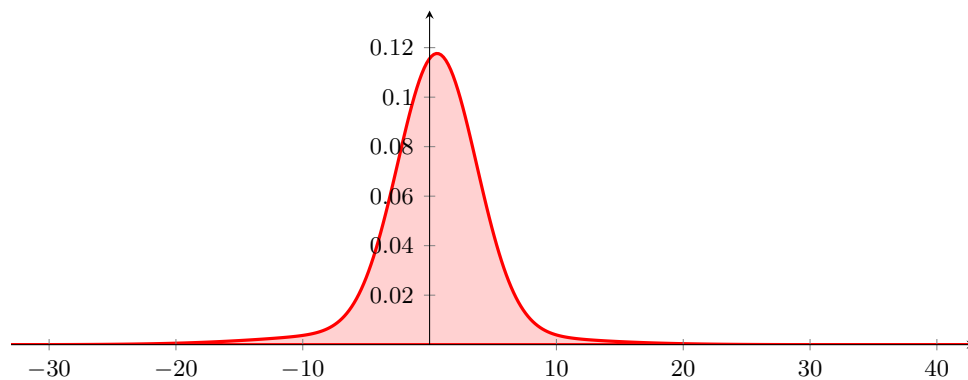


Figure 1.4.: Distribution of the monthly log returns of the S&P 500 index: a sharp central peak with tails much heavier than the normal.

Prices are not normally distributed, and their volatility is not constant. Any model that assumes otherwise is buying tractability with realism.

價格並非常態分佈，波動率也非定值；凡是假設相反的模型，都是以寫實性為代價，換取數學上的易處理性。

§1.7. Analysis of Algorithms

— Slides pp. 20–21 —

*I can calculate the motions
of the heavenly bodies,
but not the madness of people.*

— Isaac Newton (1642–1727)

*It is unworthy of excellent men
to lose hours like slaves
in the labor of computation.*

— Gottfried Wilhelm Leibniz (1646–1716)

Having seen how large the markets are and how awkward their data is, we now fix the vocabulary for the other half of the course: what it means for a pricing problem to be solvable, and at what cost.

§1.7.i. Computability and Algorithms

— Slides p. 22 —

Definition 1.7.1. An **algorithm** is a precise procedure that can be turned into a computer program.

Not every question admits one. **Uncomputable** problems are those for which no algorithm exists at all, no matter how much time we are willing to spend. Two standard examples — both, uncomfortably, about programs themselves:

- Does this program have infinite loops?
- Is this program bug free?

Everything else is **computable**, and computable problems split again according to how expensive the algorithm is: **intractable** problems admit algorithms in principle but not ones we can afford to run, while **tractable** problems admit algorithms that are practical at the sizes we care about. Figure 1.5 summarises the picture.

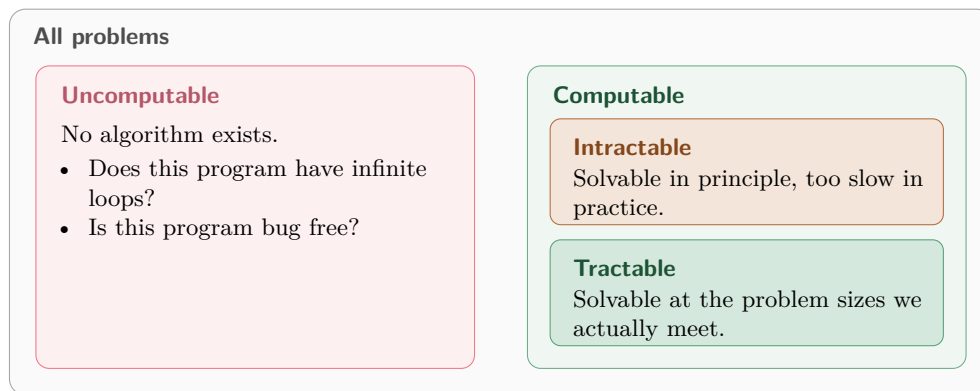


Figure 1.5.: The classification of problems. This course lives almost entirely in the bottom-right box — and much of the research effort consists of moving pricing problems into it.

Remark 1.7.1. The conventional formal criterion is how running time scales with the input size n . A problem is counted *tractable* when it admits an algorithm of *polynomial* time, $O(n^k)$ for some fixed k , and *intractable* when it admits none — every algorithm needs superpolynomial time, in most cases exponential. The rationale is practical: a polynomial bound keeps the cost manageable as n grows, whereas anything superpolynomial blows up.

§1.7.ii. Complexity

— Slides p. 23 —

To measure cost without committing to a machine or a language, we count operations rather than seconds.

Definition 1.7.2. A set of basic operations — $+$, $-$, $\times$, $/$, $\log$, x^y , e^x , $\dots$ — are assumed to take one unit of time each. The total number of such operations performed is the total work done by an algorithm, its **computational complexity**. The amount of memory space it uses is its **space complexity**.

Remark 1.7.2. Two conventions are worth making explicit.

- We concentrate on the *abstract* complexity of an algorithm instead of its detailed implementation. Constant factors from the language, compiler and hardware are deliberately ignored.
- Despite that, complexity is a good guide to an algorithm’s *actual* running time. It is the coarse measure that survives a change of machine.

§1.7.iii. Common (Asymptotic) Complexities

— Slides p. 24 —

Let n stand for the “size” of the problem — the number of elements, the number of cash flows, the number of time periods, and so on. Complexity is then expressed as a function of n ,

and the standard names are as follows.

Name	Complexity	Remark
Linear	$O(n)$	
Quadratic	$O(n^2)$	
Cubic	$O(n^3)$	
Superpolynomial	e.g. $2^{O(\sqrt{n})}$	higher than any polynomial ⁵
Exponential time	$2^{O(n)}$	the usual worst case

Table 1.2.: Common asymptotic complexities, in increasing order of growth.

Note 1.7.1. The superpolynomial entry is not a curiosity. Several pricing problems that were only known to admit exponential-time algorithms have been pushed down to $2^{O(\sqrt{n})}$ — an improvement large enough to turn an unusable algorithm into a usable one, and precisely the kind of result this course aims at.

⁵E.g., T. Dai (B82506025, R86526008, D8852600) & Lyuu (2007); Lyuu & C. Wang (F95922018) (2011); H. Chiu (R98723059) (2012).

2. Basic Financial Mathematics

— Slides pp. 25–26 —

*In the fifteenth century
mathematics was mainly concerned with
questions of commercial arithmetic and
the problems of the architect.*
— Joseph Alois Schumpeter (1883–1950)

*I'm more concerned about
the return of my money than
the return on my money.*
— Will Rogers (1879–1935)

Everything in this chapter follows from one idea: a dollar today and a dollar next year are different goods, and the interest rate is the exchange rate between them. Once that exchange rate is pinned down, valuing a stream of payments is arithmetic — and, as we will see, arithmetic whose cost we can still argue about.

§2.1. The Time Line

— Slides p. 27 —

Before any formula, fix the convention for talking about time. Time is measured in **periods**; time 0 is now, and period k is the interval running from time $k - 1$ to time k . A cash flow “at time k ” arrives at the *end* of period k .

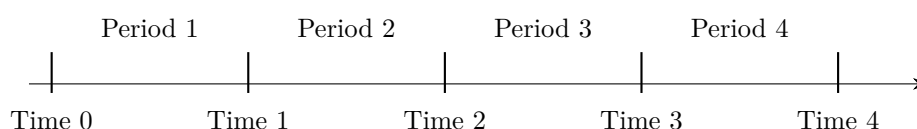


Figure 2.1.: The time line. Period k spans time $k - 1$ to time k .

§2.2. Time Value of Money

— Slides p. 28 —

Definition 2.2.1. Let r be the **interest rate** per period. The **future value** (FV) and **present value** (PV) of a single amount over n periods are related by

$$\begin{aligned} \text{FV} &= \text{PV}(1 + r)^n, \\ \text{PV} &= \text{FV} \times (1 + r)^{-n}. \end{aligned} \tag{2.1}$$

Intuition . $(1 + r)^n$ carries money forward in time and $(1 + r)^{-n}$ carries it backward. *Discounting* — the second operation — is the one we use constantly, because a security’s price is quoted today while its payoffs arrive later.

$(1 + r)^n$ 把錢往未來搬， $(1 + r)^{-n}$ 把錢往現在搬。我們不斷用到的是後者——折現 (discounting) ——因為證券的價格報在今天，收益卻在日後才到。

The idea is old: it goes back to Fibonacci (1170–1240), and was put on a modern footing by Irving Fisher (1867–1947).

§2.3. Compounding

— Slides pp. 29–30 —

Equation (2.1) is stated per *period*, but rates are quoted per *year*. Reconciling the two is what compounding conventions are for — and getting the convention wrong is a classic source of pricing errors.

§2.3.i. Periodic Compounding

Suppose the *annual* interest rate r is compounded m times per annum. Each compounding period earns r/m , so one dollar grows along

$$1 \rightarrow \left(1 + \frac{r}{m}\right) \rightarrow \left(1 + \frac{r}{m}\right)^2 \rightarrow \left(1 + \frac{r}{m}\right)^3 \rightarrow \dots$$

Hence, after n years,

$$\text{FV} = \text{PV} \left(1 + \frac{r}{m}\right)^{nm}. \quad (2.2)$$

§2.3.ii. Common Compounding Methods

Method	m		
Annual	1	Weekly	52
Semiannual	2	Daily	365
Quarterly	4	Continuous	∞
Monthly	12		

§2.3.iii. Translating between Rate and Frequency

— Slides pp. 31–32 —

An annual interest rate of r compounded m times a year is “equivalent to” an interest rate of r/m per $1/m$ year. Read in reverse: if a loan asks for a return of 1% *per month*, the *annual* interest rate will be 12% with monthly compounding.

Example 2.3.1

The annual interest rate is 10% compounded twice per annum. Each dollar will grow to be

$$\left[1 + (0.1/2)\right]^2 = 1.1025$$

one year from now. The rate is therefore “equivalent to” an interest rate of 10.25% com-

pounded once *per annum*, since

$$1 + 0.1025 = 1.1025.$$

The 10.25% in the example is the **effective annual rate**: the single annually-compounded rate that produces the same money after one year. The quoted 10% is only a **nominal rate**, and by itself it does not determine anything.

An interest rate is meaningless without the compounding frequency.

§2.3.iv. The Rule of 72

— Slides pp. 33–35 —

Let the annual interest rate be r with annual compounding. How many years T will it take for your money to double? The identity to solve is

$$(1 + r)^T = 2, \quad \text{so} \quad T = \frac{\ln 2}{\ln(1 + r)}.$$

Is there an easier way? The **rule of 72** is a heuristic that estimates T without a logarithm:

$$T \approx \frac{72}{r \text{ (\%)}}. \quad (2.3)$$

So it takes about $72/12 = 6$ years to double the GDP if the annual growth rate is 12%.

Reason. Expanding $\ln(1 + r) = r - r^2/2 + O(r^3)$ gives, for small r ,

$$\frac{1}{\ln(1 + r)} \approx \frac{1}{r(1 - r/2)} \approx \frac{1}{r} + \frac{1}{2},$$

and therefore

$$\frac{\ln 2}{\ln(1 + r)} \approx \frac{0.693}{r} + 0.3466. \quad \blacksquare$$

Remark 2.3.1. The leading constant is 0.693, i.e. 69.3 when r is measured in percent — so a “rule of 69.3” is the asymptotically correct one. The extra +0.3466 is a positive offset, so inflating the numerator to 72 compensates for it over the range of rates one actually meets. That 72 is also divisible by 2, 3, 4, 6, 8, 9, 12 is a large part of why it, rather than 70, became the folk rule.

Figure 2.2 plots the error of both variants: the true interest rate minus the approximation, in percent, against the number of years to double.

Remark 2.3.2. 這張圖把 Eq. (2.3) 反過來用：平常是「給利率 r ，估翻倍要幾年 T 」，這裡改成「給橫軸的 T （想要幾年翻倍），估需要多少利率」——把公式解成 $r \approx 72/T$ 。縱軸畫的是兩個利率估計值的差：真正在 T 年翻倍所需的 $r_{\text{真}} = 2^{1/T} - 1$ ，減去 Rule of 72 反解出的 $r_{\text{估}} = 72/T$ （兩者都乘 100 化成百分比）。所以縱軸單位是「利率的百分點」，不是「年」；橫軸的 T 在這裡是已知條件，不是被估計的量。

代入數字比較有感覺： $T = 1$ 年時，真正利率 100% 對 Rule of 72 的 72%，差 28 個百分點，極不準； $T = 8$ 年時，真正利率約 9.05% 對 9%，差只有 0.05，正是 caption 說的「超過約 8 年後極準」； $T = 20$ 年時差變成負的（約 -0.07 ），代表 Rule of 72 在長期會略為高估所需利率（“overshoot”）。

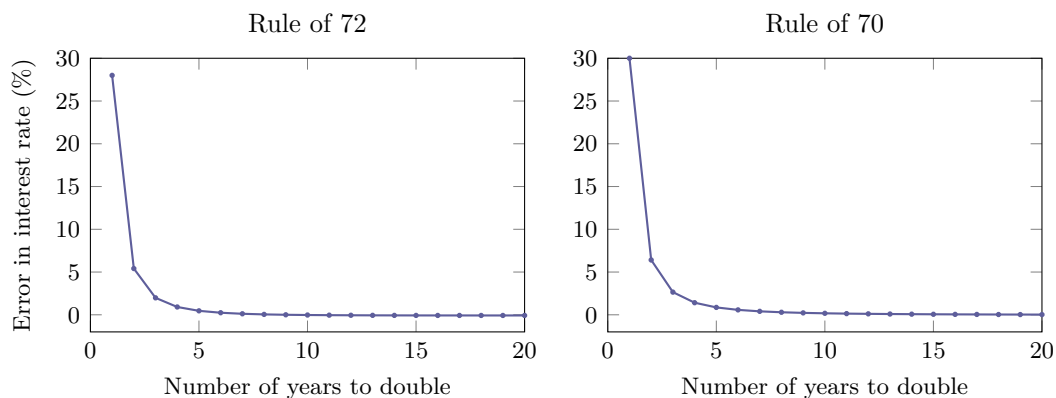


Figure 2.2.: How good is the rule of 72? Plotted is the true interest rate minus the approximation (in %). Both are badly off for very short doubling times and excellent beyond about eight years; the rule of 72 slightly *overshoots* there, the rule of 70 slightly *undershoots*.

§2.3.v. Continuous Compounding

— Slides pp. 36–37 —

Let $m \rightarrow \infty$ in Eq. (2.2), so that

$$\left(1 + \frac{r}{m}\right)^m \rightarrow e^r.$$

Then

$$FV = PV \times e^{rn}, \quad (2.4)$$

where $e = 2.71828\dots$ ¹

Continuous compounding is easier to work with, because the exponents simply add. Suppose the annual interest rate is r_1 for n_1 years and r_2 for the following n_2 years. Then the FV of one dollar will be

$$e^{r_1 n_1 + r_2 n_2}$$

after $n_1 + n_2$ years — no nested products, just a sum in the exponent.

§2.3.vi. Conversion between Compounding Methods

— Slides pp. 38–40 —

The word “equivalent” has already been used twice in quotation marks. Here is what it means.

Definition 2.3.2 (Equivalent rates). The **accumulation factor** $A(n)$ of a quoting convention is the amount one dollar grows to under it after n years:

$$A(n) = \begin{cases} e^{rn}, & r \text{ compounded continuously,} \\ \left(1 + \frac{r}{m}\right)^{mn}, & r \text{ compounded } m \text{ times per annum.} \end{cases}$$

Two quoted annual rates are **equivalent** if they turn one dollar into the same amount of money at every horizon — that is, if their accumulation factors agree as functions of n .

¹Jacob Bernoulli (1654–1705) in 1685.

Note 2.3.1. Checking one year is enough. Each accumulation factor above is a pure power of its own one-year factor, $A(n) = A(1)^n$, so $A_1(1) = A_2(1)$ already forces $A_1(n) = A_2(n)$ for every $n \geq 0$. Equivalently: two rates are equivalent exactly when they share the same **effective annual rate** $A(1) - 1$ of [Section 2.3.iii](#).

Let r_1 be the annual rate with continuous compounding and r_2 the *equivalent* rate compounded m times per annum. How are they related?

Proposition 2.3.3

The principle is that both interest rates must produce the same amount of money after one year, that is,

$$\left(1 + \frac{r_2}{m}\right)^m = e^{r_1}.$$

Therefore^a

$$\begin{aligned} r_1 &= m \ln\left(1 + \frac{r_2}{m}\right), \\ r_2 &= m(e^{r_1/m} - 1). \end{aligned}$$

^aAre they really equivalent? In what sense are they equivalent? Contributed by Mr. Chen, Tung-Li (D09922014) on February 24, 2023.

The same principle handles two periodic conventions. Suppose r_1 is the annual rate compounded m_1 times per annum and r_2 is the *equivalent* rate compounded m_2 times per annum. Then

$$\left(1 + \frac{r_1}{m_1}\right)^{m_1} = \left(1 + \frac{r_2}{m_2}\right)^{m_2}. \quad (2.5)$$

Observation . Every conversion in this subsection is the same one-line demand: *equal value after one year*. There is nothing to memorise beyond that.

§2.4. Present Value

— Slides p. 41 —

§2.4.i. The PV Formula

Definition 2.4.1. The PV of the cash flow $C_1, C_2, \dots, C_n$ at times $1, 2, \dots, n$ is

$$\text{PV} = \frac{C_1}{1+y} + \frac{C_2}{(1+y)^2} + \dots + \frac{C_n}{(1+y)^n}, \quad (2.6)$$

where y is the discount rate per period.

Remark 2.4.1. 我們想要衡量一筆包含未來的錢的現金流折算到現在價值多少，這條公式就是把單筆折現 $\text{PV} = \text{FV}(1+y)^{-n}$ 對每一筆現金流各做一次，再加起來：在 time k 收到的 C_k 要往回搬 k 期，所以除以 $(1+y)^k$ ——越晚到的錢折得越兇——加總便是整串現金流今天的總值。 y 是每期折現率，字母取自 *yield*，角色與前一章的 r 相同。

This formula and its variations are the engine behind most of financial calculations.² The difficulty is never the algebra; it is deciding what goes into it:

- What is y ?
- What are the C_i ?
- What is n ?

Note 2.4.1. Equation (2.6) is taken on faith for now. It will be justified later in the course from “sound” principles rather than assumed.

§2.4.ii. Evaluating the PV Formula

— Slides pp. 42–44 —

Even a formula this simple raises the question the previous chapter set up: how much work does it actually take? The direct transcription is [Algorithm 2.1](#).

Algorithm 2.1: Evaluating the PV in Eq. (2.6) directly

```

1  $x := 0$ ;
2 for  $i = 1, 2, \dots, n$  do
3    $x := x + C_i / (1 + y)^i$ ;
4 return  $x$ ;
```

Computing $(1 + y)^i$ by repeated multiplication costs i operations, so the algorithm takes time proportional to $\sum_{i=1}^n i = O(n^2)$ if only $+$, $-$, $\times$ and $/$ are allowed. It can be improved to $O(n)$ by applying $a^b = e^{b \ln a}$ in step 3, since we agreed to count x^y as taking one unit of time.

But there is a better route, which needs no exponentiation at all ([Algorithm 2.2](#)).

Algorithm 2.2: Evaluating the PV by Horner’s rule

```

1  $x := 0$ ;
2  $d := 1 + y$ ;
3 for  $i = n, n - 1, \dots, 1$  do
4    $x := (x + C_i) / d$ ;
5 return  $x$ ;
```

Proposition 2.4.2 (Horner’s rule)

The underlying identity is

$$\left(\cdots \left(\left(\frac{C_n}{1+y} + C_{n-1} \right) \frac{1}{1+y} + C_{n-2} \right) \frac{1}{1+y} + \cdots \right) \frac{1}{1+y},$$

due to Horner (1786–1837) in 1819.

The algorithm takes $O(n)$ time, and it is the most efficient possible in terms of the absolute number of arithmetic operations.³ It can also be computed in parallel in $O(\log n)$ time using prefix sums.⁴

²Cochrane (2005), “Asset pricing theory all stems from one simple concept [...]: price equals expected discounted payoff.”

³A. Borodin & Munro (1975).

⁴Contributed by Mr. Chen, Yi-Feng (R08922077) on February 24, 2021.

Two algorithms for the same formula, $O(n^2)$ against $O(n)$ — and the fast one is also the numerically simpler one. This is the pattern the whole course chases.

§2.5. Annuities

— Slides pp. 45–47 —

§2.5.i. Annuities Certain

Definition 2.5.1. An **annuity** pays out the same C dollars at the end of each year for n years.

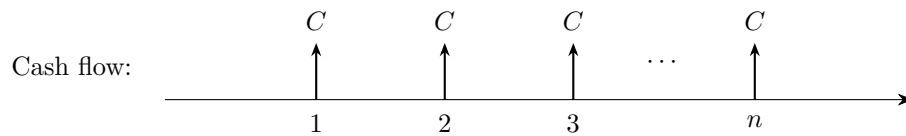


Figure 2.3.: The cash flow of an n -year annuity certain.

With a rate of r , specialising Eq. (2.6) to $C_i \equiv C$ and summing the geometric series gives

$$\text{PV} = \sum_{i=1}^n C(1+r)^{-i} = C \frac{1 - (1+r)^{-n}}{r},$$

while the FV at the end of the n th year is

$$\text{FV} = \sum_{i=0}^{n-1} C(1+r)^i = C \frac{(1+r)^n - 1}{r}.$$

The study of these sums goes back to Jan de Witt (1625–1672) in 1671 and Nicholas Bernoulli (1687–1759) in 1709.

§2.5.ii. General Annuities

Suppose instead that m payments of C dollars each are received per year (the **general annuity**), and let r be compounded m times per annum.⁵ Then the same geometric sum with period rate r/m and nm periods gives

$$\begin{aligned} \text{PV} &= \sum_{i=1}^{nm} C \left(1 + \frac{r}{m}\right)^{-i} = C \frac{1 - \left(1 + \frac{r}{m}\right)^{-nm}}{\frac{r}{m}}, \\ \text{FV} &= \sum_{i=0}^{nm-1} C \left(1 + \frac{r}{m}\right)^i = C \frac{\left(1 + \frac{r}{m}\right)^{nm} - 1}{\frac{r}{m}}. \end{aligned} \tag{2.7}$$

Remark 2.5.1. Annuity is paid at the end of every year, so note the indices of above formulas.

⁵If your r has a compounding frequency different from m , convert it with Eq. (2.5) first.

§2.6. Amortization

— Slides pp. 48–49 —

§2.6.i. The Amortization Method

Definition 2.6.1. **Amortization** is a method of repaying a loan through regular payments of interest *and* principal.

The mechanics are worth stating carefully, because every line of the schedule below follows from them:

- The size of the loan (the original balance) is reduced by the *principal* part of each payment.
- The *interest* part of each payment pays the interest incurred on the *remaining* principal balance.
- As the principal gets paid down over the term of the loan, the interest part of the payment diminishes.

An amortized loan is exactly a general annuity seen from the lender's side: the loan amount is the PV of the payment stream. Inverting Eq. (2.7) therefore gives the regular payment,

$$C = \text{loan amount} \times \frac{\frac{r}{m}}{1 - \left(1 + \frac{r}{m}\right)^{-nm}}. \quad (2.8)$$

For monthly payments, take $m = 12$.

Remark 2.6.1. 每期還款總金額一樣，只是把每期還款拆賬成上期剩餘本金造成的利息跟一部分本金。

Remark 2.6.2. 初看之下兩者的計息方式並不一樣：攤還 (amortization) 貸款的利息是對剩餘本金計算的，而年金 (annuity) 公式 (2.7) 裡根本沒有出現「餘額」，只是把每期的 C 折現後相加。兩者一致的理由是：折現本身就是「利息只算到該筆錢實際在外流通為止」的數學編碼。以下三點說的是同一件事。

(a) 這是恆等式，不是類比。記 $i \triangleq r/m$ 、 $N \triangleq nm$ ，並令 B_k 為第 k 期付款後的剩餘本金。「利息按剩餘本金計、付款先付息再沖本」這句話的全部內容就是遞迴

$$B_k = B_{k-1}(1+i) - C, \quad B_0 = \text{loan amount}.$$

把它改寫成 $B_{k-1} = (C + B_k)/(1+i)$ 並反覆代入自己，得

$$B_0 = \sum_{k=1}^N \frac{C}{(1+i)^k} + \frac{B_N}{(1+i)^N}.$$

而 *fully amortized* 的意思正是 $B_N = 0$ ，末項消失，於是貸款額 = 付款流的現值 (present value)，即 (2.7)。年金並不是用來近似攤還貸款的模型，它就是那條餘額遞迴式的封閉解。

(b) 折現如何「知道」餘額在變。把 B_0 拆成 N 筆互不相干的零息小貸款，第 k 筆的金額恰為 $C(1+i)^{-k}$ 。它獨自滾 k 期複利後成為 $C(1+i)^{-k}(1+i)^k = C$ ，剛好被第 k 期的付款一次結清，此後即離開帳本、不再生息。每一筆都只在自己尚未被償還的期間生息，合起來就是「利息只對剩餘本金計算」。關鍵在於折現因子的指數是該筆付款自己的期數 k 而非全期 N ——已經還掉的錢不再出現在後面各項裡。

(c) 兩個假設要分開。(a) 的恆等式對任意付款序列 $C_1, \dots, C_N$ 都成立，此時 $B_0 = \sum_k C_k(1+i)^{-k}$ ；「餘額計息」本身只保證這件事，與付款是否等額無關。真正額外用到的

假設是 等額付款 ($C_k \equiv C$)，它使上式成為等比級數、能求和成封閉解 (2.7)，我們也才能反解出 (2.8)。

§2.6.ii. Example: Home Mortgage

— Slides pp. 50–54 —

By paying down the principal consistently, the risk to the lender is lowered, and when the borrower sells the house, only the remaining principal is due the lender. Consider the equal-payment case, i.e., fixed-rate, level-payment, fully amortized mortgages — these are called traditional mortgages in the U.S.⁶

Remark 2.6.3. 上段中譯：藉由持續償還本金，放款人承擔的風險逐步降低；當借款人賣掉房子時，只需清償當時的剩餘本金即可。以下只考慮等額付款的情形，也就是 **固定利率** (fixed-rate)、**等額付款** (level-payment)、**完全攤還** (fully amortized) 的房貸——這類房貸在美國被稱為 **傳統房貸** (traditional mortgage)。

註腳中譯：《經濟學人》(2016)：「多數國家的銀行以短期或浮動利率房貸來壓低自身風險；美國的借款人則享有較好的條件——便宜的 30 年期固定利率房貸，而且可以免費提前清償（即無提前清償違約金）。」

Example 2.6.2

Consider a 15-year, \$250,000 loan at 8.0% interest rate. Solve Eq. (2.7) with $PV = 250000$, $n = 15$, $m = 12$ and $r = 0.08$. That is,

$$250000 = C \frac{1 - \left(1 + \frac{0.08}{12}\right)^{-15 \times 12}}{\frac{0.08}{12}},$$

which gives a monthly payment of $C = 2389.13$.

Month	Payment	Interest	Principal	Remaining principal
				250,000.000
1	2,389.13	1,666.667	722.464	249,277.536
2	2,389.13	1,661.850	727.280	248,550.256
3	2,389.13	1,657.002	732.129	247,818.128
		...		
178	2,389.13	47.153	2,341.980	4,730.899
179	2,389.13	31.539	2,357.591	2,373.308
180	2,389.13	15.822	2,373.308	0.000
Total	430,043.438	180,043.438	250,000.000	

Table 2.1.: The amortization schedule of the 15-year, \$250,000 loan at 8%.

In every month:

- The principal and interest parts add up to \$2,389.13.
- The remaining principal is reduced by the amount indicated under the **Principal** heading.⁷

⁶The *Economist* (2016), “In most countries banks minimize their risk by offering short-term or floating-rate mortgages. American borrowers get a better deal: cheap 30-year fixed-rate mortgages that can be repaid early free.”

⁷This column varies with r . Thanks to a lively class discussion on Feb 24, 2010. In fact, every column varies with r . Contributed by Ms. Wu, Japie (R01921056) on February 20, 2013.

The Principal column forms a geometric sequence.⁸

- The interest is computed by multiplying the remaining balance of the previous month by $0.08/12$.

Note also that:

- The Principal column adds up to \$250,000 — it must, since the loan is fully amortized.
- The Payment column adds up to \$430,043.438! The difference, \$180,043.438, is the total interest paid over 15 years.
- If the borrower plans to pay off the mortgage right *after* the 178th month's monthly payment, he needs to pay another \$4,730.899.⁹

Figure 2.4 shows the third bullet of Table 2.1 graphically: the payment is level, but its composition rotates from almost all interest to almost all principal.

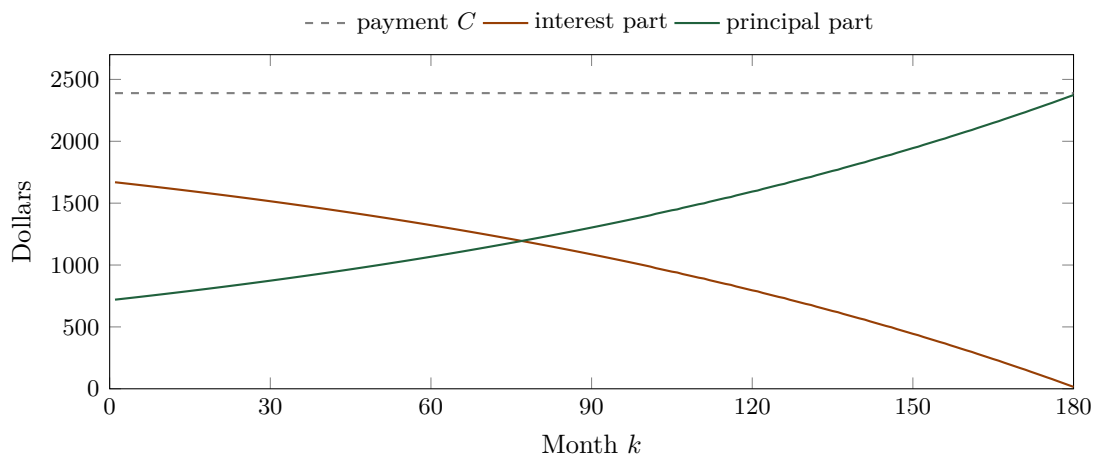


Figure 2.4.: Composition of the level monthly payment over the life of the loan, computed from the example above. The two parts always sum to C , and the principal part is a geometric sequence with ratio $1 + r/m$.

§2.6.iii. Calculating the Remaining Principal

— Slides pp. 55–57 —

Two ways to answer “how much is still owed after k payments?”

Method 1: run the schedule

- A month's principal payment = monthly payment – (previous month's remaining principal) $\times$ (monthly interest rate).
- A month's remaining principal = previous month's remaining principal – principal payment calculated above.
- Generate the amortization schedule until the particular month you are interested in.

This method is relatively slow — it costs $O(k)$ work — but it is universal in its applicability. It can, for example, accommodate prepayments and floating interest rates, neither of which has a closed form.

⁸Contributed by Mr. Sun, Ao (R05922147) on February 22, 2017.

⁹Contributed by Ms. Wu, Japie (R01921056) on February 20, 2013.

Method 2: discount the remaining cash flows

Right after the k th payment, the remaining principal is the PV of the future $nm - k$ cash flows because we have

$$B_{k-1}(1+i) - C = B_k$$

and we can solve this to obtain

$$B_k = \sum_{i=1}^{nm-k} C \left(1 + \frac{r}{m}\right)^{-i} = C \frac{1 - \left(1 + \frac{r}{m}\right)^{-nm+k}}{\frac{r}{m}}. \quad (2.9)$$

This method is much faster — $O(1)$ instead of $O(k)$ — but it is limited in applications because it makes more assumptions: a level payment, a fixed rate, and no prepayment.

Exercise 2.6.3. Use Eq. (2.9) together with Eq. (2.8) to prove that the Principal column of Table 2.1 forms a geometric sequence.

The same trade-off as in Algorithm 2.2: the closed form is faster, the loop is more general. Which one you may use is decided by the contract, not by the mathematics.

§2.7. Yields and the Internal Rate of Return

— Slides pp. 58–59 —

So far the rate was given and we computed a value. The rest of the chapter runs the other way: the price is observed in the market, and we solve for the rate it implies.

§2.7.i. Bond Equivalent Yield and Mortgage Equivalent Yield

The term **yield** denotes the return of investment. Two widely used yields are the **bond equivalent yield** (BEY) and the **mortgage equivalent yield** (MEY). Recall Eq. (2.2), $FV = PV \left(1 + \frac{r}{m}\right)^{nm}$:

- BEY corresponds to the r above that equates PV with FV when $m = 2$.
- MEY corresponds to the r above that equates PV with FV when $m = 12$.

Observation . BEY and MEY may differ numerically, but they refer to the same yield — they are the same economic fact quoted under two different compounding conventions, exactly as in Equation 2.5.

Remark 2.7.1. 殖利率 (yield)、債券等值殖利率 (bond equivalent yield, BEY)、房貸等值殖利率 (mortgage equivalent yield, MEY)。

上面兩點的意思是：市場給的事實只有「今天付 PV、 n 年後拿回 FV」，而年化利率並非事實而是報價——必須先選定一年複利幾次，才解得出

$$r = m \left[\left(\frac{FV}{PV} \right)^{1/(nm)} - 1 \right],$$

其值隨 m 而異。取 $m = 2$ 者稱 BEY，取 $m = 12$ 者稱 MEY。例如今天付 100、一年後拿回 108.16，同一筆現金流可報價為 8.16% ($m = 1$ ，即**有效年利率** (effective annual rate, EAR))、8.00% (BEY) 或 7.87% (MEY)： m 愈大，達成同樣成長所需的**名目年利**

率 (nominal rate) 愈小。

兩個名字來自報價慣例——美國公債每半年付息 ($m = 2$)，房貸每月繳款 ($m = 12$)。因此跨市場比較前務必先換算，否則會誤判：房貸的 8% MEY 相當於債券的 8.13% BEY，兩個 8% 並不一樣。

§2.7.ii. The Internal Rate of Return

Definition 2.7.1. The **internal rate of return** (IRR) is the yield y which equates an investment's PV with its price P ,

$$P = \frac{C_1}{1+y} + \frac{C_2}{(1+y)^2} + \cdots + \frac{C_n}{(1+y)^n}.$$

Remark 2.7.2. **內部報酬率** (internal rate of return, IRR)；用在債券上時通常稱 **到期殖利率** (yield to maturity, YTM)。

上述定義口語一點的說法是：期初投入 P 元，此後在各個時點分別回收 $C_1, \dots, C_n$ 元；IRR 就是讓這些回收恰好抵掉當初那筆投入的利率。說得再具體些，令 B_k 表示第 k 期回收之後尚未收回的投資餘額：期初分文未收，故 $B_0 = P$ ；此後餘額每期先以利率 y 增值，再扣掉當期收到的 C_k 。IRR 就是使餘額在最後一期恰好歸零的那個 y ：

$$B_0 = P, \quad B_k = B_{k-1}(1+y) - C_k, \quad B_n = 0.$$

把這條遞迴展開即回到上面的定義式。之所以稱為「內部」，是因為算它只用到交易本身的現金流，不必從外部指定折現率。

Table 2.1 正是這條遞迴的逐期紀錄：餘額每月以 0.08/12 滾息、再減去 2,389.13 元，到第 180 個月剛好歸零。所以 8% 就是那筆房貸對放款人而言的 IRR——我們當初是先給定利率去求月付款，而 IRR 是同一條式子反過來問：給定價格與現金流，利率是多少？

IRR 的好處是自動把時間價值算進去。例如今天付 1,000、往後三年各收回 400，解 $1000 = \sum_{k=1}^3 400(1+y)^{-k}$ 得 $y = 9.70\%$ ；若同樣的 1,200 元要到第三年才一次收到，則只有 $1.2^{1/3} - 1 = 6.27\%$ 。總金額相同，但早收到的錢報酬率高得多。

惟下述再投資假設不可輕忽：承上例，若中途收回的 400 只能以 3% 再投資，實際年化報酬僅 7.33%，而非宣稱的 9.70%。

最後， y 在定義式中同時出現在 n 個不同的次方裡，求解 y 等同於解一個 n 次多項式的根，一般沒有公式解，只能用 Section 2.8 的數值方法逼近。

Remark 2.7.3. The IRR assumes all cash flows are reinvested at the *same* rate as the internal rate of return, because

$$FV = C_1(1+y)^{n-1} + C_2(1+y)^{n-2} + \cdots + C_n.$$

So it must be used with caution. The issue disappears when there are no intermediate cash flows — which is one reason zero-coupon bonds (Section 2.9.i) are so convenient.

§2.8. Numerical Methods for IRRs

§2.8.i. The Root-Finding Problem

Define

$$f(y) \triangleq \sum_{t=1}^n \frac{C_t}{(1+y)^t} - P,$$

where P is the market price, and solve $f(y) = 0$ for a real $y \geq -1$.¹⁰

Proposition 2.8.1

If $C_t > 0$ for all t , then $f(y)$ is monotonically decreasing in y , so a unique real-number solution exists.

There is no closed form for y in general, so we need iterative methods. The three below trade off robustness against speed.

Remark 2.8.1. *Trade off A against B* 意為在 A 與 B 之間權衡取捨：想多要一點 B，就得犧牲一些 A。此處的兩個軸是**穩健性**（robustness，是否保證收斂、對起始值是否敏感）與**速度**（收斂階數），三者的對照見 Table 2.2。

二分法最穩健：只要初始區間滿足 $f(a)f(b) < 0$ 便保證收斂，但每步僅將區間減半，故最慢。牛頓法與割線法則都不保證收斂。例如 $f(x) = x^{1/3}$ 的牛頓疊代為 $x_{k+1} = x_k - 3x_k = -2x_k$ ，自任何非零起點出發都翻倍發散；而 $f(x) = x^3 - 2x + 2$ 自 $x_0 = 0$ 出發則在 $0 \leftrightarrow 1$ 之間永遠循環，一步都不前進。此外 $f'(x_k) \approx 0$ 時步長會爆炸，多根時亦可能收斂到非預期的那一個；割線法還多一個分母 $f(x_k) - f(x_{k-1})$ 趨零的風險。理論上能保證的只有局部收斂：起點充分接近單根時保證二次收斂，但定理並未告訴你「充分接近」究竟是多近。

不過就 IRR 這個問題而言，Proposition 2.8.1 其實已把風險排除了。當所有 $C_t > 0$ 時 f 不僅單調遞減，還滿足 $f''(y) = \sum_t t(t+1)C_t(1+y)^{-t-2} > 0$ 而為凸函數。凸函數的切線恆在曲線下方，故 $f(x_{k+1}) \geq 0$ ；又 f 遞減，這表示第一步之後每個迭代點都落在根的左側。此後 $f(x_k) \geq 0$ 而 $f'(x_k) < 0$ ，步長 $-f/f'$ 恆非負，迭代遂單調遞增且以根為上界，必定收斂。所以該命題不只是在說「根唯一」，更是在替牛頓法背書：對付息全為正的 IRR，牛頓法自任何 $y_0 > -1$ 出發都保證收斂。

實務上的求根程式（如 Brent's method）多半混用：先以二分法將根框住，再切換至割線或牛頓法加速，一旦迭代點跳出括號便退回二分法——保證來自二分法，速度來自另外兩者。

§2.8.ii. The Bisection Method

Start with a and b where $a < b$ and $f(a)f(b) < 0$. Then $f(\xi)$ must be zero for some $\xi \in [a, b]$. If we evaluate f at the midpoint $c \triangleq (a+b)/2$, either

- (1) $f(c) = 0$,
- (2) $f(a)f(c) < 0$, or
- (3) $f(c)f(b) < 0$.

In the first case we are done; in the second we continue the process with the new bracket $[a, c]$; and in the third we continue with $[c, b]$. The bracket is halved in the latter two cases, so after n steps ξ lies within a bracket of length $(b-a)/2^n$.

¹⁰Negative interest rates became a reality for German and Swiss bonds in 2015. In 2016, Sweden, Denmark, and Japan imposed negative interest rates on excess reserves. As many as 355 corporate bonds were issued with negative yields as of June of 2016. Even so, -100% should be a natural lower bound because why would anyone or financial institutions want to have every cent confiscated?

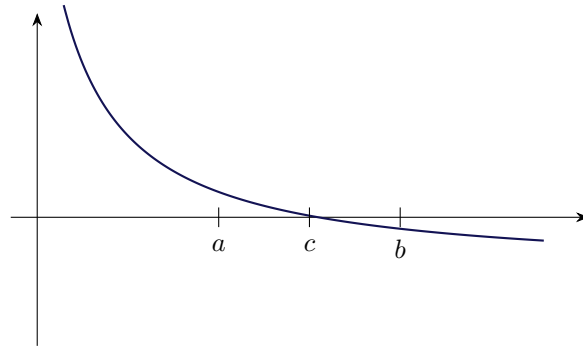


Figure 2.5.: The bisection method. f is decreasing with $f(a) > 0 > f(b)$, so the root is bracketed; evaluating at c halves the bracket.

§2.8.iii. The Newton–Raphson Method

It converges faster than the bisection method. Start with a first approximation x_0 to a root of $f(x) = 0$. Then

$$x_{k+1} \triangleq x_k - \frac{f(x_k)}{f'(x_k)}.$$

When computing yields,

$$f'(x) = - \sum_{t=1}^n \frac{t C_t}{(1+x)^{t+1}}. \quad (2.10)$$

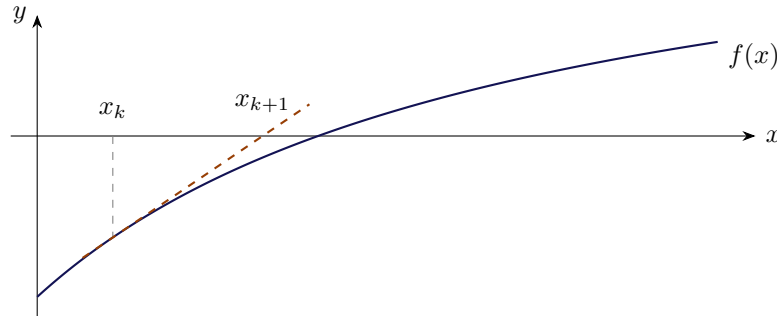


Figure 2.6.: The Newton–Raphson method. The tangent at x_k is followed to the axis to give x_{k+1} .

§2.8.iv. The Secant Method

A variant of the Newton–Raphson method: replace differentiation with difference. Start with two approximations x_0 and x_1 , then compute the $(k+1)$ st approximation with

$$x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})},$$

so $[f(x_k) - f(x_{k-1})]/(x_k - x_{k-1})$ replaces $f'(x_k)$.

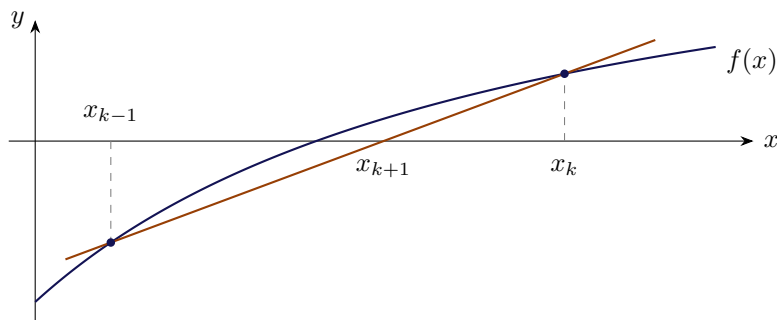


Figure 2.7.: The secant method. The chord through the two latest iterates is followed to the axis to give x_{k+1} ; compare Figure 2.6, which follows the tangent at a single iterate instead.

Its convergence rate is 1.618, slightly worse than the Newton–Raphson method’s 2. But the secant method does not evaluate $f'(x_k)$, which saves about 50% in computation efforts per iteration. For reference, the convergence rate of the bisection method is 1.

Method	Convergence rate	Needs f' ?	Convergence guaranteed?
Bisection	1	no	yes, once bracketed
Secant	1.618	no	no
Newton–Raphson	2	yes	no (see Proposition 2.8.1)

Table 2.2.: The three root-finding methods. Higher rate means fewer iterations, but each Newton–Raphson iteration costs roughly twice a secant one — and the last column is what that speed is bought with. Bisection cannot fail once $f(a)f(b) < 0$; the other two may diverge, cycle, or land on the wrong root.

Remark 2.8.2. 割線法 (secant method) 的幾何意義見 Figure 2.7：牛頓法在一個點上作切線、追到 x 軸得下一個近似值，割線法則改取最近兩個點連成的割線，追到 x 軸得下一個近似值——名稱即由此而來。公式也是這樣來的：通過 $(x_{k-1}, f(x_{k-1}))$ 與 $(x_k, f(x_k))$ 的直線斜率為差商 $[f(x_k) - f(x_{k-1})]/(x_k - x_{k-1})$ ，令該直線的 y 座標為零解出交點，就得到上式。換言之，割線法就是把牛頓法裡的 $f'(x_k)$ 換成差商——用兩點的實際落差去估斜率，而不去要求函數可微或另外寫一份導數程式。

所謂省下約 50% 的計算量，關鍵在於 $f(x_{k-1})$ 在上一輪就已經算過、可直接沿用，故每輪只需一次新的函數求值；牛頓法每輪則要各算一次 f 與 f' ，而由 (2.10) 可見算 f' 的成本與算 f 相當。若改以每次函數求值（而非每輪疊代）來衡量效率，牛頓法為 $2^{1/2} \approx 1.414$ 、割線法為 1.618：割線法反而略勝一籌。

至於 1.618 這個怪數字，來自誤差遞迴 $e_{k+1} \approx C e_k e_{k-1}$ ：設 $e_{k+1} \sim A e_k^p$ 代入可得 $p^2 = p + 1$ ，故 $p = (1 + \sqrt{5})/2$ ，恰為黃金比例。

最後可與二分法對照：兩者都持有兩個點，但二分法只看 f 的正負號並取中點，割線法則利用 f 的數值大小外推。多用了資訊所以更快，代價是失去保證——割線法不保證收斂，且當 $f(x_k) \approx f(x_{k-1})$ 時分母趨近於零，數值上會不穩定。

§2.8.v. Systems of Nonlinear Equations

It is not easy to extend the bisection method to higher dimensions — there is no natural notion of “bracketing” a root in the plane. But the Newton–Raphson method extends directly.

Let (x_k, y_k) be the k th approximation to the solution of the two simultaneous equations

$$f(x, y) = 0, \quad g(x, y) = 0.$$

The $(k+1)$ st approximation (x_{k+1}, y_{k+1}) satisfies the following *linear* equations,

$$\begin{bmatrix} \frac{\partial f(x_k, y_k)}{\partial x} & \frac{\partial f(x_k, y_k)}{\partial y} \\ \frac{\partial g(x_k, y_k)}{\partial x} & \frac{\partial g(x_k, y_k)}{\partial y} \end{bmatrix} \begin{bmatrix} \Delta x_{k+1} \\ \Delta y_{k+1} \end{bmatrix} = - \begin{bmatrix} f(x_k, y_k) \\ g(x_k, y_k) \end{bmatrix},$$

with unknowns

$$\Delta x_{k+1} \triangleq x_{k+1} - x_k, \quad \Delta y_{k+1} \triangleq y_{k+1} - y_k.$$

The above has a unique solution for $(\Delta x_{k+1}, \Delta y_{k+1})$ when the 2×2 matrix — the Jacobian — is invertible. Finally, set

$$x_{k+1} = x_k + \Delta x_{k+1}, \quad y_{k+1} = y_k + \Delta y_{k+1}.$$

Intuition . Each step replaces the nonlinear system by its linearisation at the current point and solves that exactly. In one dimension the linearisation is the tangent line of [Figure 2.6](#); here it is the tangent plane.

Remark 2.8.3. 這節處理的是「 n 條方程式、 n 個未知數，且方程式非線性」的情形，前面求 IRR 不過是 $n = 1$ 的特例。金融上最常見的來源是**校準** (calibration)：以市場上觀測到的 n 個價格，反解出模型的 n 個未知參數——例如用兩檔債券的市價反推某個兩參數利率模型的兩個參數。

二分法之所以推不到高維，是因為它的保證來自中間值定理：一維上 f 一變號就把根夾住了。但在平面上 $f(x, y) = 0$ 的解集通常是一條曲線而非一個點，我們要的是它與 $g(x, y) = 0$ 那條曲線的交點；此時並沒有「把區域對半切、且能判斷根落在哪一半」的辦法，**框住根** (bracketing) 這個概念本身就不存在。

至於那條矩陣方程式並非憑空而來，它就是一階泰勒展開後令其為零。在 (x_k, y_k) 附近，

$$f(x_{k+1}, y_{k+1}) \approx f(x_k, y_k) + \frac{\partial f}{\partial x} \Delta x_{k+1} + \frac{\partial f}{\partial y} \Delta y_{k+1},$$

g 亦然。我們希望新的一點恰好是解，於是令兩式右側皆為零、把常數項移到等號另一邊，寫成矩陣形式就是上式。

因此這整套做法與一維的牛頓法其實是同一件事。把該矩陣方程記為 $J\Delta = -F$ ，則 $\Delta = -J^{-1}F$ ，亦即 $\mathbf{x}_{k+1} = \mathbf{x}_k - J^{-1}F(\mathbf{x}_k)$ ：一維的「除以導數 $f'(x_k)$ 」在多維成了「左乘 Jacobian 的反矩陣」，而一維要求 $f'(x_k) \neq 0$ 也正對應到此處要求 J 可逆。繞這一圈的用意則在於，非線性方程組難解，線性方程組卻可直接以高斯消去法解出。

§2.9. Bonds

— Slides pp. 70–74 —

§2.9.i. Zero-Coupon Bonds

Definition 2.9.1. A **zero-coupon bond** (or **pure discount bond**) makes a single payment at maturity. By Eq. (2.1), the price of one that pays F dollars in n periods is

$$F/(1+r)^n, \tag{2.11}$$

where r is the interest rate per period.

Zero-coupon bonds can be used to meet future obligations as there is no reinvestment risk — the caveat attached to the IRR above simply does not arise.

Remark 2.9.1. 零息債券 (zero-coupon bond)，亦稱純折價債券 (pure discount bond)； F 為面值 (par value，即票面金額)， n 為到期期數。口語地說，它就是一張「中途完全不付息、到期一次付你一筆固定金額」的債券——正因為中途沒有任何票息 (coupon)，才叫「零息」。它的報酬全部來自買價與面值之間的價差：由 (2.11) 可知，只要 $r > 0$ 價格便必然低於面值，即必定折價發行，故又稱純折價債券。

上段所謂沒有再投資風險 (reinvestment risk)，指的正是先前 IRR 那條但書在此不成立：中途沒有現金流，就沒有錢需要再投資，也就無從假設再投資利率。因此零息債券的 IRR 就是你實際拿到的報酬，一分不差；這也是為何它適合用來支應未來某個時點的確定支出。

例中 20.83 這個數字採債券市場的報價慣例——以面額 100 為基準，故 20.83 意為面額 100 元的債券今天只賣 20.83 元。另可留意 $45.64\% \times 45.64\% \approx 20.83\%$ ：20 年期的價格恰為 10 年期價格的平方，因為折現是連乘而非相加。

Example 2.9.2

The interest rate is 8% compounded semiannually. A zero-coupon bond that pays the par value 20 years from now will be priced at $1/(1.04)^{40}$, that is, 20.83% of its par value.^a It will be quoted as 20.83. If the bond matures in 10 years instead of 20, its price would be 45.64.

^aOne fifth!

§2.9.ii. Level-Coupon Bonds

A **level-coupon bond** is described by its **coupon rate** and its **par value**, paid at maturity. Write F for the par value and C for the coupon.

Remark 2.9.2. 固定票息債券 (level-coupon bond，亦稱附息債券)； c 為票面利率 (coupon rate)， F 為面值 (par value)， C 為每期的票息 (coupon)。口語地說，它就是「每期付你固定的 C 元，到期那一期除了 C 之外再一次還你 F 元」的債券，現金流見 Figure 2.8。留意原文的 *paid at maturity* 修飾的是 par value 而非 coupon rate：票息是每期都付，面值則是到期才一次付清。所謂 *level* (等額) 是指每期票息固定不變，以與浮動利率債券相區別。

票面利率與票息的換算為下文的 $C = Fc/m$ ： c 是年利率，乘上面值得一年的總票息，再除以一年付息次數 m 即為每期票息。例如面值 100、票面利率 8%、每半年付息一次，則 $m = 2$ 、每期票息 $C = 100 \times 0.08/2 = 4$ 元。

最要緊的是分清 c 與 r 這兩個利率： c 由合約寫死，決定你每期收到多少錢，發行後永不改變； r 則是市場折現率，隨市場浮動，決定這串現金流今天值多少錢。兩者的大小關係恰好決定債券售價高於、等於或低於面值，即溢價 (premium)、平價 (par) 與折價 (discount) 三種情形：當 $c = r$ 時 $C = Fr/m$ ，代入 (2.12) 後前項化為 $F[1 - (1 + \frac{r}{m})^{-n}]$ ，與後項 $F(1 + \frac{r}{m})^{-n}$ 相加恰得 $P = F$ ；又 P 對 r 遞減，故

$$c = r \Rightarrow P = F \text{ (平價)}, \quad c < r \Rightarrow P < F \text{ (折價)}, \quad c > r \Rightarrow P > F \text{ (溢價)}.$$

而 $c = 0$ 即退化為前一小節的零息債券，價格恰為 (2.11)——零息債券只是本式的特例。

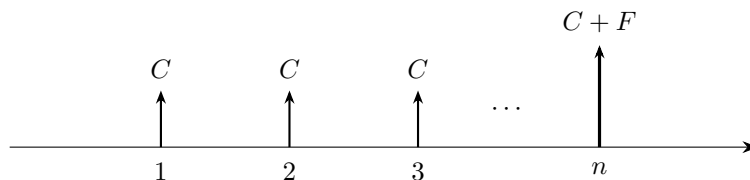


Figure 2.8.: The cash flow of a level-coupon bond.

Remark 2.9.3. Figure 2.8 隱含一項理想化的假設：今天恰好站在付息日上，故各筆現金流剛好落在 $1, 2, \dots, n$ 這些整數期上。真實的交割日多半落在兩個付息日之間，屆時每一筆現金流都要往前挪，指數由 i 變成 $\omega + i$ ——這正是 (2.15) 所處理的情形。

不過要留意，最後一期同時收到 C 與 F 並非理想化，真實債券的最後一次付息本就與本金同日支付。這一點在 (2.15) 中依然成立：末筆票息 ($i = n - 1$) 與 F 的折現指數同為 $\omega + n - 1$ ，可見放寬假設後兩者仍同時發生。

Coupon bonds can be thought of as a matching package of zero-coupon bonds, at least theoretically.^a

^a“You see, Daddy didn’t bake the cake, and Daddy isn’t the one who gets to eat it. But he gets to slice the cake and hand it out. And when he does, little golden crumbs fall off the cake. And Daddy gets to eat those,” wrote Tom Wolfe (1931–2018) in *Bonfire of the Vanities* (1987).

Remark 2.9.4. 這條 moral 的具體意思是：一張 n 期的付息債券，可視為 n 張面額 C 、到期日分別為 $1, 2, \dots, n$ 的零息債券，再加上一張面額 F 、到期日為 n 的零息債券，所組成的投資組合。下方的定價公式 (2.12) 正是把這個拆解逐項寫出來：第一項是 n 張票息零息債券的現值之和，第二項則是面值那一張的現值。

所謂 *at least theoretically*，是指這種拆解未必真能在市場上執行；但在美國公債市場它確實可行。自 1985 年起的 **STRIPS** 制度 (Separate Trading of Registered Interest and Principal of Securities) 允許交易商將一張公債的每一期票息與本金拆開、各自登記為獨立的零息債券單獨買賣，亦允許反向重組回原債券。雙向皆可執行，套利便會迫使「整張債券的價格」等於「各零息碎片價格之和」——這條 moral 因而不只是心算技巧，而是由無套利條件所保證。

這個觀點日後還有一個更要緊的用處：拆開之後每一張零息債券各有自己的到期日，理應各用該期限的即期利率折現。(2.12) 對所有期別一律套用單一的 r ，其實是一種簡化，正確的做法是改用一整條利率期限結構。

至於註腳的典故，出自 Tom Wolfe 的小說 *Bonfire of the Vanities* (1987)：書中債券業務員 Sherman McCoy 的妻子向女兒解釋爸爸的工作——蛋糕不是爸爸烤的，也不是爸爸吃的，但由爸爸負責切開、分送給大家；切的時候會掉下金色的碎屑，那些碎屑就歸爸爸。放在此處是一語雙關：交易商正是把債券「切片」(strip) 後分銷，自己既不創造現金流也不領取現金流，賺的就是買賣價差那些「碎屑」。

Splitting the cash flow of Figure 2.8 into an annuity of coupons plus a zero-coupon bond for the par value, and applying Eq. (2.7) and Eq. (2.11), gives the pricing formula

$$\begin{aligned}
 P &= \sum_{i=1}^n \frac{C}{\left(1 + \frac{r}{m}\right)^i} + \frac{F}{\left(1 + \frac{r}{m}\right)^n} \\
 &= C \frac{1 - \left(1 + \frac{r}{m}\right)^{-n}}{\frac{r}{m}} + \frac{F}{\left(1 + \frac{r}{m}\right)^n}.
 \end{aligned} \tag{2.12}$$

Here n is the number of cash flows, m the number of payments per year, and r the annual rate compounded m times per annum. Note $C = Fc/m$ when c is the annual coupon rate.

Note 2.9.1. Because the sum collapses to a closed form, the price P can be computed in $O(1)$ time — no loop over the n coupons is needed.

§2.9.iii. Yields to Maturity

Definition 2.9.3. The **yield to maturity** (YTM) is the r that satisfies Eq. (2.12) with P being the bond price.

Example 2.9.4

For a 15% BEY, a 10-year bond with a coupon rate of 10% paid semiannually sells for

$$5 \times \frac{1 - [1 + (0.15/2)]^{-2 \times 10}}{0.15/2} + \frac{100}{[1 + (0.15/2)]^{2 \times 10}} = 74.5138$$

percent of par. So 15% is the yield to maturity if the bond sells for 74.5138.^a

^aNote that the yield 15% exceeds the coupon rate 10%.

Remark 2.9.5. **到期殖利率** (yield to maturity, YTM) 的名稱可拆成兩半來理解：*yield* 指報酬率，*to maturity* 則指這個報酬率是在「一路持有到到期日」的前提下算出來的。換言之它回答的是：若今天以價格 P 買進、此後每期領取票息、到期領回面值且中途不賣出，這串現金流相當於年利率多少？

之所以要把持有期間寫進名字裡，是因為債券還有別的「殖利率」，各自假設不同的持有期間。**當期收益率** (current yield) = Cm/P 只比較票息收入與價格，完全不管到期時的資本利得或損失——以上例而言其值為 $10/74.5138 = 13.42\%$ ，低於 YTM 的 15%，差額正是價格將由 74.51 回升至面值 100 的那段資本利得。**贖回殖利率** (yield to call) 則假設持有至發行人提前贖回之日而非到期日；若中途賣出，實際得到的又是另一個持有期報酬率。既然數字取決於你假設持有多久，名字裡自然得講明是哪一種。

至於它與前面 IRR 的關係：YTM 就是把 (2.12) 反解 r ，亦即債券這筆投資的 IRR，只是債券市場習慣改稱 YTM。因此 IRR 的但書一併繼承：YTM 要成為你實際拿到的報酬，除了持有到期、發行人不違約之外，還須能把每一期收到的票息都以 r 再投資出去。零息債券沒有中途票息、無從再投資，其 YTM 恰為實際報酬——這正是前面說它「沒有再投資風險」的意思。

儘管有這些但書，YTM 仍是債券市場的標準報價，因為它把價格、票息與到期日壓縮成單一數字，使票面利率與年期各異的債券得以直接比較。

§2.9.iv. Price Behavior

— Slides pp. 75–79 —

(1) Direction

Bond prices fall when interest rates rise, and vice versa. This is immediate from Eq. (2.12) — every term is decreasing in r — and yet “only 24 percent answered the question correctly.”¹¹

¹¹CNN, December 21, 2001.

Remark 2.9.6. 價格與市場利率反向的道理在於：票息 C 與面值 F 已由合約寫死，市場利率變動改變的不是你會收到多少錢，而是那些錢今天值多少。利率上升時，新發行的債券給得更多，你手上這張給得少的自然得降價才賣得掉；利率下跌則反之。註腳所引 2001 年 CNN 的調查中僅 24% 的人答對，正是因為多數人誤把「利率上升」聯想成「債券更值錢」——真正在變動的是折現率，而非票息。

(2) Premium, par, and discount

A level-coupon bond sells¹²

- at a **premium** (above its par value) when its coupon rate c is above the market interest rate r ;
- at **par** (at its par value) when its coupon rate is equal to the market interest rate;
- at a **discount** (below its par value) when its coupon rate is below the market interest rate.

Bonds selling at par, at a premium and at a discount are called **par bonds**, **premium bonds** and **discount bonds** respectively.

Yield (%)	Price (% of par)
7.5	113.37
8.0	108.65
8.5	104.19
9.0	100.00
9.5	96.04
10.0	92.31
10.5	88.79

Table 2.3.: A 9% coupon bond. The bond trades at par exactly when the yield equals the coupon rate (bold row), at a premium above, at a discount below.

Remark 2.9.7. 溢價 (premium)、平價 (par) 與折價 (discount)；依售價分類者分別稱為平價債券 (par bonds)、溢價債券 (premium bonds) 與折價債券 (discount bonds)。三者的定義與判準已見 Remark 2.9.2： $c = r$ 時 $P = F$ ，且 P 對 r 遞減，故 $c > r$ 時售價高於面值 (溢價)、 $c < r$ 時低於面值 (折價)。

以下補上溢價與折價的金額。記 $a \triangleq [1 - (1 + \frac{r}{m})^{-n}] / \frac{r}{m}$ 為年金現值因子，則代入 $C = Fc/m$ 後，(2.12) 可寫成 $P = \frac{Fc}{m} a + F(1 + \frac{r}{m})^{-n}$ 。在同一式中把 c 換成 r ，右側恰化簡為 F ，於是有恆等式 $F = \frac{Fr}{m} a + F(1 + \frac{r}{m})^{-n}$ ——這正是平價的情形。兩式相減，末項相消，故

$$P - F = \frac{F(c - r)}{m} a = \sum_{i=1}^n \frac{F(c - r)}{m} \frac{1}{(1 + \frac{r}{m})^i},$$

末式是把 $a = \sum_{i=1}^n (1 + \frac{r}{m})^{-i}$ 展開的結果，意思一目了然：第 i 期你比市場行情多領 (或少領) $F(c - r)/m$ 元，把這 n 筆差額各自折現回今天再加總，就是溢價 (或折價) 的金額。

以 Table 2.3 的 9% 債券驗證：殖利率 8% 時每期多領 0.5 元， $a = 17.292$ ，故 $P - F = 0.5 \times 17.292 = 8.65$ ，恰為表中 108.65 與面值 100 之差。另可留意，無論溢價或折價，到

¹²Consult the text for proofs.

期時價格必然回到面值，故溢價債券的價格會隨時間逐步下滑至 F ——那是溢價的攤銷，並非虧損。

Remark 2.9.8. 在此說明如何讀 Table 2.3。表中描述的是一張 15 年期、票面利率 $c = 9\%$ 、每半年付息一次的債券，即 $n = 30$ 、 $m = 2$ 、每期票息 $C = Fc/m = 4.5$ 元；整張表自始至終都是這同一張債券，票面利率固定為 9% 不變。變動的只有左欄的市場殖利率 r ，右欄則是把該 r 代入 (2.12) 算得的價格，並依債券市場慣例以面額 100 為基準表示（故 100.00 即等於面值）。例如第一列 $r = 7.5\%$ ： $a = [1 - (1.0375)^{-30}]/0.0375 = 17.829$ ， $P = 4.5 \times 17.829 + 100 \times (1.0375)^{-30} = 113.37$ ，與表中相符。粗體列即 $r = c = 9\%$ 的平價情形；其上方各列殖利率較低故溢價，下方各列殖利率較高故折價——留意表格說明裡的 *above / below* 指的是粗體列的上下方，而非殖利率的高低。這七列也就是 Figure 2.9 曲線上標出的七個點。

(3) Convexity

The price–yield relationship is not merely decreasing; it is *convex*. Concretely, take any yield change of size $\Delta > 0$. The price *gain* from a fall of Δ is larger than the price *loss* from a rise of that same Δ ,

$$P(r - \Delta) - P(r) > P(r) - P(r + \Delta),$$

and the gap widens as Δ grows.

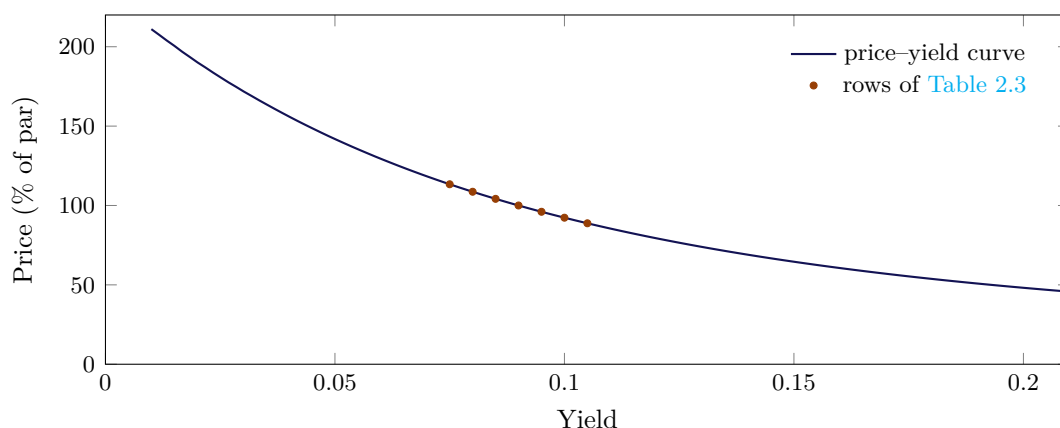


Figure 2.9.: Convexity of the price–yield curve, for the 15-year, 9% coupon bond paid semiannually that generates Table 2.3. The seven tabulated prices are marked; they lie on the curve.

Remark 2.9.9. 凸性即 $\partial^2 P / \partial r^2 > 0$ ：(2.12) 的每一項皆形如 $A(1 + \frac{r}{m})^{-i}$ ，其二階導數 $\frac{i(i+1)A}{m^2}(1 + \frac{r}{m})^{-i-2} > 0$ ，逐項為凸，總和自然為凸。其實際意涵是漲跌不對稱：由 Table 2.3，殖利率自 9% 降至 8% 價格漲 8.65，升至 10% 卻只跌 7.69，且幅度愈大這個不對稱愈明顯。換言之利率無論往哪個方向變動同樣幅度，持有人賺的總是多於賠的，故凸性本身即具價值，市場會為凸性較高的債券付出較高的價格——這是日後存續期間與凸性避險的基礎。

§2.10. Day Count Conventions

— Slides pp. 80–83 —

Every formula above assumed cash flows land on exact period boundaries. Real bonds settle on arbitrary calendar dates, so we need a rule converting two dates into a number of days. The rule is a *convention*, not a fact, and different markets use different ones.

§2.10.i. Actual/Actual

The first “actual” refers to the actual number of days in a month; the second refers to the actual number of days in a year.

Example 2.10.1

The number of days between June 17, 1992, and October 1, 1992, is 106: 13 days in June, 31 days in July, 31 days in August, 30 days in September, and 1 day in October.

Remark 2.10.1. 此處的「幾天」指的是兩個日期之間經過的天數，即「結束日減起始日」，而非兩端所涵蓋的日曆日數。以上例而言，6 月只算 13 天（6/18 至 6/30，不含起始日 6/17），10 月則算 1 天（含結束日 10/1），故為「不算頭、算尾」。

不過要留意，改採「算頭不算尾」會得到同一個答案（6 月 14 天、10 月 0 天，合計仍是 106），因為兩者算的都是半開區間的長度。真正會算錯的是兩端都算（得 107）與兩端都不算（得 105）。

之所以採此慣例，是為了讓天數具有可加性（additivity）：任取中間日期 B ，恆有「 A 到 B 」的天數加「 B 到 C 」的天數等於「 A 到 C 」的天數。若兩端都算，銜接點 B 便會被重複計入，串接多段期間時就會逐段多算一天。下一小節的 (2.13) 也正是「結束日減起始日」的形式，可見兩種慣例在這一點上完全一致——差別只在每月、每年各算幾天，不在端點怎麼取。

§2.10.ii. 30/360

Each month has 30 days and each year 360 days. In general, the number of days from date (y_1, m_1, d_1) to date (y_2, m_2, d_2) is

$$360 \times (y_2 - y_1) + 30 \times (m_2 - m_1) + (d_2 - d_1). \quad (2.13)$$

Example 2.10.2

The number of days between June 17, 1992, and October 1, 1992, is now 104: 13 days in June, 30 days in July, 30 days in August, 30 days in September, and 1 day in October.

Remark 2.10.2. (2.13) 看似繁瑣，其實只是「把每個日期換算成一個序號，再相減」。既然每月一律 30 天、每年一律 360 天，日期 (y, m, d) 在這套虛擬曆法上的序號就是 $360y + 30m + d$ ；兩個日期的序號相減即得該式，三項分別是「差幾年 $\times 360$ 」、「再差幾個月 $\times 30$ 」與「最後差幾號」。

關鍵在於後兩項允許為負，不必手動借位。以上例的 6/17 至 10/1 為例， $30 \times (10 - 6) + (1 - 17) = 120 - 16 = 104$ ；那個 -16 看似奇怪，但代數已自動處理了跨月的借位。

如此的好處是完全不必查曆法——不管某月是 28、29、30 還是 31 天，任兩日期的間隔

都能純以算術求得，且每個「月」所孳生的利息一律相等。代價則是這套曆法裡沒有 31 號，故 d_1 或 d_2 為 31 時須先改成 30，即下文的調整。

If d_1 or d_2 is 31, we must change it to 30 *before* applying formula (2.13).¹³ Hence:

- There are 3 days between February 28 and March 1.
- There are 2 days between February 29 and March 1.
- There are 29 days between March 1 and March 31.

Remark 2.10.3. An equivalent formula to (2.13) without any adjustment is (check it)

$$360 \times (y_2 - y_1) + 30 \times (m_2 - m_1 - 1) + \max(30 - d_1, 0) + \min(d_2, 30).$$

There are many variations on the “30/360” convention regarding 31, February 28, and February 29.^a

^aKosowski & Neftci (2015).

Remark 2.10.4. 上式的四項各有明確意義，合起來正是「起始月的殘餘 + 中間的整月 + 結束月已過的部分」：

- $\max(30 - d_1, 0)$ ：起始月從 d_1 號到月底還剩幾天。若 $d_1 = 31$ 則為 0，等於把 31 號視同月底。
- $30 \times (m_2 - m_1 - 1)$ ：夾在起訖兩月之間的完整月份，每月 30 天。
- $\min(d_2, 30)$ ：結束月從月初到 d_2 號已過幾天。若 $d_2 = 31$ 則取 30。
- $360 \times (y_2 - y_1)$ ：整年的部分。

以 6/17 至 10/1 驗證： $\max(30 - 17, 0) = 13$ （6 月）、 $30 \times (10 - 6 - 1) = 90$ （7、8、9 三個整月）、 $\min(1, 30) = 1$ （10 月），合計 104——恰好就是前例「13 天 + 30 + 30 + 30 + 1 天」那句話的代數寫法。

所謂 *without any adjustment*，是指 31 號的處理已被 $\max$ 與 $\min$ 吸收，不必如前文那樣先改數字再套公式，因此可直接寫成程式。至於 *check it*：令 $d' \triangleq \min(d, 30)$ 表示調整後的日，並注意恆等式 $\max(30 - d_1, 0) = 30 - d'_1$ ，則

$$\underbrace{30(m_2 - m_1 - 1)}_{\text{少算了一個月}} + \underbrace{(30 - d'_1)}_{\max(30-d_1,0)} + \underbrace{d'_2}_{\min(d_2,30)} = 30(m_2 - m_1) + (d'_2 - d'_1),$$

右側加上 $360(y_2 - y_1)$ 後正是調整後的 (2.13)，故兩式等價。

末句提醒的變體差異不可輕忽：本節採用的是歐式的 30E/360，對 d_1 與 d_2 一律截至 30，規則對稱；而美式的 30/360 (Bond Basis) 在 $d_2 = 31$ 但 $d_1 < 30$ 時並不把 d_2 改為 30，對二月底的處理也另有規定，算出的天數因而可能不同。故實務上合約必須載明採用哪一種慣例，否則利息金額會對不起來。

§2.11. Full Price, Accrued Interest, and the Clean Price

— Slides pp. 84–91 —

§2.11.i. The Full Price

In reality, the settlement date may fall on any day between two coupon payment dates. Let

$$\omega \triangleq \frac{\text{no. of days between the settlement and the next coupon payment date}}{\text{no. of days in the coupon period}}. \quad (2.14)$$

¹³This is the simplest of all the “30/360” variations (called the “30E/360” convention), used mainly in the Eurobond market (Kosowski & Neftci, 2015).

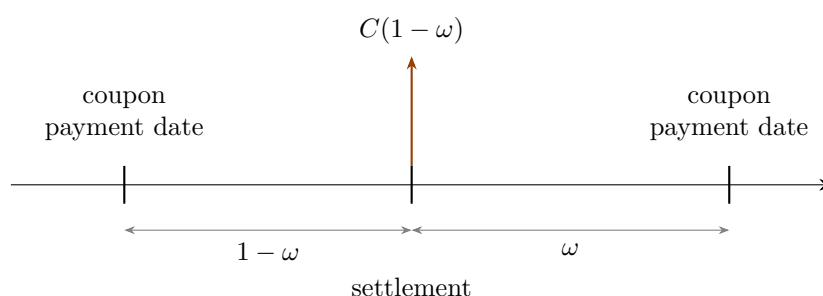


Figure 2.10.: A settlement date falling between two coupon payment dates. ω is the fraction of the coupon period still to run.

Discounting each remaining cash flow over its true (fractional) number of periods, the price is now calculated by

$$\begin{aligned} \text{PV} &= \frac{C}{\left(1 + \frac{r}{m}\right)^\omega} + \frac{C}{\left(1 + \frac{r}{m}\right)^{\omega+1}} + \cdots \\ &= \sum_{i=0}^{n-1} \frac{C}{\left(1 + \frac{r}{m}\right)^{\omega+i}} + \frac{F}{\left(1 + \frac{r}{m}\right)^{\omega+n-1}}. \end{aligned} \quad (2.15)$$

This is the **full price**, also called the **dirty price** or **invoice price**.

Remark 2.11.1. 交割日 (settlement date) 是債券易手之日：賣方交出債券、買方付錢，此後所有尚未發生的現金流都歸買方。於是問題成了「這些剩餘現金流在交割日這一刻值多少」，而 (2.15) 算的正是這個數，稱為**全價** (full price)，亦稱**含息價** (dirty price) 或**發票價格** (invoice price)。

麻煩在於交割日多半不落在**付息日** (coupon payment date) 上，各筆現金流因而不位於整數期。(2.14) 的 ω 正是用來度量這個零頭：分子是交割日到下一個付息日的天數，分母是整個付息期間的天數，故 ω 表示「本期還剩多少比例沒走完」， $1 - \omega$ 則是已走完的部分（見 Figure 2.10）。

有了 ω ，(2.15) 做的事便與 (2.12) 相同——把每筆剩餘現金流各自折現回交割日再加總——只是每筆的期數都多帶一個 ω 。 $i = 0$ 是最近的那筆票息，只需再等 ω 期； $i = 1, \dots, n-1$ 是其後各筆，每筆再晚一整期；末筆票息與 F 同日支付，故指數同為 $\omega + n - 1$ 。取 $\omega = 1$ （交割日恰在付息日之後）即化回 (2.12)，可見後者只是本式的特例。

以 Example 2.11.1 的數字 ($C = 5$ 、 $F = 100$ 、 $n = 4$ 、 $\frac{r}{m} = 0.015$ 、 $\omega = 1/3$) 逐項展開：

i	距今期數 $\omega + i$	現金流	折現值
0	1/3	5	4.9752
1	4/3	5	4.9017
2	7/3	5	4.8293
3	10/3	5	4.7579
—	10/3	100	95.1583
合計			114.6225

而淨價加應計利息為 $111.2891 + 3.3333 = 114.6224$ ，兩者相符（尾差來自四捨五入）——這就是該例所謂「可用 (2.15) 驗證殖利率為 3%」的意思。

§2.11.ii. Accrued Interest and the Clean Price

The quoted price in the U.S./U.K. does *not* include the accrued interest; it is called the **clean price** or **flat price**. The buyer pays the invoice price: the quoted price *plus* the **accrued**

interest (AI). AI equals

$$C \times \frac{\text{no. of days from the last coupon payment to the settlement date}}{\text{no. of days in the coupon period}} = C \times (1 - \omega).$$

Hence the yield to maturity is the r satisfying Eq. (2.15) when PV is the invoice price:

$$\text{clean price} + \text{AI} = \sum_{i=0}^{n-1} \frac{C}{\left(1 + \frac{r}{m}\right)^{\omega+i}} + \frac{F}{\left(1 + \frac{r}{m}\right)^{\omega+n-1}}.$$

Intuition . The clean price is quoted precisely so that it does *not* saw-tooth: the full price jumps down by C on every coupon date, and stripping out the accrued interest removes that artefact, leaving a number that moves only with the yield.

Remark 2.11.2. invoice price 是整個 bond 剩餘的價值，包含賣方還沒領到的那一份 coupon，所以 coupon 應該照比例 $(1 - \omega)$ 分給賣方，也就是說 full price 扣掉 AI (分給賣方的利息) 才是 clean price (對買方來說 bond 對他真正有的價值)。

Remark 2.11.3. 先說清楚全價 (full price) 是什麼：它就是 (2.15) 算出的那個數，也就是買方在交割日實際要掏出來的錢。它的毛病在於，縱使殖利率與債券本身都毫無變化，這個數字仍會隨付息週期上下起伏。以 $F = 100$ 、 $c = r = 10\%$ 、 $m = 2$ (故 $C = 5$) 為例，全價恰為 $100 \times 1.05^{1-\omega}$ ：剛付完息時 ($\omega = 1$) 為 100，其後隨下一筆票息逼近而一路爬升，至付息前夕 ($\omega \rightarrow 0$) 達 105；票息一經發放，該筆現金流即自剩餘現金流中消失，全價隨之驟降回 100。如此週而復始，全價的時間序列遂呈鋸齒狀。於是看到「甲債券報 105、乙債券報 100」時，你分不出甲究竟真的比較貴，還是只是離付息日比較近而已。

市場的對策是把全價拆成兩塊：隨持有天數線性累積的那部分稱為應計利息 (accrued interest, AI)，扣掉它之後的餘額稱為淨價 (clean price)，亦稱 flat price——注意此處的 flat 與先前的平價 (par) 毫無關係，切勿混淆。承上例，淨價 $= 100 \times 1.05^{1-\omega} - 5(1-\omega)$ ，整個付息週期內都停在 100 附近 ($\omega = 1/2$ 時為 99.97)，鋸齒已被濾掉。美、英市場報的就是淨價，它只隨殖利率與信用狀況變動。

AI 之所以等於 $C \times (1 - \omega)$ ，是因為票息按持有天數比例分攤：賣方持有了本期的 $1 - \omega$ ，就該分得這一期票息的 $1 - \omega$ 。但整筆 C 會在下一個付息日全額付給買方，故買方須在交割時先把屬於賣方的那一份補上——這便是買方實付發票價格 (invoice price) = 報價 + AI 的由來，而它正好等於全價。

於是最容易出錯的一點是：反解殖利率時，(2.15) 左邊的 PV 必須填發票價格而非報價，這正是本小節末那條 “clean price + AI = ...” 的用意。

§2.11.iii. Example (“30/360”)

Example 2.11.1

A bond with a 10% coupon rate and paying interest semiannually, with clean price 111.2891. The settlement date is July 1, 1993, and the maturity date is March 1, 1995.

There are 60 days between July 1, 1993, and the next coupon date, September 1, 1993, so $\omega = 60/180$. The accrued interest is

$$(10/2) \times \left(1 - \frac{60}{180}\right) = 3.3333$$

per \$100 of par value.

The yield to maturity is 3%. This can be verified by Eq. (2.15) with $\omega = 60/180$, $n = 4$, $m = 2$, $F = 100$, $C = 5$, $PV = 111.2891 + 3.3333$ and $r = 0.03$.

Remark 2.11.4. 疑問：例中驗證 YTM 時，不由分說直接取 $F = 100$ ；但若不知道這張債券的面額究竟是多少，YTM 不是就算不出來了嗎？

先說 $F = 100$ 是怎麼來的。它並非假設這張債券的面額恰好是 100 元，而是因為債券價格一律以面額的百分比報價：所謂 clean price 111.2891 意即「面額的 111.2891%」，而例中也明言 AI 是 *per \$100 of par value*。本章三處都採此慣例——零息債券例的 “quoted as 20.83”、Table 2.3 的欄位 “% of par”，以及本例。取 $F = 100$ 只是挑一個讓報價直接等於數值的單位罷了。

更要緊的是，YTM 根本不依賴面額。把 (2.15) 兩邊同除以 F 並代入 $C = Fc/m$ ，得

$$\frac{PV}{F} = \sum_{i=0}^{n-1} \frac{c/m}{(1 + \frac{r}{m})^{\omega+i}} + \frac{1}{(1 + \frac{r}{m})^{\omega+n-1}},$$

式中 F 已完全消失，只剩「價格佔面額的比例」、票面利率 c 、 ω 與 n 。面額加倍時票息與價格同步加倍，解出的 r 不變；故你不必知道面額究竟是 100、1,000 還是 1,000,000，只要知道價格相對於面額是多少——而那正是報價所提供的資訊。

真正需要用到實際面額的只有結算金額：買方應匯出 $\frac{\text{報價} + \text{AI}}{100} \times \text{實際面額}$ 。以本例而言，若買的是 1,000,000 面額，實付 $1.146224 \times 1,000,000 = 1,146,224$ 元，但殖利率仍是 3%。

§2.11.iv. Price Behavior (2) Revisited

Previously we said a bond sells at par if the yield to maturity equals the coupon rate. But that assumed the settlement date is *on* a coupon payment date.

Suppose the settlement date for a bond selling at par¹⁴ falls between two coupon payment dates. Then its yield to maturity is *less* than the coupon rate.¹⁵ The reason: exponential growth *to* C is replaced by linear growth, hence overpaying the accrued interest.

Remark 2.11.5. 疑問：前面才說「殖利率等於票面利率 $\Leftrightarrow$ 賣在平價」，但這只在交割日是在付息日才會是對的，為何交割日一落在兩個付息日之間，平價的債券殖利率反而低於票面利率？原文只丟下一句「指數成長被線性成長取代」與「見課本習題」。

情境：關鍵在「賣在平價」講的是哪個價——註腳已註明是報價（即淨價）等於面值，但買方實付的是淨價加 AI。於是真正該問的是：這筆實付金額，與「殖利率真的等於 c 時本應支付的金額」相比，究竟是多還是少？

脈絡：先講清楚前提——「賣在平價」是假設而非推導結果，註腳已言明它指報價（淨價）等於面值。於是買方實付

$$\underbrace{F}_{\text{淨價 (依假設等於面額)}} + \underbrace{C(1-\omega)}_{\text{AI}} = F + F \frac{c}{m} (1-\omega) = F \left[1 + \frac{c}{m} (1-\omega) \right],$$

此處用到票息 $C = Fc/m$ 由合約決定、與殖利率無關。（留意 F 在此身兼二職：它既是 (2.15) 裡的面額，又因假設而等於報價。）

¹⁴The quoted price equals the par value.

¹⁵See Exercise 3.5.6 of the textbook for proof.

接著問：若殖利率真的等於票面利率 c ，本應付多少？把 (2.15) 提出 $(1 + \frac{c}{m})^{1-\omega}$ ，得

$$PV = \left(1 + \frac{c}{m}\right)^{1-\omega} \left[\sum_{k=1}^n \frac{C}{\left(1 + \frac{c}{m}\right)^k} + \frac{F}{\left(1 + \frac{c}{m}\right)^n} \right],$$

中括號恰為 (2.12) 在殖利率等於 c 時的值，即 F ；故本應支付 $F(1 + \frac{c}{m})^{1-\omega}$ 。

令 $x \triangleq 1 - \omega$ 表示本期已走完的比例，兩者遂可並排寫成

$$\underbrace{F\left(1 + \frac{c}{m}\right)^x}_{\text{本應支付}} \quad \text{與} \quad \underbrace{F\left[1 + \frac{c}{m}x\right]}_{\text{慣例實付}}.$$

後者正是前者在 $x = 0$ 與 $x = 1$ 兩端之間的直線內插——兩式在 $x = 0$ 時同為 F （剛付完息，AI = 0），在 $x = 1$ 時同為 $F(1 + \frac{c}{m})$ （付息前夕，AI = C ），兩端完全吻合，差別只在中間。而 $(1 + \frac{c}{m})^x$ 為凸函數，凸函數的圖形在兩端點之間必位於該連線之下（見 Figure 2.11），故對 $0 < x < 1$ 恆有

$$\left(1 + \frac{c}{m}\right)^x < 1 + \frac{c}{m}x.$$

亦即實付金額高於「殖利率等於 c 時應付的金額」；又價格對殖利率遞減，買方付得愈多代表殖利率愈低，故實際殖利率必然低於票面利率。

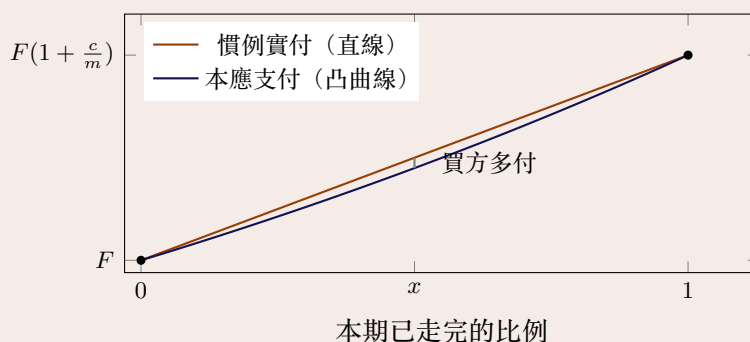


Figure 2.11.: 慣例的線性累積與正確的指數累積。兩者在期初與期末完全一致，中間則因直線位於凸曲線之上而使買方略為多付。（為看得清楚，圖中的 c/m 遠大於實際。）

代個數字看規模—— $F = 100$ 、 $c = 10\%$ 、 $m = 2$ 、 $\omega = 1/2$ 時，本應付 $100 \times 1.05^{0.5} = 102.4695$ ，慣例卻要付 $100 + 2.5 = 102.5$ ，多付約 0.03；偏差在期中最大，兩端則歸零。

歸根結底，這是 AI 採「按天數線性分攤」所付出的代價。該約定之所以線性，是為了讓買賣雙方不必對任何利率有共識就能算出同一個數字（公式裡根本沒有 r ）；換來的偏差就記在這裡。這也正是下方 moral 的意思：平價、溢價、折價那些漂亮的敘述，都是「交割日恰在付息日上」這個乾淨模型底下的結論。

The tidy statements about premium, par and discount are statements about a model with clean period boundaries. Day count conventions are where that model meets the calendar — and the tidy statements need amending.

3. Bond Price Volatility

— Slides pp. 92–93 —

Lecture 2
September 17,
2026

*Well, Beethoven, what is this?*¹

— Attributed to Prince Anton Esterházy

*You mean to tell me that
the success of my program and my re-election
hinges on the Federal Reserve
and a bunch of f—king bond traders?*

— Bill Clinton to Robert Rubin, January 7, 1993

The previous chapter answered the question “what is a bond worth?”. Given a yield y we can compute a price P , and given a price we can solve back for the yield. This chapter answers the next question, which is the one a risk manager actually loses sleep over: *if the yield moves, how much does the price move?*

The whole chapter is an exercise in Taylor expansion. Write the price as a function $P(y)$ of the yield; then

$$\Delta P = \frac{\partial P}{\partial y} \Delta y + \frac{1}{2} \frac{\partial^2 P}{\partial y^2} (\Delta y)^2 + \dots$$

The first-order term, suitably normalised, is called **duration**; the second-order term, suitably normalised, is called **convexity**. Everything else — Macaulay duration, modified duration, effective duration, dollar duration, $D\%$, effective convexity — is one of these two numbers in a different set of units, or computed by a different method. Keeping that map in mind is the best defence against the terminology.

Duration is the first derivative of price with respect to yield; convexity is the second. Everything in this chapter is bookkeeping around those two facts.

存續期間 (duration) 就是價格對殖利率的一階導數，凸性 (convexity) 就是二階導數；本章其餘的一切，都只是圍繞這兩件事的記帳工作。

§3.1. Price Volatility

— Slides pp. 94–95 —

Volatility here measures how bond prices respond to interest rate changes. It is the key input to the risk management of interest rate-sensitive securities: a desk that holds \$1 billion of bonds needs to know, in dollars, what a one-basis-point move costs it.

The question to make precise is this: *what is the sensitivity of the percentage price change to changes in interest rates?* Note the two deliberate choices hidden in that sentence. We measure the response in *percentage* terms, so that bonds of different sizes can be compared, and we measure it against a change in the *yield*, which is the single number we have agreed to summarise the whole discount curve by.

¹Mass in C major, Op. 86 (1808).

Definition 3.1.1 (Price volatility). The **price volatility** of a security with price P and yield y is

$$-\frac{\frac{\partial P}{\partial y}}{P}. \quad (3.1)$$

Intuitively, for a small yield move Δy ,

$$\frac{\Delta P}{P} \approx -\text{price volatility} \times \Delta y. \quad (3.2)$$

Remark 3.1.1. (3.1) 的兩個設計都有理由。

為何要除以 P ？ $\partial P/\partial y$ 的單位是「元 per 單位殖利率」，面額 100 的債券和面額 1,000 的同一檔債券會得到差十倍的數字。除以 P 之後得到的是百分比變化對殖利率變化的敏感度，與部位大小無關，才能拿不同債券互相比較。

為何要加負號？價格對殖利率遞減 ($\partial P/\partial y < 0$)，加上負號後，**價格波動性** (price volatility) 對一般債券是正數，講「這檔債券的波動性是 8」比講「是 -8 」自然。代價是 (3.2) 裡必須把負號再寫回去。

Note 3.1.1. (3.1) is a *local* statement: it is exact only in the limit $\Delta y \rightarrow 0$. How badly it fails for a finite Δy is precisely what convexity, in Section 3.10, will measure.

§3.2. Price Volatility of a Level-Coupon Bond

— Slides p. 96 —

For a level-coupon bond the derivative can be computed in closed form. Recall from the previous chapter that with n periods to maturity, a coupon C per period, par value F , and a *periodic* yield y ,

$$P = \frac{C}{y} \left[1 - (1+y)^{-n} \right] + F(1+y)^{-n}. \quad (3.3)$$

Proposition 3.2.1

The price volatility of a level-coupon bond is

$$-\frac{(C/y)n - (C/y^2)[(1+y)^{n+1} - (1+y)] - nF}{(C/y)[(1+y)^{n+1} - (1+y)] + F(1+y)}. \quad (3.4)$$

Here F is the par value and C is the coupon payment per period.

Proof. Differentiate (3.3) term by term:

$$\frac{\partial P}{\partial y} = -\frac{C}{y^2} + \frac{C}{y^2}(1+y)^{-n} + \frac{C}{y}n(1+y)^{-n-1} - nF(1+y)^{-n-1}.$$

Multiplying through by $(1+y)^{n+1}$ clears every negative power:

$$(1+y)^{n+1} \frac{\partial P}{\partial y} = \frac{C}{y}n - \frac{C}{y^2}[(1+y)^{n+1} - (1+y)] - nF,$$

which is the numerator of (3.4). Multiplying (3.3) by the same factor,

$$(1+y)^{n+1}P = \frac{C}{y} \left[(1+y)^{n+1} - (1+y) \right] + F(1+y),$$

which is the denominator. The quotient is therefore $(\partial P/\partial y)/P$, and (3.4) is its negative, as (3.1) requires. ■

Two observations about the formula.

- **It simplifies with $C = Fc/m$.** If the bond pays an annual coupon rate c with m payments per year, then $C = Fc/m$, and F divides out of numerator and denominator alike. What is left depends only on c/m , y and n — as it must, since price volatility is a percentage quantity and cannot depend on how big the bond is.
- **It is positive.** For the bond above,

$$-\frac{\partial P/\partial y}{P} > 0,$$

because P is a sum of positive cash flows discounted by $(1+y)^{-i}$, every one of which falls as y rises. Higher yield, lower price: the sign is not an accident but a restatement of the discounting formula.

§3.3. Macaulay Duration

— Slides p. 97 —

Definition 3.3.1 (Macaulay duration). The **Macaulay duration**² (MD) is a weighted average of the times to an asset's cash flows. The weights are the cash flows' PVs divided by the asset's price. Formally, for cash flows $C_1, \dots, C_n$ at times $1, \dots, n$,

$$\text{MD} \triangleq \frac{1}{P} \sum_{i=1}^n \frac{C_i}{(1+y)^i} i. \quad (3.5)$$

The definition really is an average. Put

$$w_i \triangleq \frac{1}{P} \frac{C_i}{(1+y)^i},$$

the fraction of today's price that is accounted for by the i th cash flow. Since $P = \sum_i C_i(1+y)^{-i}$ we have $\sum_i w_i = 1$, and (3.5) reads $\text{MD} = \sum_i w_i i$. So MD is the *centre of mass* of the cash flows when each is given the weight of its present value.

Remark 3.3.1. 麥考利存續期間 (Macaulay duration, MD) 在問一句話：這檔債券的錢，平均要等幾期才拿得到？

i 是第 i 筆現金流距今幾期（不是到期日，到期日是 n ）。直接把 $1, \dots, n$ 平均，等於不管每期收到的錢多寡，通通當同一回事；但債券到期那期通常連本金帶票息一起付，金額遠大於中途單純的票息，不該跟其他期一樣輕。所以要用每筆現金流的現值佔總價格的比重 w_i 來加權——金額愈大、時間愈近，權重愈大，MD 就愈被拉向那一期。這就是 (3.5) 是**加權平均** (weighted average)、 w_i 叫**現值權重** (PV weight) 的原因。

下面例子裡 $w_5 = 0.744$ ，於是 $\text{MD} = 4.34$ ：比單純平均的 3 大得多（因為多數價值壓在最後一期），又比到期日 5 略小（因為中途先收到票息）。把每期現值想成掛在時間軸上的砝碼，MD 就是這根槓桿的平衡點，即**重心** (centre of mass)，見 Figure 3.1。

²Frederick Macaulay (1882–1970) in 1938.

這個數字有用，是因為錢等得愈久、折現折得愈兇，價格對殖利率也就愈敏感——所以「平均要等幾期」同時也是「殖利率一動，價格會動多少」的度量，這正是下一小節 (3.6) 要證明的事。

單位是「期」，換算成年要除以每年付息次數 k (Section 3.5)。

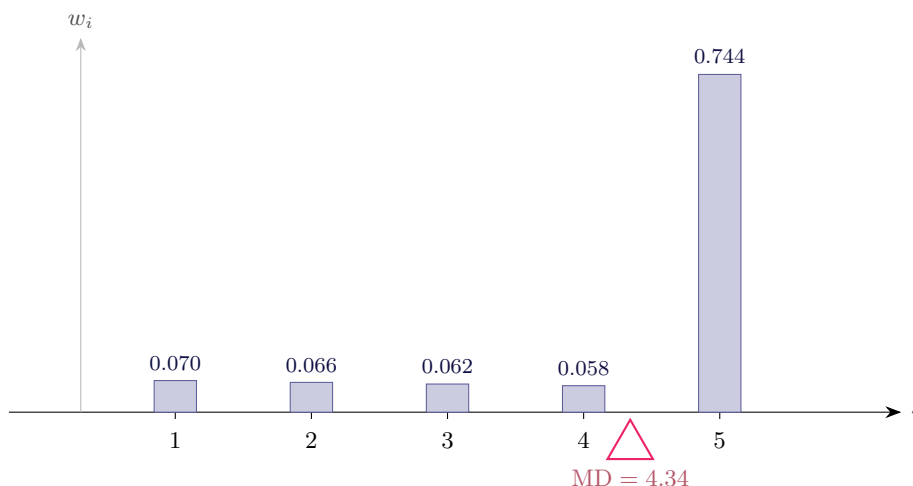


Figure 3.1.: Macaulay duration as a centre of mass. The bars are the PV weights w_i of a five-period bond with $F = 100$, period coupon rate 8% and yield $y = 6\%$; the balance point sits at MD = 4.34 periods, short of the maturity $n = 5$ because the four coupons pull it left.

Example 3.3.2 (Not on the slides)

A five-period bond with $F = 100$, coupon $C = 8$ per period and periodic yield $y = 6\%$ has

$$P = \sum_{i=1}^4 \frac{8}{1.06^i} + \frac{108}{1.06^5} = 108.4247,$$

with PV weights $w_1, \dots, w_5 = 0.0696, 0.0657, 0.0620, 0.0584, 0.7443$. Hence

$$\text{MD} = 1(0.0696) + 2(0.0657) + 3(0.0620) + 4(0.0584) + 5(0.7443) = 4.3422 \text{ periods.}$$

Almost three quarters of the price is the final principal repayment, which is why MD sits so close to $n = 5$.

Question 3.3.3. What if $C_i = (1 + c)^i$ for some constant c , letting $n \rightarrow \infty$ and assuming $c > y$?^a

^aContributed by Mr. Chen, Yu-Hsing (B06901048, R11922045) on March 3, 2023.

Answer. Write $q \triangleq (1 + c)/(1 + y) > 1$, so that $C_i(1 + y)^{-i} = q^i$. Then

$$P = \sum_{i=1}^n q^i = \frac{q(q^n - 1)}{q - 1} \xrightarrow{n \rightarrow \infty} \infty,$$

and the unnormalised numerator $\sum_{i=1}^n i q^i$ diverges faster still. Their ratio behaves like n for large n , so MD $\rightarrow \infty$ as well.

The example is not a curiosity but a warning. The weighted-average reading of (3.5) presumes that the weights w_i sum to a finite number; when the coupons grow faster than the discount

factor shrinks, the price itself is infinite and there is nothing left to average. Any security whose cash flows grow at a rate exceeding its own discount rate has no finite price and no finite duration. ■

§3.3.i. MD as a Measure of Price Volatility

— Slides p. 98 —

Here is the fact that makes MD worth defining: it is the bridge between the “average waiting time” picture of (3.5) and the “price sensitivity” picture of (3.1).

Theorem 3.3.4

The Macaulay duration, in periods, equals

$$\text{MD} = -(1+y) \frac{\partial P}{\partial y} \frac{1}{P}. \quad (3.6)$$

Proof. From $P = \sum_{i=1}^n C_i(1+y)^{-i}$,

$$\frac{\partial P}{\partial y} = - \sum_{i=1}^n i C_i (1+y)^{-i-1} = - \frac{1}{1+y} \sum_{i=1}^n i C_i (1+y)^{-i}.$$

Multiply both sides by $-(1+y)/P$:

$$-(1+y) \frac{\partial P}{\partial y} \frac{1}{P} = \frac{1}{P} \sum_{i=1}^n \frac{C_i}{(1+y)^i} i = \text{MD}. \quad \blacksquare$$

Intuition . (3.6) says $\text{MD} = (1+y) \times \text{price volatility}$. Two quantities defined for entirely different reasons — one a statement about *time*, the other a statement about *sensitivity* — turn out to differ only by the factor $1+y$.

The reason is visible in the proof: differentiating $(1+y)^{-i}$ brings down exactly the factor i , which is the time index. Long cash flows are discounted more steeply, so a security’s sensitivity to y is governed by how far out its money sits.

§3.3.ii. MD of a Level-Coupon Bond

— Slides p. 99 —

Proposition 3.3.5

The MD of a level-coupon bond is

$$\text{MD} = \frac{1}{P} \left[\sum_{i=1}^n \frac{iC}{(1+y)^i} + \frac{nF}{(1+y)^n} \right]. \quad (3.7)$$

It can be simplified to

$$\text{MD} = \frac{c(1+y)[(1+y)^n - 1] + ny(y-c)}{cy[(1+y)^n - 1] + y^2}, \quad (3.8)$$

where c is the *period* coupon rate.

(3.7) is just (3.5) with the cash flows of a level-coupon bond written out: a coupon C in each of periods $1, \dots, n$, plus the par value F at period n .

For (3.8), write $C = Fc$ and let $A(n) \triangleq \sum_{i=1}^n (1+y)^{-i} = [1 - (1+y)^{-n}]/y$ be the ordinary annuity factor, so (3.3) reads $P = F[cA(n) + (1+y)^{-n}]$. The bracket in (3.7) factors the same way:

$$\sum_{i=1}^n \frac{iC}{(1+y)^i} + \frac{nF}{(1+y)^n} = F \left[c \sum_{i=1}^n \frac{i}{(1+y)^i} + \frac{n}{(1+y)^n} \right].$$

So the numerator and the denominator P of (3.7) carry the *same* factor F , and it cancels — this is why F disappears. C disappears too, but only because it was never independent: it enters solely through $C = Fc$, and once F is gone all that is left of it is the rate c . Summing the remaining arithmetic–geometric series and simplifying gives (3.8), which — exactly as with (3.4) — depends only on c , y and n , never on the bond's size.

Two consequences are worth stating separately.

- **A coupon bond has MD < n.** If $c > 0$ then some of the weight w_i sits at times $i < n$, so the centre of mass in Figure 3.1 is pulled to the left of n .
- **A zero-coupon bond has MD = n.** With $c = 0$ there is a single cash flow, $w_n = 1$, and the average of one number is that number. Its duration *is* its term to maturity.

§3.4. What MD Does and Does Not Mean

§3.4.i. When the Formulas Apply

— Slides p. 100 —

Note 3.4.1. Formulas (3.6) and (3.7) hold only if the coupon C , the par value F , and the maturity n are all independent of the yield y — that is, only if the cash flow is independent of yields.

This is a far stronger assumption than it looks, and the slides' argument for why deserves unpacking.

Suppose the market yield declines. Mechanically, (3.8) then reports a *longer* MD: lower y discounts the distant cash flows less severely, so the weights w_i shift towards large i and the centre of mass moves right. For a plain bond that is the end of the story, and a longer MD does mean greater price volatility.

But now suppose the security's maturity *itself* shortens when yields fall. Then the actual price volatility, as originally defined in (3.1), may *decrease* at the very moment the formula says the duration lengthened. Formula and reality point in opposite directions.

Remark 3.4.1. 白話翻譯一下上面這段在講什麼。

MD 的公式 ((3.6)、(3.7)) 都偷偷假設了一件事： C 、 F 、 n 這些現金流的內容不會因為 y 變動而跟著變——也就是「不管殖利率變成怎樣，這張債券該付多少錢、哪一期付，永遠不變」。一般公司債、公債確實如此，但可贖回債券 (callable bond) 跟房貸抵押證券不是：殖利率一降，發行人可能提前贖回、屋主可能提前還款 (prepayment) 再融資，現金流本身就跟著 y 一起變了。

現在把「公式算出來的」跟「實際上發生的」分開看：

- 公式算出來的：套用 (3.8)，殖利率 y 下降 → 折現折得比較輕 → 較晚那幾期的權重 w_i 變大 → MD (加權平均的期數) 變長。這一步純粹是代公式，跟債券實際上會不

會被贖回無關。

- 實際上發生的：如果這張債券在殖利率下降時被提前贖回或提前還款，它的壽命反而縮短了。壽命一短，價格對利率的敏感度（(3.1) 那個真正的定義）通常也跟著下降。

為什麼壽命短，敏感度就低？最乾淨的例子是零息債： $P = F(1+y)^{-n}$ ，算一下 (3.1) 就得到價格波動性 $= n/(1+y)$ ——敏感度就是到期期數 n 除以 $1+y$ ， n 愈小，敏感度愈低，一目了然。直覺上也對： n 愈小，這筆錢愈快回到你手上， y 的變動只需要「折現少數幾期」，累積效果自然小； n 愈大，同一個 y 的變動要連乘（或連加，在指數上）更多次，效果就被放大。一般附息債也是同樣道理：Section 3.3.i 已經證明 MD 就是 $(1+y) \times$ 價格波動性，而 MD 本身是現金流到期期數的加權平均，到期日一旦提前，這個加權平均自然跟著縮小，價格波動性也跟著等比例縮小。

§3.4.ii. How Not To Think about MD

— Slides p. 101 —

The MD has its origin in measuring the length of time a bond investment is outstanding, and the name still carries that connotation. But it should be seen mainly as measuring *price volatility*, for the simple reason that the temporal reading breaks down in cases the volatility reading handles without complaint.

- **Duration can be longer than maturity.** A leveraged inverse floater — a note whose coupon is, say, $20\% - 2 \times$ the reference rate — behaves like a long position in a fixed-rate bond financed by a short position in a floater. Its price sensitivity can exceed that of *any* fixed-rate bond of the same maturity.
- **Duration can be negative.** An interest-only (IO) strip of a mortgage pool loses value when rates fall, because prepayments cut short the very interest payments it owns. Its price *rises* with yields, so $\partial P / \partial y > 0$ and its duration is negative. Figure 3.2 on p. 49 shows exactly such a price curve.

Neither of these makes sense under the maturity interpretation: no security is outstanding for longer than its maturity, and none is outstanding for a negative length of time. Many, if not most, duration-related terms can only be comprehended as measuring volatility.

Remark 3.4.2. 簡單說：MD 本質上是「利率敏感度」(sensitivity)，「加權平均到期時間」只是現金流固定不變時剛好等於敏感度的副產品；現金流一旦隨 y 變化，MD 就可能跑出時間該有的範圍。

- **槓桿反向浮動利率債券 (leveraged inverse floater)**：票息 $= 20\% - 2 \times$ 參考利率。等同於「多一張固定利率債 (fixed-rate bond)、用借來的錢放空一張浮動利率債 (floater)」——浮動債每期重設票息，久期 (duration) 近乎 0，固定債久期則正常。用槓桿多買固定債、空浮動債對沖，等於把固定債的利率敏感度用槓桿倍數放大，MD 自然能超過同期限的一般定息債。
- **IO strip (interest-only strip, 純利息證券)**：只分到房貸池的利息，不分本金。 y (連帶房貸利率) 一下降，屋主**提前還款** (prepayment, 也叫 refinance 轉貸) 的比例上升；本金一旦提前還完，後面本該收的利息就直接消失，不只是折現折得少。這個「現金流消失」的損失蓋過折現效果，導致價格跟 y 反向變動， $\partial P / \partial y > 0$ ，MD 變負。

Read “duration” as “price sensitivity measured in units of time”, never as “time”.

把 duration 讀成「以時間為單位表示的價格敏感度」，而不是「時間」。

§3.5. Conversion to Annual Units

— Slides p. 102 —

So far everything has been measured in *periods*, with y the periodic yield. Market conventions quote years and annual yields, so both formulas need converting. Let k be the compounding frequency per annum and let y now denote the *annual* yield, so that the periodic yield is y/k and cash flow i arrives at time i/k years.

For the MD to be year-based, modify (3.7) to

$$\frac{1}{P} \left[\sum_{i=1}^n \frac{i}{k} \frac{C}{(1 + \frac{y}{k})^i} + \frac{n}{k} \frac{F}{(1 + \frac{y}{k})^n} \right]. \quad (3.9)$$

Nothing has changed except that each time index i has been re-expressed as i/k years. (n itself is still a count of *periods*, not years; only the time label in the numerator has been converted to i/k and n/k years, while the discounting exponent stays in periods.) Formula (3.6) likewise becomes

$$\text{MD} = -\left(1 + \frac{y}{k}\right) \frac{\partial P}{\partial y} \frac{1}{P}. \quad (3.10)$$

By definition, then,

$$\text{MD (in years)} = \frac{\text{MD (in periods)}}{k}.$$

Remark 3.5.1. 注意 MD(in periods) 是指每期利率 $\frac{y}{k}$ 算出來的 md，然後 MD(in year) 只是把時間的單位從期換到年，利率還是一樣。三個式子講的是同一件事，只是單位不同，但實務上最常出錯的就是單位。

(3.9) 這一步很直覺： i/k 、 n/k 只是把「第 i 期」重新標成「第 i/k 年」， k 是常數可以直接從加總裡拿出來，所以整個式子自然就是 $\text{MD}(\text{periods})/k$ ，沒有懸念。

(3.10) 看起來像是另外冒出來的一個公式，但它不是重新定義出來湊答案的，而是可以從 (3.6) 用連鎖律 (chain rule) 硬算出來，結果剛好也等於 $\text{MD}(\text{periods})/k$ 。推導如下。

把每期的殖利率記成 $y_p = y/k$ (也就是 (3.6) 裡原本那個 y ，這裡改名是為了不跟年殖利率 y 搞混)。因為 P 其實是 y_p 的函數， y 只是透過 $y_p = y/k$ 間接影響 P ，連鎖律給出

$$\frac{\partial P}{\partial y} = \frac{\partial P}{\partial y_p} \cdot \frac{\partial y_p}{\partial y} = \frac{1}{k} \frac{\partial P}{\partial y_p}.$$

直覺上： y 動 1 個單位，真正拿去折現的 $y_p = y/k$ 只被拖著動 $1/k$ 個單位，所以價格對「年殖利率 y 」的敏感度，天生就比對「期殖利率 y_p 」的敏感度小了 k 倍——這就是那個 $1/k$ 的來源，跟前面的係數完全無關。

把這個關係代進 (3.10)：

$$-\left(1 + \frac{y}{k}\right) \frac{\partial P}{\partial y} \frac{1}{P} = -(1 + y_p) \cdot \frac{1}{k} \frac{\partial P}{\partial y_p} \cdot \frac{1}{P} = \frac{1}{k} \underbrace{\left[-(1 + y_p) \frac{\partial P}{\partial y_p} \frac{1}{P} \right]}_{= \text{MD}(\text{periods}), \text{ 即 (3.6)}} = \frac{\text{MD}(\text{periods})}{k}.$$

注意係數 $(1 + y/k)$ 只是原本的 $(1 + y_p)$ 換了個寫法而已，並沒有跟著 k 做任何調整；真正帶出 $1/k$ 的，是求導變數從 y_p 換成 y 時連鎖律多乘出來的那個因子。兩者是獨立的兩件事，只是恰好都跟 k 有關，才會讓最後結果乾淨地等於 $\text{MD}(\text{periods})/k$ ——跟直接把加總式除以 k ((3.9) 的做法) 殊途同歸，這正是三個式子彼此一致、講的是同一件事的原因。

§3.6. Modified Duration

— Slides p. 103 —

The factor $1 + y$ in (3.6) is a nuisance if what one actually wants is the sensitivity itself. Dividing it out gives the quantity traders quote.

Definition 3.6.1 (Modified duration).

$$\text{modified duration} \triangleq -\frac{\partial P}{\partial y} \frac{1}{P} = \frac{\text{MD}}{1 + y}. \quad (3.11)$$

So modified duration is exactly the price volatility (3.1) of the first section, under a new name. By the Taylor expansion,

$$\text{percent price change} \approx -\text{modified duration} \times \text{yield change}. \quad (3.12)$$

Note 3.6.1. Taylor-expanding P to first order, $P(y + \Delta y) \approx P(y) + \frac{\partial P}{\partial y} \Delta y$; dividing by $P(y)$ and substituting (3.11) gives exactly (3.12) — the same computation that produced (3.2). “price volatility” and “modified duration” are two names for that one term.

Note 3.6.2. Modified duration equals MD under continuous compounding. With continuous compounding $P = \sum_i C_i e^{-yt_i}$, so $\partial P / \partial y = -\sum_i t_i C_i e^{-yt_i}$ and the ratio $-(\partial P / \partial y) / P$ is the PV-weighted average of the t_i with no correction factor at all. The $1 + y$ in (3.11) is an artifact of discrete compounding; it is the discrete analogue of the fact that $\frac{d}{dy} e^{-yt} = -t e^{-yt}$ exactly, whereas $\frac{d}{dy} (1 + y)^{-i} = -i(1 + y)^{-i-1}$ picks up one extra factor of $(1 + y)^{-1}$.

§3.6.i. Example

— Slides p. 104 —

Example 3.6.2

Consider a bond whose modified duration is 11.54 with a yield of 10%. If the yield increases instantaneously from 10% to 10.1%, the approximate percentage price change is

$$-11.54 \times 0.001 = -0.01154 = -1.154\%.$$

Remark 3.6.1. 注意單位：殖利率從 10% 升到 10.1%，變化量是 0.1 個百分點，也就是 $\Delta y = 0.001$ ，即十個基點 (basis point, bp)，不是 0.1。

§3.6.ii. Modified Duration of a Portfolio

— Slides p. 105 —

Proposition 3.6.3

By calculus, the modified duration of a portfolio equals

$$\sum_i \omega_i D_i, \quad (3.13)$$

where D_i is the modified duration of the i th asset and ω_i is the market value of that asset expressed as a percentage of the market value of the portfolio.

Proof. Let the portfolio value be $V = \sum_i P_i$ and $\omega_i = P_i/V$. Then

$$-\frac{1}{V} \frac{\partial V}{\partial y} = -\frac{1}{V} \sum_i \frac{\partial P_i}{\partial y} = \sum_i \frac{P_i}{V} \left(-\frac{1}{P_i} \frac{\partial P_i}{\partial y} \right) = \sum_i \omega_i D_i. \quad \blacksquare$$

§3.7. Effective Duration

— Slides p. 106 —

Yield changes may alter the cash flow, or the cash flow may be too complex for simple formulas — exactly the callable bonds and mortgage securities that broke (3.7) in Section 3.4. We need a general numerical formula for volatility, one that asks the pricing model rather than a closed form.

Definition 3.7.1 (Effective duration).

$$\text{effective duration} \triangleq \frac{P_- - P_+}{P_0(y_+ - y_-)}, \quad (3.14)$$

where P_- is the price if the yield is decreased by Δy , P_+ is the price if the yield is increased by Δy , P_0 is the initial price, y is the initial yield, and Δy is small.

So $y_{\pm} = y \pm \Delta y$ and $y_+ - y_- = 2\Delta y$: (3.14) is the **central difference** approximation to $-(\partial P / \partial y) / P$. The prices P_- , P_0 , P_+ are obtained by re-running the pricing model three times — with the cash flows allowed to change in response to the yield, which is the whole point.

— Slides p. 107 —

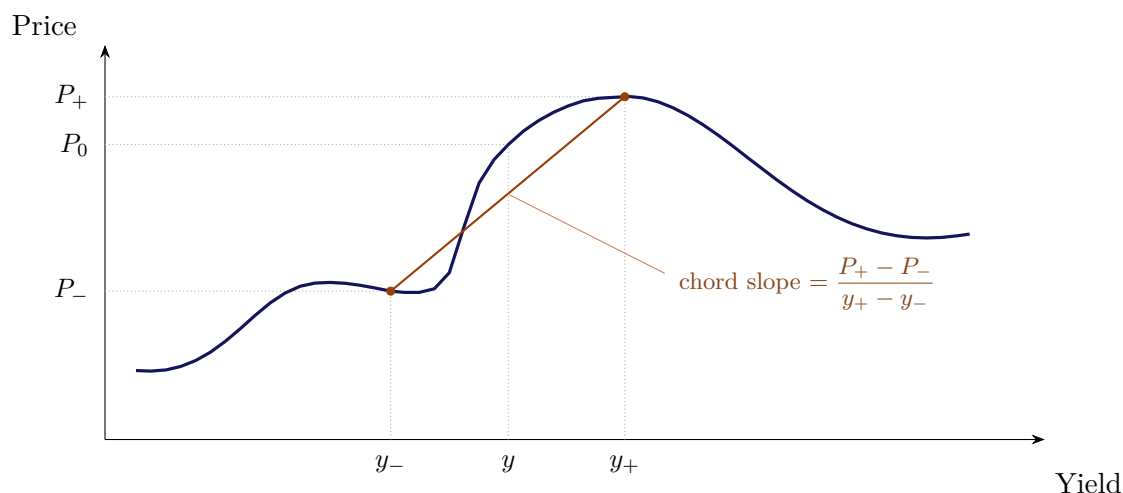


Figure 3.2.: Effective duration reads the slope of the chord through (y_-, P_-) and (y_+, P_+) , not of the tangent at y . The curve is traced from the plot shown in class. It is deliberately badly behaved — a plateau on the left, a steep rise through the bracket, a peak just past y_+ — and it is *increasing* in the yield over $[y_-, y_+]$, so this security's duration there is negative. No closed-form duration formula can handle such a price curve.

Remark 3.7.1. 這張圖是全章脈絡最容易斷掉的地方，值得說清楚它在講什麼。

圖中的價格曲線隨殖利率上升 ($P_- < P_0 < P_+$)，而且還不是單調的——左邊有個平台、右邊過了 y_+ 就掉下來。這種形狀在 MBS、**純息分割債券** (interest-only strip, IO)、可贖回債券上都真實存在，公式 (3.7) 對它完全無能為力。

(3.14) 的作法是：不去碰導數，改成把定價模型跑三次，用割線斜率近似切線斜率。代入這張圖的數字， $P_- - P_+ < 0$ ，所以 effective duration 是負的——正好印證 Section 3.4 裡「duration 可以是負的」那句話。

另外注意 (3.14) 用的是**中央差分** (central difference)，把 y_+ 與 y_- 對稱地放在 y 兩側。這麼做的誤差是 $O((\Delta y)^2)$ ，而單邊差分只有 $O(\Delta y)$ ——下一頁的替代公式就是拿精度換計算量。

— Slides p. 108 —

One can compute the effective duration of just about any financial instrument: all it takes is a pricing routine that can be called three times. An alternative is to use the one-sided difference

$$\frac{P_0 - P_+}{P_0 \Delta y}, \quad (3.15)$$

which is more economical — two model runs instead of three, and P_0 is usually already known — but theoretically less accurate.

Note 3.7.1. The accuracy claim is the standard one from numerical differentiation. Expanding $P(y \pm \Delta y)$ about y ,

$$\frac{P_- - P_+}{2\Delta y} = -\frac{\partial P}{\partial y} - \frac{1}{6} \frac{\partial^3 P}{\partial y^3} (\Delta y)^2 + \dots, \quad \frac{P_0 - P_+}{\Delta y} = -\frac{\partial P}{\partial y} - \frac{1}{2} \frac{\partial^2 P}{\partial y^2} \Delta y + \dots$$

The central difference loses the first-order error term to symmetry: its leading error is $O((\Delta y)^2)$ against $O(\Delta y)$ for the one-sided version. Halving Δy buys a factor of four in one case and only a factor of two in the other.

§3.8. The Practices: Duration in Percentage Terms

— Slides p. 109 —

Duration is usually expressed in percentage terms — call it $D_\%$ — for quick mental calculation.³ The percentage price change expressed in percentage terms is then approximated by

$$-D_\% \times \Delta r \quad (3.16)$$

when the yield increases instantaneously by $\Delta r\%$.

Example 3.8.1

Suppose $D_\% = 10$ and $\Delta r = 2$. Price will drop by 20%, as $10 \times 2 = 20$.

Exercise 3.8.2. $D_\%$ in fact equals the original duration. Prove it!

Proof. Let D be the modified duration, so that $\Delta P/P \approx -D \Delta y$ with Δy in decimals. Writing the yield change in percentage points, $\Delta r = 100 \Delta y$, and the price change in percent, $100 \Delta P/P$,

³Neftci (2008): “Market professionals do not like to use decimal points.”

gives

$$100 \frac{\Delta P}{P} \approx -D (100 \Delta y) = -D \Delta r.$$

Comparing with (3.16) yields $D_{\%} = D$. The two factors of 100 — one on each side — cancel exactly, which is why the convention costs nothing. ■

Remark 3.8.1. 換句話說， $D_{\%}$ 不是新的量，只是換一套讀數的習慣：把殖利率變化以「百分點」計、價格變化以「百分比」計之後，同一個 duration 數字可以直接相乘，不必先換算成小數。前一頁 -11.54×0.001 那種容易錯的運算，改寫成 $-11.54 \times 0.1 = -1.154$ （百分比）就直觀多了。

§3.9. Hedging

— Slides p. 110 —

Hedging offsets the price fluctuations of the position to be hedged by the hedging instrument in the opposite direction, leaving the total wealth unchanged.

Percentages are the wrong unit for that job: two positions of different size with equal and opposite *modified* durations do not cancel. What must cancel is the change in *dollars*.

Definition 3.9.1 (Dollar duration).

$$\text{dollar duration} \triangleq \text{modified duration} \times \text{price} = -\frac{\partial P}{\partial y}. \quad (3.17)$$

The approximate dollar price change is then

$$\text{price change} \approx -\text{dollar duration} \times \text{yield change}. \quad (3.18)$$

One can hedge a bond portfolio with a dollar duration D by bonds with a dollar duration $-D$: the two first-order terms cancel and the combined position is, to first order, insensitive to y .

Intuition. Modified duration is a *rate*; dollar duration is a *quantity*. Hedging is an accounting identity about quantities, so the hedge ratio must be computed in dollar durations.

修正存續期間是比率，**金額存續期間** (dollar duration) 是金額。**避險** (hedging) 是關於金額的會計恆等式，所以**避險比例** (hedge ratio) 必須用金額存續期間來算。

Note 3.9.1. The hedge is only first-order and only instantaneous. It fails for large moves — that is the second-order term, i.e. convexity — and it fails when the two legs' yields do not move together, which no single- y framework can express.

§3.10. Convexity

— Slides p. 111 —

Duration is the first derivative. The second derivative gets its own name.

Definition 3.10.1 (Convexity).

$$\text{convexity (in periods)} \triangleq \frac{\partial^2 P}{\partial y^2} \frac{1}{P}. \quad (3.19)$$

Exercise 3.10.2. The convexity of a level-coupon bond is positive. Prove it!

Proof. Differentiate $P = \sum_{i=1}^n C_i(1+y)^{-i}$ twice:

$$\frac{\partial^2 P}{\partial y^2} = \sum_{i=1}^n i(i+1) C_i (1+y)^{-i-2} > 0,$$

since every $C_i \geq 0$ and at least one is strictly positive. Dividing by $P > 0$ leaves the sign unchanged. The argument uses nothing about level coupons: *any* security with non-negative, yield-independent cash flows has positive convexity. ■

For a bond with positive convexity, the price rises more for a rate decline than it falls for a rate increase of equal magnitude. That asymmetry is the content of Figure 3.3: the true price curve bends away from its own tangent, and it bends *upwards*, so the tangent understates the price on both sides.

— Slides p. 112 —

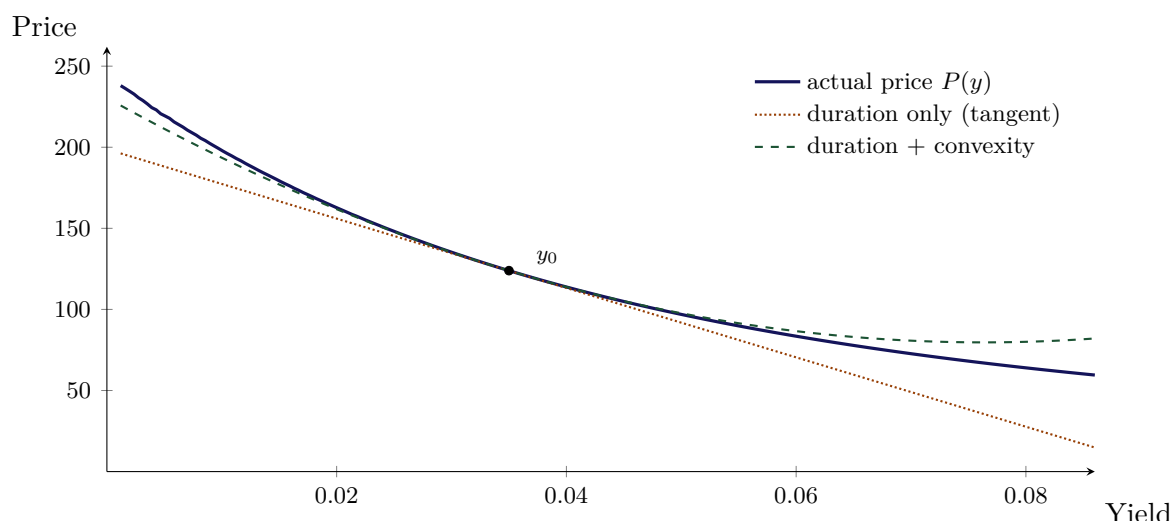


Figure 3.3.: Price versus yield for a level-coupon bond ($n = 30$, $C = 4.8$, $F = 100$), expanded about $y_0 = 3.5\%$. The tangent — the duration-only approximation — lies below the true curve everywhere, understating the gain from a rate fall and overstating the loss from a rate rise. Adding the convexity term recovers almost all of the discrepancy over the plotted range.

Remark 3.10.1. 把這張圖讀成三句話：

- 實線是真正的價格曲線，對殖利率遞減且向上凸。
- 點線是只用 duration 的線性近似，也就是曲線在 y_0 的切線。由於曲線是凸的，切線恆在曲線下方：殖利率跌時低估漲幅，殖利率漲時高估跌幅——兩邊都對持有人有利，這就是「正凸性是好事」的圖解。
- 虛線加進了凸性的二次項，在相當寬的範圍內幾乎與實線重合。

注意切線與真實曲線的落差隨 $|\Delta y|$ 呈平方成長： $\Delta y = 1\%$ 時幾乎看不出來， $\Delta y = 4\%$ 時已經差了幾十塊。這就是為什麼小幅度時可以只用 duration，大幅度時非補上凸性不可。

§3.10.i. Consequences and Units

— Slides p. 113 —

So between two bonds with the same price and duration, the one with a higher convexity is more valuable.⁴

Suppose there are k periods per annum. Convexity measured in periods and convexity measured in years are related by

$$\text{convexity (in years)} = \frac{\text{convexity (in periods)}}{k^2}. \quad (3.20)$$

Note 3.10.1. The k^2 is not a new convention but a consequence of the k in $\text{MD}(\text{years}) = \text{MD}(\text{periods})/k$. Convexity is a *second* derivative with respect to the yield, so re-expressing the yield in annual units divides it twice. Any quantity that is an r th derivative with respect to y picks up a factor k^{-r} .

Remark 3.10.2. 那個腳註的「問題」在哪裡？

表面上的推論是：兩檔債券今天價格相同、duration 相同，但一檔凸性較高。殖利率不論漲跌，凸性高的那檔表現都不會比較差（漲得多、跌得少）。既然如此，它應該比較值錢——但它們今天價格明明一樣。這聽起來像是白吃的午餐，也就是**套利**（arbitrage）機會。

Christensen 與 Sørensen（1994）點出的正是這個矛盾：在**無套利**（no-arbitrage）的**期限結構**（term structure）模型裡，這種情形不可能持續存在。真正的機制是凸性要付錢買——市場會把凸性高的債券的殖利率壓低（或說價格推高），高凸性帶來的好處恰好被較低的收益率抵銷。

更根本的問題出在推論的隱含前提：整條殖利率曲線平行移動。這個假設本身就與無套利相容不了（若曲線只能平行移動，就能建構出套利部位）。一旦改用真正的期限結構模型，「同價同 duration 但凸性較高」的兩檔債券根本不會同時存在於市場上。

§3.11. Use of Convexity

— Slides p. 114 —

The approximation $\Delta P/P \approx -\text{duration} \times \text{yield change}$ works for small yield changes. For larger yield changes, keep the next term of the Taylor expansion:

$$\begin{aligned} \frac{\Delta P}{P} &\approx \frac{\partial P}{\partial y} \frac{1}{P} \Delta y + \frac{1}{2} \frac{\partial^2 P}{\partial y^2} \frac{1}{P} (\Delta y)^2 \\ &= -\text{duration} \times \Delta y + \frac{1}{2} \times \text{convexity} \times (\Delta y)^2. \end{aligned} \quad (3.21)$$

Recall [Figure 3.3](#): the dashed curve is (3.21), and the gap between it and the dotted tangent is the convexity term doing its work.

Note 3.11.1. Because convexity is positive for an ordinary bond, the second term is positive whichever way y moves. The correction always *raises* the predicted price relative to the duration-only estimate — which is exactly the asymmetry stated on p. 111 of the slides.

⁴Do you spot a problem here (Christensen & Sørensen, 1994)?

§3.12. The Practices: Convexity in Percentage Terms

— Slides p. 115 —

Convexity is likewise usually expressed in percentage terms — call it $C_{\%}$ — for quick mental calculation. The percentage price change expressed in percentage terms is approximated by

$$-D_{\%} \times \Delta r + C_{\%} \times (\Delta r)^2/2 \quad (3.22)$$

when the yield increases instantaneously by $\Delta r\%$.

Example 3.12.1

Price will drop by 17% if $D_{\%} = 10$, $C_{\%} = 1.5$ and $\Delta r = 2$, because

$$-10 \times 2 + \frac{1}{2} \times 1.5 \times 2^2 = -20 + 3 = -17.$$

Exercise 3.12.2. $C_{\%}$ equals convexity divided by 100. Prove it!

Proof. Start from (3.21) with Δy in decimals and substitute $\Delta y = \Delta r/100$:

$$100 \frac{\Delta P}{P} \approx -D(100 \Delta y) + \frac{1}{2} \text{conv} \frac{(100 \Delta y)^2}{100} = -D \Delta r + \frac{1}{2} \frac{\text{conv}}{100} (\Delta r)^2.$$

Comparing with (3.22) gives $D_{\%} = D$ as before and $C_{\%} = \text{conv}/100$. ■

Remark 3.12.1. 為什麼 duration 換算係數是 1，凸性卻是 1/100？因為 duration 項對 Δy 是一次的，兩側各乘 100 恰好相消；凸性項對 Δy 是二次的，左邊只乘了 100、右邊卻乘了 100^2 ，於是多出一個 100 要除掉。一般而言， r 階項的百分比版本等於原值除以 100^{r-1} 。

§3.13. Effective Convexity

— Slides p. 116 —

Just as effective duration replaced the closed-form first derivative by a difference quotient, effective convexity replaces the second.

Definition 3.13.1 (Effective convexity).

$$\text{effective convexity} \triangleq \frac{P_+ + P_- - 2P_0}{P_0(0.5 \times (y_+ - y_-))^2}, \quad (3.23)$$

where P_- is the price if the yield is decreased by Δy , P_+ the price if it is increased by Δy , P_0 the initial price, y the initial yield, and Δy small.

Since $0.5 \times (y_+ - y_-) = \Delta y$, the denominator is just $P_0(\Delta y)^2$, and (3.23) is the standard central second difference $[P_+ + P_- - 2P_0]/(\Delta y)^2$ normalised by P_0 . Effective convexity is most relevant when a bond's cash flow is interest rate sensitive — the callable bonds and mortgage securities for which (3.19) cannot even be written down, and for which the convexity often turns out *negative*.

How to choose the right Δy is a delicate matter.

§3.14. A Numerical Caveat: Second Differences in Floating Point

— Slides p. 117 —

The last sentence is not a throwaway remark, and the slides make the point with a deliberately trivial example. Take $f(x) = x^2$, whose second derivative at $x = 1$ is exactly 2, and form the central second difference⁵

$$\frac{(1 + \Delta x)^2 + (1 - \Delta x)^2 - 2}{(\Delta x)^2}.$$

In exact arithmetic the numerator is $2(\Delta x)^2$ and the quotient is 2 for *every* Δx — there is no truncation error at all, since f is a quadratic. In floating point the answer is a disaster.

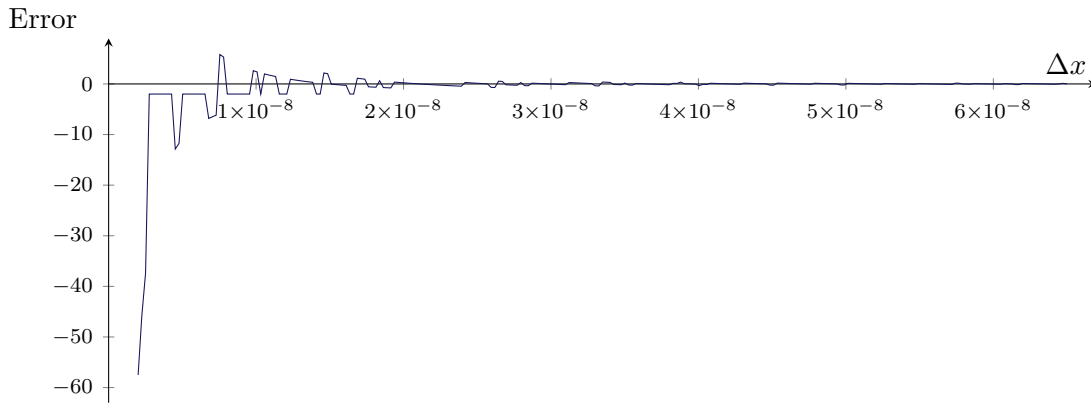


Figure 3.4.: The difference between $[(1 + \Delta x)^2 + (1 - \Delta x)^2 - 2]/(\Delta x)^2$ and its exact value 2, computed in IEEE double precision. The flat stretches at -2 are the cases where the numerator underflows to exactly zero; the spikes are the cases where it survives as one or two units in the last place.

Remark 3.14.1. 這個現象叫做**災難性抵銷** (catastrophic cancellation)，值得一步一步看清楚。

取 $\Delta x \approx 10^{-8}$ 。在雙精度浮點數中，1 附近的**機器精度** (machine epsilon) 是 $\varepsilon = 2^{-52} \approx 2.22 \times 10^{-16}$ ，也就是說 1 附近能分辨的最小間隔就是這個數量級。

而 $(1 + \Delta x)^2 + (1 - \Delta x)^2 - 2$ 的真值是 $2(\Delta x)^2 \approx 2 \times 10^{-16}$ ——恰好落在 ε 的同一個數量級。換句話說，我們要算的整個答案都藏在最後一個位元裡。多數情況下捨入直接把它抹成 0，商數變成 0，誤差就是 $0 - 2 = -2$ （圖中那些平坦段）；偶爾殘留一兩個 ulp，除以 $(\Delta x)^2 \approx 10^{-16}$ 之後就被放大成 ± 10 甚至 -57 的巨大誤差（圖中的尖峰）。

推論： Δy 不是愈小愈好。把**截斷誤差** (truncation error) 與**捨入誤差** (round-off error) 寫在一起，設 u 為相對捨入誤差的量級：

$$\underbrace{\frac{1}{12}|P^{(4)}|(\Delta y)^2}_{\text{截斷誤差，隨}\Delta y\text{遞減}} + \underbrace{\frac{4u|P|}{(\Delta y)^2}}_{\text{捨入誤差，隨}\Delta y\text{遞增}}.$$

兩者相權，二階差分的最佳步長約為 $\Delta y \sim \varepsilon^{1/4} \approx 1.2 \times 10^{-4}$ ；一階（中央）差分則是 $\Delta y \sim \varepsilon^{1/3} \approx 6 \times 10^{-6}$ 。所以 (3.23) 的 Δy 取 1 個 basis point (10^{-4}) 左右通常是合理的，取 10^{-8} 則會得到一堆雜訊。

⁵The slide writes the target as $d^2 f(x)^2/dx^2$; the quantity actually approximated is $d^2 f/dx^2 = 2$.

This numerical issue is common in financial engineering but does not have general solutions yet.⁶

§3.15. Interest Rates and Bond Prices: Which Determines Which?

— Slides p. 118 —

Question 3.15.1. Do interest rates determine bond prices, or do bond prices determine interest rates?^a

^aContributed by Mr. Wang, Cheng (R01741064) on March 5, 2014.

If you have one, you have the other. Given a price, the yield is the root of (3.3); given a yield, the price is its value. So they are just two names given to the same thing: the **cost of fund**.

The difference is one of habit rather than of substance:

- Traders most likely work with prices.
- Banks most likely work with interest rates.

Price and yield are one quantity in two coordinate systems. Duration and convexity are the derivatives that translate between them.

價格與殖利率是同一個量的兩套座標；duration 與 convexity 就是在兩者之間換算的導數。

⁶See pp. 872ff of the slides.

4. Term Structure of Interest Rates

— Slides pp. 119–120

*Why is it that the interest of money is lower,
when money is plentiful?*

— Samuel Johnson (1709–1784)

If you have money, don't lend it at interest.

*Rather, give [it] to someone
from whom you won't get it back.*

— Thomas Gospel 95

Every formula so far has carried a single interest rate. One y discounted all of a bond's cash flows in (2.6); one y was perturbed to get duration and convexity in the last chapter. That was a convenient fiction, and this chapter retires it.

Money borrowed for three months and money borrowed for thirty years do not cost the same, and the pattern of rates across maturities — the **term structure** — is itself the object to be measured. The plan is:

- (1) say what the term structure is and what its standard shapes are;
- (2) replace the single y by a whole schedule of **spot rates** $S(1), S(2), \dots$, one per maturity, and show why yield-to-maturity discounting was inconsistent all along;
- (3) extract those spot rates from observed bond prices (**bootstrapping**), and see where that procedure runs into the data;
- (4) use the curve to measure credit risk (**spreads**) and relate it back to the yield curve we started from.

§4.1. What the Term Structure Is

— Slides p. 121

The term structure is concerned with how interest rates change with maturity. Concretely, the set of yields to maturity for bonds form the term structure — subject to two conditions that do all the work:

- The bonds must be of equal quality.
- They differ solely in their terms to maturity.

The term structure is fundamental to the valuation of fixed-income securities.

Remark 4.1.1. 那兩個條件是整個概念的成立前提，值得說明白。

我們想分離出來的是純粹因為到期日不同而產生的利率差異。若拿一檔 AAA 公司債和一檔垃圾債來比，兩者殖利率的差距主要來自**信用風險**而非期限，畫出來的曲線就不是期限結構，而是信用等級的混合物。所以要求「品質相同」（實務上通常取同一發行體，例如美國公債）與「只有到期日不同」。

真實世界裡這兩個條件從來不會完全成立——流動性、稅負、可否質押、新券舊券（on-the-run / off-the-run）都會造成差異。這正是後面 Section 4.7 所列問題的來源。

— Slides p. 122

Three names get used loosely and are worth separating once and for all:

- The term “**term structure**” often refers exclusively to the yields of *zero-coupon* bonds. This is the convention adopted here, and the reason will become clear in [Section 4.4](#): only a zero has a single, unambiguous date attached to it.
- A **yield curve** plots the yields to maturity of *coupon* bonds against maturity.
- A **par yield curve** is constructed from bonds trading near par.

Note 4.1.1. The three curves are different objects computed from the same market. A coupon bond’s yield blends the rates of every date on which it pays, so the yield curve is a smeared version of the term structure; the par yield curve fixes the smearing in a standard way by insisting the bond trade at par, which pins its coupon to its own yield. Most published “yield curves”, including the U.S. Treasury series in [Figure 4.1](#), are par yield curves.

§4.1.i. The Yield Curve of U.S. Treasuries

— Slides p. 123 —

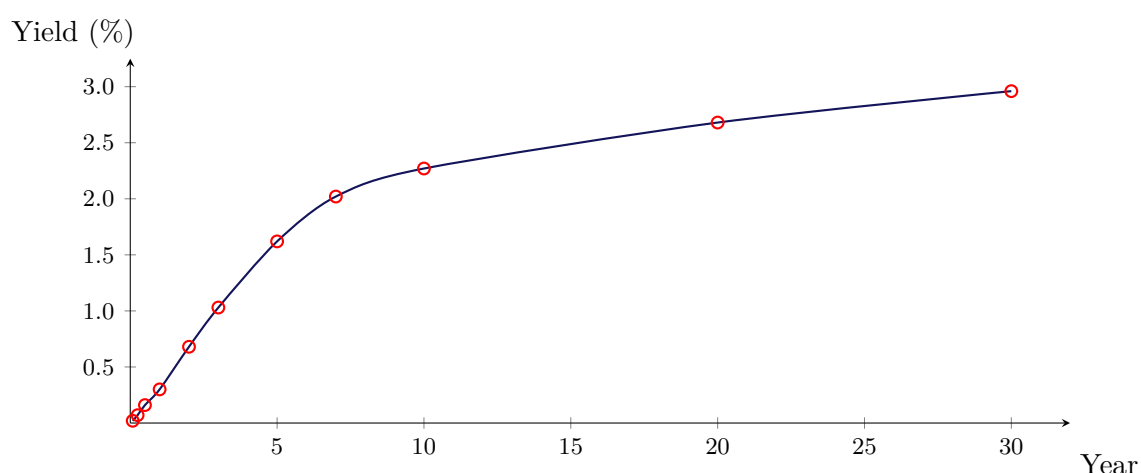


Figure 4.1.: Yield curve of U.S. Treasuries as of July 24, 2015, as shown in class. The circles are the eleven quoted maturities — 1 and 3 and 6 months, then 1, 2, 3, 5, 7, 10, 20 and 30 years — and the line is an interpolation through them. Note how unevenly the quotes are spaced: eight of the eleven sit inside the first ten years.

Remark 4.1.2. 這張圖有兩個訊息，一個明顯、一個要留意。

明顯的是形狀：短端幾乎貼地（1 個月只有 0.02%），長端接近 3%，是標準的**正常**（normal）曲線。這是 2015 年的美國——金融海嘯後聯準會把短率壓在零附近已近七年。

要留意的是橫軸上的點有多稀疏。市場只在十一個到期日上報價，其中八個擠在前十年；10 年到 20 年之間整整十年沒有任何資料點。圖上那條線是畫出來的，不是觀察到的——這個看似無害的細節，就是 [Figure 4.4](#) 的全部內容。

§4.1.ii. Four Typical Shapes

— Slides p. 124 —

- A **normal** yield curve is upward sloping.
- An **inverted** yield curve is downward sloping.
- A **flat** yield curve is flat.
- A **humped** yield curve is upward sloping at first but then turns downward sloping.

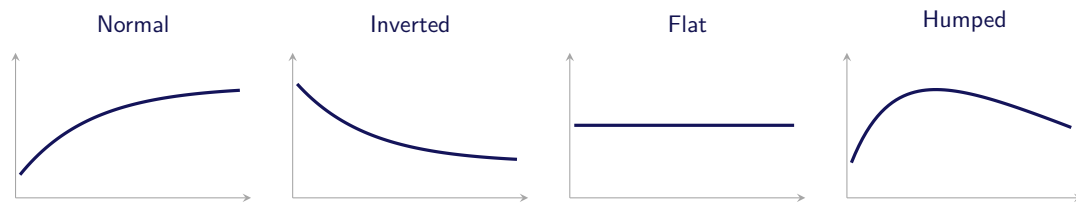


Figure 4.2.: The four shapes named in class. The horizontal axis is maturity and the vertical axis is yield in every panel.

Remark 4.1.3. 這四種形狀不只是分類，市場普遍把它們讀成對未來的訊息：

- **正常** (normal)：最常見。長天期資金較貴，通常解讀為經濟擴張、或投資人對長期持有要求額外補償。
- **倒掛** (inverted)：**短率** (short rate) 高於長率。歷史上多次出現在景氣衰退之前，因此被當作預警訊號——市場預期央行日後必須降息。
- **平坦** (flat)：常是正常與倒掛之間的過渡狀態。
- **駝峰** (humped)：中期利率最高，兩端較低。

要強調的是：這些都是經驗上的關聯，不是本課程要證明的定理。曲線形狀為何如此，是 [Section 4.7](#) 之後幾節（預期理論等）才會處理的問題；此處只需認得這四個名字。

§4.2. Spot Rates

— Slides p. 125 —

Definition 4.2.1 (Spot rate). The i -period **spot rate** $S(i)$ is the yield to maturity of an i -period zero-coupon bond. The PV of one dollar i periods from now is by definition

$$[1 + S(i)]^{-i}. \quad (4.1)$$

Three immediate remarks, all of them from the slides:

- (4.1) is the price of an i -period zero-coupon bond, by (2.11). There is nothing to prove: a zero has one cash flow, so its yield and its discount rate are the same number by construction.
- The one-period spot rate $S(1)$ is called the **short rate**.
- The **spot rate curve** — that is, the term structure, per our convention — is the plot of spot rates against maturity:

$$S(1), S(2), \dots, S(n).$$

Intuition . A spot rate is a *price*, not a forecast. $S(5) = 4\%$ says that one dollar delivered five periods from now trades today at 1.04^{-5} dollars; it says nothing about what the short rate will be in five periods' time.

即期利率 (spot rate) 是價格，不是預測。 $S(5) = 4\%$ 的意思是「五期後的一塊錢今天值 1.04^{-5} 元」，而不是「五期後的短率會是 4% 」。

§4.3. What Is Wrong with Yield-to-Maturity Discounting

— Slides p. 126 —

In the bond price formula (2.6),

$$\sum_{i=1}^n \frac{C}{(1+y)^i} + \frac{F}{(1+y)^n},$$

every cash flow is discounted at the same yield y . Now consider two riskless bonds with *different* yields to maturity, the difference arising purely because their cash flows differ:

$$PV_1 = \sum_{i=1}^{n_1} \frac{C}{(1+y_1)^i} + \frac{F}{(1+y_1)^{n_1}}, \quad (4.2)$$

$$PV_2 = \sum_{i=1}^{n_2} \frac{C}{(1+y_2)^i} + \frac{F}{(1+y_2)^{n_2}}. \quad (4.3)$$

— Slides p. 127 —

The yield-to-maturity methodology discounts their *contemporaneous* cash flows with different rates. But shouldn't they be discounted at the same rate?

Remark 4.3.1. 這一頁的投影片只丟出一個問句就結束了，但這是本章的樞紐，值得補完。

把 (4.2) 與 (4.3) 的第 i 期 (i 同時小於 n_1 與 n_2) 拿出來對照：兩檔債券都在同一天付出同樣的 C 元，兩者都無風險，可是前者用 $(1+y_1)^{-i}$ 折現、後者用 $(1+y_2)^{-i}$ 折現。既然 $y_1 \neq y_2$ ，同一天、同樣金額、同樣無風險的一塊錢，被賦予了兩個不同的現值。

這違反**一價法則** (law of one price)：同一件商品不能有兩個價格。而且這不只是美感問題——真有兩個價格就有**套利** (arbitrage) 空間：從便宜的那邊買進未來的一塊錢、往貴的那邊賣出，鎖住無風險利潤。

病灶在哪裡？ y 是「把一整串現金流壓成一個數字」的內部報酬率，它同時被當成第 1 期到第 n 期的折現率使用。但這串現金流的時間分佈（票息高低、到期長短）會影響這個平均值，於是不同的債券得到不同的 y 。換句話說， y 是債券的屬性，而折現率應該是日期的屬性。

解方就在下一節：每個日期給它自己的折現率 $S(i)$ ，與是哪一檔債券付這筆錢無關。**期限結構** (term structure) 平坦時 ($S(1) = S(2) = \dots$) 兩種作法才會一致；這也解釋了為什麼 y 一直「差不多堪用」——它是對真實折現曲線的一個加權平均近似。

§4.4. Spot Rate Discount Methodology

— Slides p. 128 —

The repair is to stop thinking of a bond as one instrument with one yield. A cash flow $C_1, C_2, \dots, C_n$ is equivalent to a *package of zero-coupon bonds*, with the i th bond paying C_i dollars at time i .

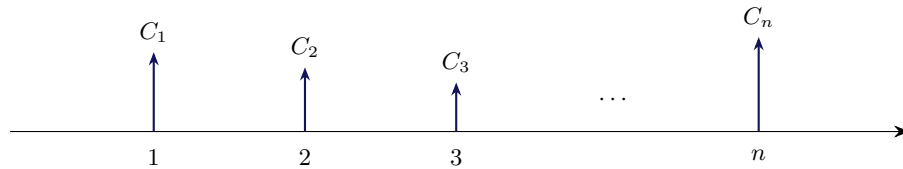


Figure 4.3.: A cash flow seen as a package of zero-coupon bonds. Each arrow is a separate security, to be priced by its own spot rate.

— Slides p. 129 —

Pricing each arrow at the spot rate matching *its own* date and adding up, a level-coupon bond has the price

$$P = \sum_{i=1}^n \frac{C}{[1 + S(i)]^i} + \frac{F}{[1 + S(n)]^n}. \quad (4.4)$$

- This pricing method incorporates information from the term structure.
- It discounts each cash flow at the matching spot rate.

The discount rate belongs to the *date*, not to the *bond*.
折現率屬於日期，不屬於債券。

Note 4.4.1. Compare (4.4) with (2.6). They have the same shape; the only change is that the single y has been replaced by a schedule $S(1), \dots, S(n)$. Every complaint of the previous section evaporates: two riskless bonds now discount their common time- i dollar by the same factor, because that factor no longer knows which bond it came from.

§4.5. Discount Factors

— Slides p. 130 —

It is worth giving those common factors a name, because they, rather than the rates, are what the pricing formula actually uses.

Definition 4.5.1 (Discount factors). In general, any riskless security having a cash flow $C_1, C_2, \dots, C_n$ should have a market price of

$$P = \sum_{i=1}^n C_i d(i), \quad (4.5)$$

where

$$d(i) \triangleq [1 + S(i)]^{-i}, \quad i = 1, 2, \dots, n,$$

are called the **discount factors**.

- $d(i)$ is the PV of one dollar i periods from now.
- The above formula will be justified later in the course, rather than assumed.
- The discount factors are often interpolated to form a continuous function called the **discount function**.

Remark 4.5.1. 為什麼折現因子 (discount factor) 要多取一個名字？因為 (4.5) 對 $d(i)$ 是線性的，對 $S(i)$ 卻不是。

$d(i)$ 是價格（今天買一張「 i 期後付一元」的憑證要花多少錢）， $S(i)$ 只是把同一件事換成年化報酬率的說法。凡是關於「一籃子現金流值多少」的敘述，用 d 寫出來都是加權求和，線性代數就能處理；換成 S 就會冒出 $[1 + S(i)]^{-i}$ 這種非線性的東西。後面的 bootstrapping、避險、以及由債券價格反推曲線的迴歸，全都是在 d 的座標下進行的。

另外， d 這個向量本身帶有無套利 (no-arbitrage) 的結構：合理的期限結構必須滿足 $1 > d(1) > d(2) > \dots > 0$ （利率為正時，愈晚的錢愈不值錢）。這種條件用 d 寫既簡單又直接。

§4.6. Extracting Spot Rates from the Yield Curve

— Slides p. 131 —

The market does not quote $S(1), S(2), \dots$ directly; beyond the very short end it quotes coupon bonds. Spot rates must therefore be *extracted*, and the extraction is recursive.

- Start with the short rate $S(1)$. Note that short-term Treasuries are zero-coupon bonds, so $S(1)$ can simply be read off.
- Compute $S(2)$ from the two-period coupon bond price P by solving

$$P = \frac{C}{1 + S(1)} + \frac{C + 100}{[1 + S(2)]^2}. \quad (4.6)$$

The point is that (4.6) contains exactly one unknown: $S(1)$ is already known, so the first term is a number, and $S(2)$ can be solved for in closed form.

— Slides p. 132 —

Inductively, we are given the market price P of the n -period coupon bond and

$$S(1), S(2), \dots, S(n-1).$$

Then $S(n)$ can be computed from (4.4), repeated here:

$$P = \sum_{i=1}^n \frac{C}{[1 + S(i)]^i} + \frac{F}{[1 + S(n)]^n}.$$

The running time can be made to be $O(n)$. The procedure is called **bootstrapping**.

Remark 4.6.1. 拔靴法 (bootstrapping) 的名字來自 “to pull oneself up by one’s bootstraps”（拉著自己的鞋帶把自己提起來）：每一步都只能靠前面已經解出來的結果，沒有外部資訊可用。 $S(1)$ 撐起 $S(2)$ ， $S(1), S(2)$ 撐起 $S(3)$ ，如此一路往上。

§4.6.i. Why $O(n)$

The naive reading of the induction costs $O(n^2)$: at step n one re-sums $n-1$ discounted coupons. The saving comes from noticing that the sum only ever grows by one term — provided we accumulate *discount factors* rather than discounted coupons, since each bond has its own coupon.

Write $A(k) \triangleq \sum_{i=1}^k d(i)$ for the running annuity factor. The n -period bond with coupon C_n and par F satisfies

$$P_n = C_n A(n-1) + (C_n + F) d(n),$$

so that

$$d(n) = \frac{P_n - C_n A(n-1)}{C_n + F}, \quad S(n) = d(n)^{-1/n} - 1, \quad A(n) = A(n-1) + d(n). \quad (4.7)$$

Each step is $O(1)$ work, so the whole curve costs $O(n)$.

Algorithm 4.1: Bootstrapping the spot rate curve

Input: prices $P_1, \dots, P_n$ and coupons $C_1, \dots, C_n$ of bonds maturing in $1, \dots, n$ periods;
par value F

Output: spot rates $S(1), \dots, S(n)$

```

1  $A := 0$ ;
2 for  $k = 1, 2, \dots, n$  do
3    $d := (P_k - C_k A) / (C_k + F)$ ;
4    $S(k) := d^{-1/k} - 1$ ;
5    $A := A + d$ ;
6 return  $S(1), \dots, S(n)$ ;
```

Note 4.6.1. The k th root costs one unit of time under the convention of [Section 1.7.ii](#), where x^y was declared a basic operation. Without that convention the count would carry an extra factor for the exponentiation — a good illustration of how much a complexity claim depends on the cost model agreed in advance.

§4.6.ii. A Worked Example

Example 4.6.1 (Not on the slides)

Suppose annual periods, $F = 100$, and that the market quotes *par* bonds — bonds priced at 100, so that each one's coupon rate equals its own yield — at

$$y_1 = 3.5\%, \quad y_2 = 4.0\%, \quad y_3 = 4.3\%, \quad y_4 = 4.5\%.$$

For a par bond, $P_k = F$ and $C_k = Fy_k$, so (4.7) simplifies to $d(k) = [1 - y_k A(k-1)] / (1 + y_k)$. Running [Algorithm 4.1](#):

k	y_k	$A(k-1)$	$d(k)$	$S(k)$
1	3.5%	0	0.966184	3.5000%
2	4.0%	0.966184	0.924378	4.0100%
3	4.3%	1.890561	0.880830	4.3204%
4	4.5%	2.771391	0.837596	4.5301%

Checking the second row against (4.6) directly: $100 - 4/1.035 = 96.135266$, and $(104/96.135266)^{1/2} - 1 = 4.0100\%$ as claimed.

Remark 4.6.2. 這張表有兩件事值得盯著看。

第一， $S(k)$ 一律高於同期的 y_k ($k = 1$ 時相等)，而且差距隨到期日拉長而擴大：1、10、20、30 個基點。這不是巧合，而是 [Section 4.9](#) 要證明的定理——曲線正常時必然如此。

第二， $d(k)$ 嚴格遞減 ($0.966 > 0.924 > 0.881 > 0.838$)，符合前一節說的無套利要求。若某組報價算出來的 d 不遞減，那不是曲線奇特，而是資料有問題。

§4.7. Some Problems

— Slides p. 133 —

Algorithm 4.1 is clean; the data are not.

- Treasuries of the same maturity might be selling at different yields (the **multiple cash flow problem**).
- Some maturities might be missing from the data points (the **incompleteness problem**).
- Treasuries might not be of the same quality.
- Interpolation and fitting techniques are needed in practice to create a smooth spot rate curve.¹

Remark 4.7.1. 逐條說明：

(一) **多重現金流問題** (multiple cash flow problem)。同一到期日可能有好幾檔公債，票息不同。依上一節的分析，票息不同的債券本來就會有不同的 y ——這正是我們一開始要拋棄 y 的理由。麻煩在於它們解出來的 $S(n)$ 也會互相矛盾，於是必須決定採信哪一檔，或做某種平均。

(二) **不完整問題** (incompleteness problem)。Algorithm 4.1 要求每一期 $1, 2, \dots, n$ 都有一檔債券，現實中不可能。Figure 4.1 裡 10 年到 20 年之間沒有任何報價，半年期的格點更是大半空著。缺的格子只能猜。

(三) **品質不齊**。**新發行券** (on-the-run) 流動性遠優於**舊券** (off-the-run)，殖利率因而偏低；稅負待遇、能否作為附買回擔保品也都造成差異。這些價差與期限無關，卻全被算進曲線裡。

(四) **插補** (interpolation)。前三項的結果是：實務上得到的不是一條曲線，而是一團帶雜訊且有缺口的點。要變成可用的 $d(\cdot)$ ，必須挑一個插補或擬合方法——而那個腳註的力道很重：通常沒有經濟學上的理由。選哪條曲線，往往只是選哪一種數學上的平滑性。

§4.7.i. Which One?

— Slides p. 134 —

The incompleteness problem is worth seeing rather than reading about. Here are the same eleven observations of Figure 4.1, joined three different ways.

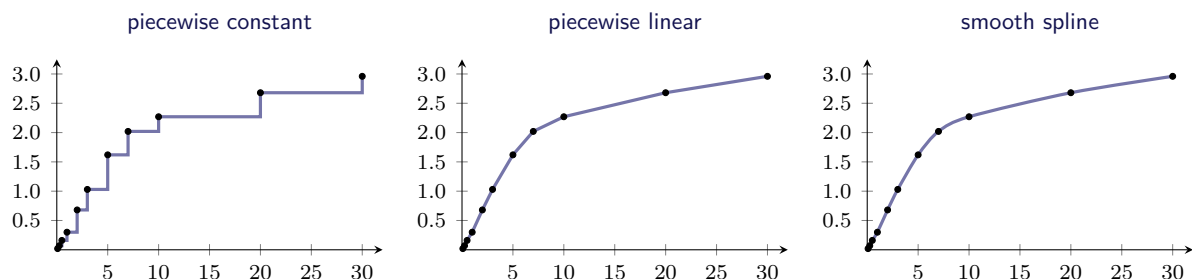


Figure 4.4.: Which one? Three curves through the identical eleven quotes of Figure 4.1; the horizontal axis is the maturity in years and the vertical axis the yield in percent, as before. Every one of them reprices all observed bonds exactly; they disagree about everything in between.

¹Often without economic justifications.

The data fix the curve only where the data are. Everywhere else the curve is an assumption — and any security priced off that stretch inherits it.

資料只能決定有資料的地方；其餘部分是假設。凡是用那段曲線定價的商品，都繼承了這個假設。

§4.8. Yield Spread and Static Spread

Everything so far assumed riskless cash flows. A corporate bond promises the same shape of cash flow but may not deliver, and the market charges for that. The two measures below quantify the charge; they differ in whether they use the term structure or ignore it.

§4.8.i. Yield Spread

— Slides p. 135 —

Consider a risky bond with the cash flow $C_1, C_2, \dots, C_n$ and selling for P .

- Calculate the IRR of the risky bond.
- Calculate the IRR of a riskless bond with comparable maturity.
- The **yield spread** is their difference.

§4.8.ii. Static Spread

— Slides p. 136 —

Were the risky bond riskless, it would fetch

$$P^* = \sum_{t=1}^n \frac{C_t}{[1 + S(t)]^t}. \quad (4.8)$$

But as risk must be compensated, in reality $P < P^*$.

Definition 4.8.1 (Static spread). The **static spread** is the amount s by which the spot rate curve has to shift in parallel to price the risky bond:

$$P = \sum_{t=1}^n \frac{C_t}{[1 + s + S(t)]^t}. \quad (4.9)$$

Unlike the yield spread, the static spread explicitly incorporates information from the term structure.

Remark 4.8.1. 兩者差在哪裡，用一句話講：**殖利率價差** (yield spread) 拿風險債券的 IRR 去減另一檔債券的 IRR；**靜態價差** (static spread) 則是把整條無風險曲線平移到剛好能解釋這個價格。

殖利率價差的毛病承襲自 Section 4.3：兩個 IRR 各自是各自現金流的加權平均，票息與期限只要不同，比較就不乾淨。靜態價差把每一期的現金流都對到它自己的 $S(t)$ ，再問「所有期別同時加上多少，價格才對得上」，因此拿掉了現金流形狀造成的干擾。

兩式的關係也值得看：(4.9) 右邊對 s 嚴格遞減， $s = 0$ 時值為 $P^* > P$ ，故解存在且唯一。期限結構平坦時 ($S(t) \equiv S$)，(4.9) 退化成 $P = \sum C_t / (1 + S + s)^t$ ，此時靜態價差恰好等於殖利率價差——兩者的差異完全來自曲線的斜率。

「靜態」二字則是留給後面的伏筆：它假設現金流固定不變。一旦債券附有贖回權或提前還款權 (Section 3.13 那些商品)，現金流會隨利率改變，就需要**選擇權調整價差** (option-adjusted spread, OAS)。

Example 4.8.2 (Not on the slides)

Take the spot curve bootstrapped above, $S(1) = 3.5000\%$, $S(2) = 4.0100\%$, $S(3) = 4.3204\%$, and a three-period risky bond paying 6, 6, 106. Then (4.8) gives $P^* = 104.7114$. Suppose it actually trades at $P = 95$.

- Its IRR solves $95 = \sum_t C_t(1+y)^{-t}$, giving $y = 7.9380\%$. Against the comparable riskless three-period par yield of 4.3%, the *yield spread* is 363.8 basis points.
- Solving (4.9) gives a *static spread* of $s = 364.7$ basis points.

The two differ by about a basis point here because the curve is only mildly sloped over three periods; on a steep curve, or a long bond, the gap is far larger.

§4.9. Of Spot Rate Curve and Yield Curve

— Slides p. 137 —

We began with the yield curve and replaced it with the spot rate curve. How do the two relate? Write y_i for the yield to maturity of the i -period coupon bond.

- $S(k) \geq y_k$ if $y_1 < y_2 < \dots$ (yield curve is normal).
- $S(k) \leq y_k$ if $y_1 > y_2 > \dots$ (yield curve is inverted).
- $S(k) \geq y_k$ if $S(1) < S(2) < \dots$ (spot rate curve is normal).
- $S(k) \leq y_k$ if $S(1) > S(2) > \dots$ (spot rate curve is inverted).
- If the yield curve is flat, the spot rate curve coincides with the yield curve.

All five follow from one small lemma, which is worth isolating because it is the precise sense in which “a yield is an average of spot rates”.

Lemma 4.9.1

Let a security with non-negative cash flows $C_1, \dots, C_k$, not all zero, be priced by (4.5), and let y_k be its yield to maturity. Then

$$\min_{1 \leq i \leq k} S(i) \leq y_k \leq \max_{1 \leq i \leq k} S(i).$$

Proof. Put $f(y) \triangleq \sum_{i=1}^k C_i(1+y)^{-i}$, which is strictly decreasing for $y > -1$, and let $\bar{S} \triangleq \max_i S(i)$. Since $(1+\bar{S})^{-i} \leq [1+S(i)]^{-i}$ for every i ,

$$f(\bar{S}) \leq \sum_{i=1}^k C_i [1+S(i)]^{-i} = P = f(y_k),$$

and f decreasing gives $y_k \leq \bar{S}$. The other inequality is symmetric. ■

The third, fourth and fifth bullets are immediate:

- If the spot rate curve is normal then $\max_{i \leq k} S(i) = S(k)$, so $y_k \leq S(k)$.
- If it is inverted then $\min_{i \leq k} S(i) = S(k)$, so $y_k \geq S(k)$.
- If the curve is flat, $\min = \max$ and y_k is squeezed to the common value; spot and yield curves coincide.

Remark 4.9.1. 用一句話記住這個引理：殖利率必落在該債券所觸及的各期即期利率的最大值與最小值之間。

把 Section 4.6 的例子代進來驗證： $S(1) = 3.50\%$ 、 $S(2) = 4.01\%$ 、 $S(3) = 4.32\%$ 是遞增的，所以三期債的 y_3 必須 $\leq S(3) = 4.32\%$ ；實際上 $y_3 = 4.30\%$ 。而且票息愈高，早期現金流的權重愈大， y_k 被往下拉得愈多，與 $S(k)$ 的差距就愈大——零息債（權重全在最後一期）則兩者相等。

至於前兩條（以殖利率曲線的形狀為條件），投影片與引理同屬一個機制，但方向相反、需要額外的歸納論證；下一頁正是要說明這個反向才是脆弱的一環。

§4.9.i. Shapes

— Slides p. 138 —

- The spot rate curve often has the same shape as the yield curve.
 - If the spot rate curve is inverted (normal, resp.), then the yield curve is inverted (normal, resp.).
- But this is only a trend, not a mathematical truth.²

Note 4.9.1. The asymmetry is worth stating plainly, because the slides put theorem and trend in the same list.

The lemma above is a *theorem*: given the shape of the spot curve, the position of y_k relative to $S(k)$ is settled at every k individually. The claim on this page is different — it is about the *shape* of the whole yield curve, i.e. about how y_k compares with y_{k+1} , and that comparison involves two different bonds with two different sets of weights. Nothing forces the weighting to behave monotonically, and the textbook exhibits a spot curve that is monotone while the induced yield curve is not.

The practical reading: use the spot curve to price, and treat the yield curve's shape as a summary for conversation, not as an input.

§4.10. Forward Rates

— Slides p. 139 —

The yield curve contains information regarding future interest rates currently “expected” by the market. To extract it, compare two ways of holding money from time 0 to time j .

- **The maturity strategy.** Invest \$1 for j periods to end up with $[1 + S(j)]^j$ dollars at time j .
- **The rollover strategy.** Invest \$1 in bonds for i periods and at time i invest the proceeds in bonds for another $j - i$ periods, where $j > i$. This will have

$$[1 + S(i)]^i [1 + S(i, j)]^{j-i}$$

dollars at time j , where $S(i, j)$ denotes the $(j - i)$ -period spot rate i periods from now.

The first number is known today. The second is not: $S(i, j)$ is a rate that will only be quoted at time i . The question the market implicitly answers is what value of $S(i, j)$ would make the two strategies tie.

— Slides p. 140 —

²See a counterexample in the text.

Definition 4.10.1 (Forward rate). When $S(i, j)$ equals

$$f(i, j) \triangleq \left[\frac{(1 + S(j))^j}{(1 + S(i))^i} \right]^{1/(j-i)} - 1, \quad (4.10)$$

we will end up with the same $[1 + S(j)]^j$ dollars. The $f(i, j)$ are the **(implied) forward (interest) rates** — more precisely, the $(j - i)$ -period forward rate i periods from now.

(4.10) is just the tie condition solved for the unknown:

$$[1 + S(i)]^i [1 + f(i, j)]^{j-i} = [1 + S(j)]^j \iff [1 + f(i, j)]^{j-i} = \frac{(1 + S(j))^j}{(1 + S(i))^i}.$$

As expected, $f(0, j) = S(j)$: with $i = 0$ the rollover strategy *is* the maturity strategy.

Intuition . $[1 + S(k)]^k$ is the price of moving a dollar from time 0 to time k . (4.10) divides the cost of travelling to j by the cost of travelling to i , leaving the cost of the leg from i to j — then annualises it by taking the $(j - i)$ th root.

$[1 + S(k)]^k$ 是把一塊錢從 0 搬到 k 的「價格」。(4.10) 用走到 j 的成本除以走到 i 的成本，剩下的就是 i 到 j 那一段的成本，再開 $(j - i)$ 次方還原成每期利率。

§4.10.i. The Time Line

— Slides p. 141 —

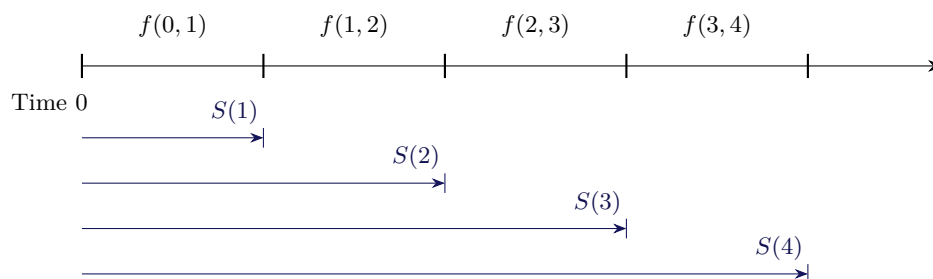


Figure 4.5.: The time line. The spot rates below run from today out to each maturity; the one-period forward rates above cover the individual legs between consecutive dates.

Remark 4.10.1. 這張圖是整節的骨架，值得盯著看清楚兩者的分工。

下半部的 $S(k)$ 都從 time 0 出發——每一條都是「從今天走到第 k 期」的平均利率，所以彼此重疊。上半部的 $f(k, k + 1)$ 則彼此不重疊，各自只負責一小段。

用速度打比方： $S(k)$ 是走完前 k 期的平均速度， $f(k, k + 1)$ 是第 $k + 1$ 期的瞬時速度。平均速度由瞬時速度累積而成——這正是 (4.11) 要說的事；而平均速度上升，必然是因為當下的瞬時速度高於既有的平均，這正是 Section 4.11 的不等式。

§4.10.ii. Forward Rates and Future Spot Rates

— Slides p. 142 —

Note 4.10.1. We did not assume any *a priori* relation between $f(i, j)$ and the future spot rate $S(i, j)$ — that is the subject of the term structure theories in [Section 4.15](#). We merely looked for the future spot rate that, *if realized*, will equate the two investment strategies.

The $f(i, i+1)$ are the **instantaneous forward rates** or **one-period forward rates**.

Remark 4.10.2. 這個區別是本章最常被誤解的地方，務必分清楚。

遠期利率 (forward rate) $f(i, j)$ 是今天就能從 (4.10) 算出來的數字，只用到今天觀察得到的 $S(i)$ 與 $S(j)$ ；它是一個**恆等式** (identity) 的產物，不是預測。 $S(i, j)$ 則是 i 期之後才會知道的隨機變數。

「market expects」那句話加了引號，正是這個意思： f 是市場定價隱含的數字，至於它是否等於市場對 $S(i, j)$ 的期望值，是一個需要額外假設的理論命題 ([Section 4.15](#) 的無偏預期理論就是這樣一個假設)，而不是推導出來的結果。

[Section 4.13](#) 會更進一步： $f(i, j)$ 不只是算出來的數字，而是今天就能用現有債券鎖定的真實利率。

§4.11. Spot Rates and Forward Rates

— Slides p. 143 —

Proposition 4.11.1

When the spot rate curve is normal, the forward rate dominates the spot rates,

$$f(i, j) > S(j) > \cdots > S(i).$$

When the spot rate curve is inverted, the forward rate is dominated by the spot rates,

$$f(i, j) < S(j) < \cdots < S(i).$$

Proof. Suppose the curve is normal, so $S(i) < S(j)$ for $i < j$. From (4.10),

$$[1 + f(i, j)]^{j-i} = \frac{(1 + S(j))^j}{(1 + S(i))^i} > \frac{(1 + S(j))^j}{(1 + S(j))^i} = [1 + S(j)]^{j-i},$$

where the inequality replaces $S(i)$ in the denominator by the larger $S(j)$. Taking $(j-i)$ th roots gives $f(i, j) > S(j)$. The inverted case reverses every inequality. ■

Example 4.11.2 (Not on the slides)

Take the curve bootstrapped in [Section 4.6](#), namely $S(1) = 3.5000\%$, $S(2) = 4.0100\%$, $S(3) = 4.3204\%$, $S(4) = 4.5301\%$. The one-period forward rates are

$$f(0, 1) = 3.5000\%, \quad f(1, 2) = 4.5225\%, \quad f(2, 3) = 4.9440\%, \quad f(3, 4) = 5.1617\%.$$

Each sits above the spot rate for the same terminal date, and the forward curve rises faster than the spot curve — the marginal rate outruns the average it is pulling up.

— Slides p. 144 —

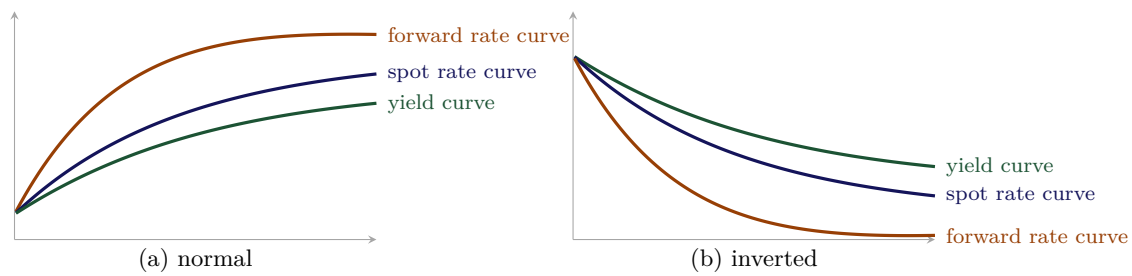


Figure 4.6.: The three curves never cross: forward above spot above yield when the curve is normal, and exactly the reverse when it is inverted. Horizontal axis maturity, vertical axis rate.

Remark 4.11.1. 三條曲線的順序不必死記，用邊際量與平均量的關係一次想通。

把 f 看成邊際利率（第 k 期那一小段的價格）， S 是它們的平均（連續複利下 (4.15) 就字面上是算術平均）， y 則是把票息也算進來的再一次平均——因為票息落在較早、利率較低的日期， y 被往下拉得比 S 更多。

微積分裡的老規矩：平均上升時，邊際必在平均之上。所以曲線正常（平均遞增）時 $f > S$ ；曲線倒掛（平均遞減）時 $f < S$ 。 y 與 S 的先後則由 Section 4.9 的引理保證。兩者串起來就是圖上的順序。

§4.12. Forward Rates $\equiv$ Spot Rates $\equiv$ Yield Curve

— Slides p. 145 —

The three curves are not merely ordered; they carry exactly the same information. The FV of \$1 at time n can be derived in two ways.

- Buy n -period zero-coupon bonds and receive

$$[1 + S(n)]^n.$$

- Buy one-period zero-coupon bonds today and a series of such bonds at the forward rates as they mature. The FV is

$$[1 + S(1)][1 + f(1, 2)] \cdots [1 + f(n - 1, n)].$$

— Slides p. 146 —

Theorem 4.12.1

Since the two are identical,

$$S(n) = \left\{ [1 + S(1)][1 + f(1, 2)] \cdots [1 + f(n - 1, n)] \right\}^{1/n} - 1. \quad (4.11)$$

Hence the forward rates — specifically the one-period forward rates — determine the spot rate curve.

Other equivalencies can be derived similarly, such as

$$f(T, T+1) = \frac{d(T)}{d(T+1)} - 1. \quad (4.12)$$

Proof of (4.12). By definition $d(k) = [1 + S(k)]^{-k}$, so

$$\frac{d(T)}{d(T+1)} = \frac{(1 + S(T+1))^{T+1}}{(1 + S(T))^T} = 1 + f(T, T+1)$$

by (4.10) with $i = T$, $j = T+1$. ■

Spot rates, forward rates, discount factors and the yield curve are four coordinate systems for one object. Any one of them can be computed from any other; none contains extra information.

即期利率、遠期利率、折現因子、殖利率曲線，是同一個東西的四套座標。任一套都能算出其餘三套，沒有誰多帶了資訊。

Example 4.12.2 (Not on the slides)

With the same curve as above, (4.11) gives

$$(1.035 \times 1.045225 \times 1.049440 \times 1.051617)^{1/4} - 1 = 4.5301\% = S(4),$$

and (4.12) gives $d(1)/d(2) - 1 = 0.966184/0.924378 - 1 = 4.5225\% = f(1, 2)$.

§4.13. Locking in the Forward Rate

— Slides p. 147 —

So far $f(n, m)$ is an arithmetic construct. The following trade turns it into a rate one can actually contract at today, using nothing but zero-coupon bonds that trade now.

- Buy one n -period zero-coupon bond for $1/(1 + S(n))^n$ dollars.
- Sell $(1 + S(m))^m / (1 + S(n))^n$ m -period zero-coupon bonds.
- No net initial investment, because the cash inflow equals the cash outflow, both being $1/(1 + S(n))^n$.
- At time n there will be a cash inflow of \$1.
- At time m there will be a cash outflow of $(1 + S(m))^m / (1 + S(n))^n$ dollars.

Why the initial cost is zero. Each m -period zero sells for $(1 + S(m))^{-m}$, so selling $(1 + S(m))^m / (1 + S(n))^n$ of them raises

$$\frac{(1 + S(m))^m}{(1 + S(n))^n} \times \frac{1}{(1 + S(m))^m} = \frac{1}{(1 + S(n))^n},$$

which is exactly what the n -period zero costs. ■

— Slides p. 148 —

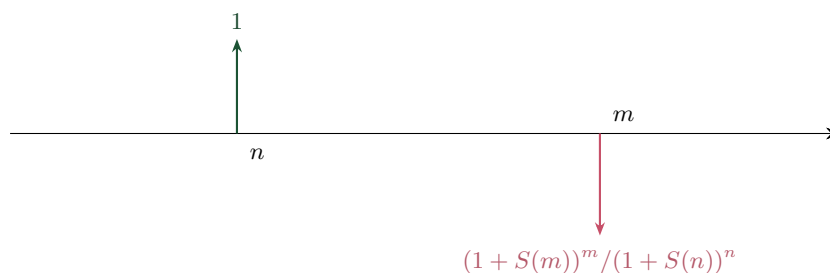


Figure 4.7.: The cash flow locked in today. Nothing changes hands at time 0; one dollar arrives at time n and is repaid at time m .

This implies the interest rate between times n and m equals $f(n, m)$, because by (4.10)

$$\frac{(1 + S(m))^m}{(1 + S(n))^n} = (1 + f(n, m))^{m-n}.$$

Example 4.13.1 (Not on the slides)

Lock in $f(1, 3)$ on the curve above. Buy one 1-period zero for $d(1) = 0.966184$ and sell $K = (1.043204)^3 / 1.035 = 1.096901$ three-period zeros, raising $K \times d(3) = 1.096901 \times 0.880830 = 0.966184$ — the same amount. The resulting cash flow is $+1$ at time 1 and -1.096901 at time 3, and $1.096901 = (1 + 4.7330\%)^2$, so the locked rate is $f(1, 3) = 4.7330\%$.

§4.13.i. Forward Loans

— Slides p. 149 —

We had generated the cash flow of a type of forward contract called the **forward loan**. Agreed upon today, it enables one to

- borrow money at time n in the future, and
- repay the loan at time $m > n$ with an interest rate equal to the known forward rate $f(n, m)$.

Question 4.13.2. Can the spot rate curve be arbitrarily drawn?^a

^aContributed by Mr. Dai, Tian-Shyr (B82506025, R86526008, D88526006) in 1998.

Answer. No. Since every forward loan can be manufactured from traded zeros, the forward rates implied by the curve are rates one can actually transact at — so they must themselves be free of arbitrage.

Suppose $f(n, n+1) < 0$ for some n , which by (4.12) means $d(n) < d(n+1)$. Then sell the $(n+1)$ -period zero and buy the n -period zero: this raises $d(n+1) - d(n) > 0$ today, pays \$1 at time n , and costs \$1 at time $n+1$. Hold the dollar received at time n as cash until time $n+1$ and the later obligation is met exactly. The result is a strictly positive amount today against no future liability — an arbitrage.

So the curve must satisfy

$$1 > d(1) \geq d(2) \geq \dots, \quad \text{equivalently} \quad (1 + S(n))^n \text{ is non-decreasing in } n.$$

In particular an inverted curve cannot fall arbitrarily fast; Section 4.14 will record the continuous-time version of the same bound. ■

Remark 4.13.1. 上面那段論證偷偷用了一個假設：現金可以無成本地保管。正因為存著鈔票的報酬率是 0，任何低於 0 的遠期利率才會被套利掉。

現實中這個假設並不完全成立——大額現金的保管、保險與運送都要錢，而且央行可以對準備金課負利率。2014 年之後的歐元區與日本公債確實出現過負的即期與遠期利率，而且持續多年。所以正確的說法是：曲線不能任意畫，但下界不是 0，而是 $-(\text{持有現金的成本})$ 。

§4.13.ii. Synthetic Bonds

— Slides p. 150 —

Read the locking trade as an identity between portfolios. We had seen that

$$\text{forward loan} = n\text{-period zero} - [1 + f(n, m)]^{m-n} \times m\text{-period zero}.$$

Thus

$$n\text{-period zero} = \text{forward loan} + [1 + f(n, m)]^{m-n} \times m\text{-period zero}.$$

We have created a **synthetic** zero-coupon bond with forward loans and other zero-coupon bonds. This is useful if the n -period zero is unavailable or illiquid.

Note 4.13.1. This is the first appearance of a move that will run through the rest of the course: an identity between cash flows is an identity between *prices*, so any instrument that can be replicated must cost what its replicating portfolio costs. Here it fills a hole in the maturity spectrum — exactly the incompleteness problem of [Section 4.7](#), attacked by construction rather than by interpolation.

§4.14. Spot and Forward Rates under Continuous Compounding

— Slides p. 151 —

Every formula above simplifies under continuous compounding, because products of gross rates become sums of rates. Recall (2.4): one dollar grows to e^{rn} rather than $(1+r)^n$.

- The pricing formula:

$$P = \sum_{i=1}^n C e^{-iS(i)} + F e^{-nS(n)}. \quad (4.13)$$

- The market discount function:

$$d(n) = e^{-nS(n)}. \quad (4.14)$$

- The spot rate is an *arithmetic average* of forward rates:

$$S(n) = \frac{f(0, 1) + f(1, 2) + \cdots + f(n-1, n)}{n}. \quad (4.15)$$

Remark 4.14.1. 把 (4.15) 與離散版的 (4.11) 並排看，就知道連續複利的好處在哪裡。

離散版是**幾何平均** (geometric mean)：1 + $S(n)$ 等於 n 個 $1 + f$ 相乘再開 n 次方。連續版是**算術平均** (arithmetic mean)： $S(n)$ 就是 n 個 f 直接相加除以 n 。

原因只有一個： $e^a e^b = e^{a+b}$ 。指數把乘法換成加法，於是「利率的累積」變成單純的求

和，微分與積分也才用得上——這就是為什麼從這裡之後的理論工作幾乎一律採用連續複利。

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- The formula for the forward rate:

$$f(i, j) = \frac{jS(j) - iS(i)}{j - i}. \quad (4.16)$$

- The one-period forward rate:

$$f(j, j+1) = -\ln \frac{d(j+1)}{d(j)}. \quad (4.17)$$

Compare (4.16) with (4.10) and (4.17) with (4.12): in each case a ratio of gross growth factors has turned into a difference of rates. Both follow from the tie condition written continuously, $e^{iS(i)}e^{(j-i)f(i,j)} = e^{jS(j)}$, which on taking logarithms reads $iS(i) + (j-i)f(i,j) = jS(j)$.

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Definition 4.14.1 (Instantaneous forward rate curve).

$$f(T) \triangleq \lim_{\Delta T \rightarrow 0} f(T, T + \Delta T) = S(T) + T \frac{\partial S}{\partial T}. \quad (4.18)$$

Proof. By (4.16) with $i = T$ and $j = T + \Delta T$,

$$f(T, T + \Delta T) = \frac{(T + \Delta T)S(T + \Delta T) - TS(T)}{\Delta T},$$

which is the difference quotient of the function $T \mapsto TS(T)$. Its limit is $\frac{d}{dT}[TS(T)] = S(T) + TS'(T)$. ■

Two consequences are recorded on the slide:

- $f(T) > S(T)$ if and only if $\partial S / \partial T > 0$, i.e. the spot rate curve is normal. This is the continuous-time restatement of Section 4.11, and (4.18) makes it immediate: $f - S = T \partial S / \partial T$.
- If $S(T) < -T(\partial S / \partial T)$, then $f(T) < 0$.

Remark 4.14.2. 第二條正是前面「曲線不能任意畫」的連續版本。

$f(T) \geq 0$ 等價於 $\frac{d}{dT}[TS(T)] \geq 0$ ，也就是累積利率 $TS(T)$ 必須遞增——與離散版 $d(n)$ 遞減、 $(1 + S(n))^n$ 遞增是同一句話（因為 $d(T) = e^{-TS(T)}$ ）。

改寫成 $S'(T) \geq -S(T)/T$ 就看得出限制有多具體：倒掛的曲線每往下走一點，都受到目前水準除以年期的約束。 $S = 2\%$ 、 $T = 5$ 時，斜率不得低於 -0.4% 每年； T 愈長，允許的下滑愈平緩。

§4.14.i. An Example

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Example 4.14.2

Let the interest rates be continuously compounded, and suppose the spot rate curve is^a

$$S(T) \triangleq 0.08 - 0.05 e^{-0.18T}.$$

Then by (4.18) the forward rate curve is

$$\begin{aligned} f(T) &= S(T) + T S'(T) \\ &= 0.08 - 0.05 e^{-0.18T} + 0.009 T e^{-0.18T}, \end{aligned}$$

since $S'(T) = 0.009 e^{-0.18T}$.

^aJ. Hull & White (1994).

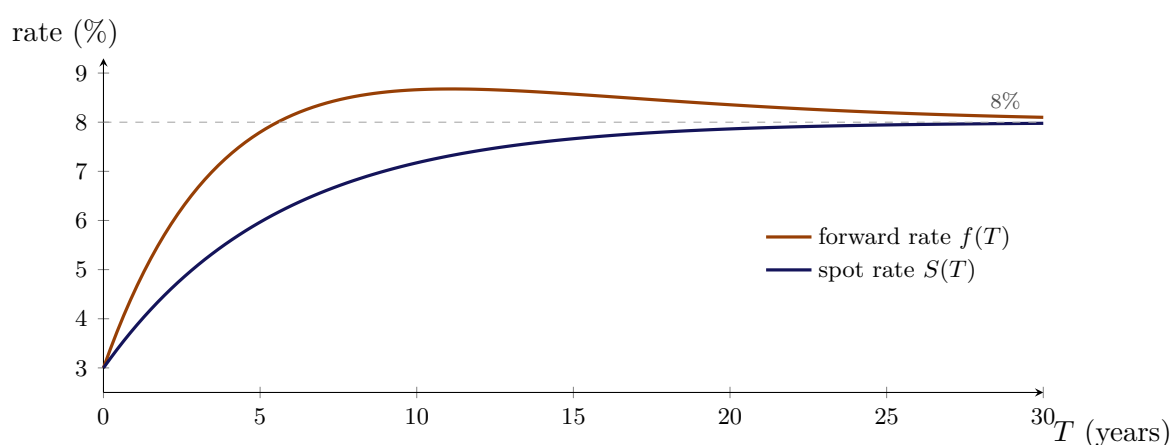


Figure 4.8.: The Hull–White spot curve and the forward curve it implies. The spot curve rises monotonically to its asymptote of 8%; the forward curve overshoots it, peaking at 8.68% near $T = 11.1$ before falling back. Both start at 3%, since $f(0) = S(0)$.

Remark 4.14.3. 遠期曲線衝過長期水準再折返，這件事值得解釋——它不是畫錯了。

$S(T)$ 是 $[0, T]$ 上所有瞬時利率的平均。既然平均要從 3% 一路爬到 8%，而起點被壓在 3%，那麼中段的瞬時利率就必須高於 8%，才拉得動整條平均線。等到平均值接近 8%、不再需要被拉動時， f 才降回 8%。

這正是「邊際領先平均」的標準圖像，也是實務上常說的：遠期曲線是即期曲線的放大版——即期曲線的任何斜率，在遠期曲線上都被乘上 T 放大了（見 (4.18)）。

§4.15. Term Structure Theories

Everything up to here was arithmetic: given prices, the forward rates follow by identity. The remaining question is economic — what, if anything, does $f(i, j)$ say about the *future* spot rate $S(i, j)$? Answering it requires an assumption, and the candidate assumptions are the term structure theories.

§4.15.i. Unbiased Expectations Theory

Definition 4.15.1 (Unbiased expectations theory). The forward rate equals the average future spot rate,

$$f(a, b) = E[S(a, b)]. \quad (4.19)$$

- It does *not* imply that the forward rate is an accurate predictor for the future spot rate.
- It implies the maturity strategy and the rollover strategy produce the same result at the horizon “on average”.

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The theory ties the shape of today’s curve to the market’s expectations.

Proposition 4.15.2

Under (4.19), a normal spot rate curve is due to the fact that the market expects the future spot rate to rise. Conversely, the spot rate is expected to fall if and only if the spot rate curve is inverted.

Proof. First, $f(j, j+1) > S(j+1)$ if and only if $S(j+1) > S(j)$. Indeed, by (4.10),

$$1 + f(j, j+1) = \frac{(1 + S(j+1))^{j+1}}{(1 + S(j))^j},$$

so $f(j, j+1) > S(j+1)$ is equivalent to $(1 + S(j+1))^j > (1 + S(j))^j$, i.e. to $S(j+1) > S(j)$. Substituting (4.19) gives

$$E[S(j, j+1)] > S(j+1) > \cdots > S(1) \quad \text{if and only if} \quad S(j+1) > \cdots > S(1). \quad \blacksquare$$

Remark 4.15.1. 第一個小結論不需要任何理論——它純粹是 (4.10) 的代數重寫，也就是 Section 4.11 的特例。理論的作用只在最後一步：把算出來的 $f(j, j+1)$ 換成期望值 $E[S(j, j+1)]$ 。

所以這條命題的內容其實是：在無偏預期理論 (unbiased expectations theory) 之下，「殖利率曲線正常」與「市場預期短率將上升」是同一件事。這也是新聞上「倒掛的殖利率曲線預示降息」那句話的理論根據——但別忘了它建立在假設之上，而下一小節正是要說明這一類假設有多脆弱。

§4.15.ii. A “Bad” Expectations Theory

— Slides p. 157 —

Consider the natural-sounding strengthening: the expected returns³ on all possible riskless bond strategies are equal for all holding periods. It turns out to be internally inconsistent.

Over a *two*-period horizon, buying a two-period bond and rolling over one-period bonds must tie:

$$(1 + S(2))^2 = (1 + S(1)) E[1 + S(1, 2)]. \quad (4.20)$$

After rearrangement,

$$\frac{1}{E[1 + S(1, 2)]} = \frac{1 + S(1)}{(1 + S(2))^2}. \quad (4.21)$$

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Now consider two *one*-period strategies.

³More precisely, the one-plus returns.

- Strategy one buys a two-period bond for $(1 + S(2))^{-2}$ dollars and sells it after one period. Its price then will be $(1 + S(1, 2))^{-1}$, so the expected return is

$$E[(1 + S(1, 2))^{-1}] / (1 + S(2))^{-2}.$$

- Strategy two buys a one-period bond with a return of $1 + S(1)$.

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The theory says the returns are equal:

$$\frac{1 + S(1)}{(1 + S(2))^2} = E\left[\frac{1}{1 + S(1, 2)}\right].$$

Combining this with (4.21) yields

$$E\left[\frac{1}{1 + S(1, 2)}\right] = \frac{1}{E[1 + S(1, 2)]}. \quad (4.22)$$

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Proposition 4.15.3

(4.22) is impossible save for a certain economy — that is, unless $S(1, 2)$ is known today.

Proof. Jensen's inequality states that $E[g(X)] > g(E[X])$ for any nondegenerate random variable X and strictly convex function g (i.e. $g''(x) > 0$). Use

$$g(x) \triangleq (1 + x)^{-1},$$

for which $g''(x) = 2(1 + x)^{-3} > 0$ whenever $x > -1$. With $X = S(1, 2)$,

$$E\left[\frac{1}{1 + S(1, 2)}\right] > \frac{1}{1 + E[S(1, 2)]} = \frac{1}{E[1 + S(1, 2)]},$$

which contradicts (4.22) unless $S(1, 2)$ is degenerate, i.e. deterministic. ■

Remark 4.15.2. 這個結果的教訓比它的算式重要，值得講清楚。

錯在哪裡？不在於某一個式子算錯，而在於這個理論要求太多：它同時宣稱所有期間長度、所有策略的期望報酬都相等。兩期視角給出 (4.20)，一期視角給出另一條式子，兩者合起來就逼出 (4.22)——一個只有在未來利率毫無不確定性時才成立的等式。

根源是「期望值」與「取倒數」不可交換。債券價格是利率的凸函數 ($1/(1+x)$ 凸)，所以「先取期望再折現」與「先折現再取期望」必然不同，差額就是 **Jensen 不等式** (Jensen's inequality) 留下的缺口。這與 Section 3.10 的凸性其實是同一件事的兩個面貌：那裡凸性讓持有人「漲賺得比跌賠得多」，這裡凸性讓 $E[1/(1+X)] > 1/E[1+X]$ 。

出路是只挑一種視角來要求，別全都要——這就是下一小節的**局部預期理論** (local expectations theory)。

§4.15.iii. Local Expectations Theory

— Slides p. 161 —

Definition 4.15.4 (Local expectations theory). The expected rate of return of any bond over a *single period* equals the prevailing one-period spot rate:

$$\frac{E[(1 + S(1, n))^{-(n-1)}]}{(1 + S(n))^{-n}} = 1 + S(1) \quad \text{for all } n > 1. \quad (4.23)$$

This theory is the basis of many interest rate models.

Note 4.15.1. The repair is exactly the one the previous subsection pointed to. (4.23) imposes the equal-expected-return condition over *one* period only, and says nothing about longer horizons; there is therefore no second equation to clash with the first, and no Jensen contradiction arises.

Read (4.23) as: today's price of every bond is the expectation of its price next period, discounted at the known one-period rate. That is precisely the recursive structure a tree or lattice model needs, which is why the theory reappears later in the course as the *risk-neutral* pricing convention.

§4.16. Duration, in Practice

— Slides p. 162 —

We had assumed parallel shifts in the spot rate curve — the caveat flagged back in Section 3.6.ii. Now that the curve is a genuine object with n separate coordinates, that assumption can be relaxed.

To handle more general shifts, define a vector $[c_1, c_2, \dots, c_n]$ that characterizes the shift:

- Parallel shift: $[1, 1, \dots, 1]$.
- Twist: $[1, 1, \dots, 1, -1, \dots, -1]$, $[1.8, 1.6, 1.4, 1, 0, -1, -1.4, \dots]$, etc.
- ...

At least one c_i should be 1 as the reference point.

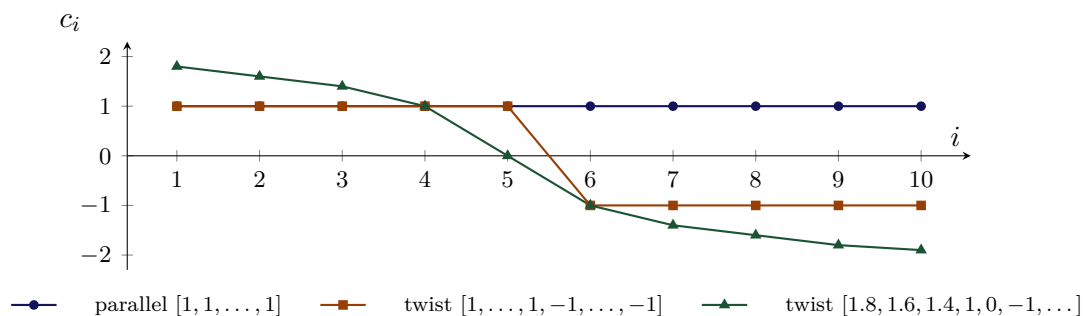


Figure 4.9.: Shift vectors. Each describes how much the spot rate at maturity i moves per unit of the common factor y ; a parallel shift is the special case in which every maturity moves alike.

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Definition 4.16.1 (Duration under a general shift). Let

$$P(y) \triangleq \sum_i \frac{C_i}{(1 + S(i) + y c_i)^i} \quad (4.24)$$

be the price associated with the cash flow $C_1, C_2, \dots$, and define duration as

$$-\frac{\partial P(y)/P(0)}{\partial y} \bigg|_{y=0} \quad \text{or} \quad -\frac{P(\Delta y) - P(-\Delta y)}{2P(0) \Delta y}.$$

Modified duration equals the above when

$$[c_1, c_2, \dots, c_n] = [1, 1, \dots, 1], \quad S(1) = S(2) = \dots = S(n).$$

Remark 4.16.1. 這一節把第三章的 duration 放回正確的位置，值得把三件事對齊。

(一) y 換了角色。(4.24) 裡的 y 不再是殖利率，而是一個抽象的衝擊大小；真正決定各期利率怎麼動的是向量 $[c_i]$ 。 y 動一單位，第 i 期的即期利率就動 c_i 單位。

(二) 為什麼要有「至少一個 $c_i = 1$ 」。 y 與 $[c_i]$ 只以乘積 yc_i 出現，把 $[c_i]$ 全部乘以 2、 y 全部除以 2，模型完全一樣。固定某個 $c_i = 1$ 就是挑一個參考期別來定尺度，讓算出來的 duration 有明確的單位：「當那一期的利率動 1 時，價格動幾個百分比」。

(三) 第二個公式就是中央差分。它與 (3.14) 的 effective duration 是同一個數值技巧，只是這裡擾動的是整條曲線而非單一殖利率——連同 Section 3.13 之後那段關於 Δy 取多大的討論，也照樣適用。

實務上把不同的 $[c_i]$ 各算一個 duration，就得到**關鍵利率存續期間** (key rate duration)：取 $[c_i]$ 為只在某一期為 1、其餘為 0 的向量，便可量出部位對曲線上各個點的個別暴險，而不是只有一個總數。一個 duration 為零的部位，仍可能對**扭轉** (twist) 有巨大暴險。

§4.17. Some Loose Ends on Dates

— Slides p. 164 —

Every formula in this chapter has indexed time by integers $1, 2, \dots, n$. Converting those to actual dates raises a list of unglamorous questions that nonetheless have to be answered before any of it can be coded:

- Holidays.
- Weekends.
- Business days ($T + 2$, etc.).
- Shall we treat a year as 1 year whether it has 365 or 366 days?

Note 4.17.1. These are the same concerns as the day count conventions of Section 2.10, now compounded by the settlement calendar. None of them is deep, and all of them are mandatory: a discount factor $d(i)$ is only as well defined as the date i names, and two desks using different holiday calendars will bootstrap two different curves from the same quotes.

5. Fundamental Statistical Concepts

— Slides pp. 165–166 —

*There are three kinds of lies:
lies, damn lies, and statistics.*

— Misattributed to Benjamin Disraeli (1804–1881)

*If 50 million people believe a foolish thing,
it's still a foolish thing.*

— George Bernard Shaw (1856–1950)

*One death is a tragedy,
but a million deaths are a statistic.*

— Josef Stalin (1879–1953)

Everything so far has been deterministic. A bond's cash flows were known, the curve was observed, and the only uncertainty admitted — in the term structure theories of [Section 4.15](#) — was smuggled in through an expectation operator that was never defined.

This chapter supplies the vocabulary that the rest of the course runs on. Nothing in it is deep; what matters is that every later model, from the binomial tree to Black–Scholes to Monte Carlo pricing, is assembled from these pieces. Three of them will do most of the work:

- **Conditional expectation**, which is how a model says “given what we know today”.
- **The normal distribution**, which is what we will assume *log* returns follow.
- **The lognormal distribution**, which is what prices then follow — and the reason ([5.20](#)) below turns into the Black–Scholes formula later on.

Note 5.0.1. The epigraphs are a warning worth keeping. [Figure 1.4](#) already showed that real returns are *not* normal: the tails are far too heavy. We use the normal distribution anyway, because it is the only one that stays tractable under the operations we need. The discipline recommended by the Modelers' Hippocratic Oath in [Section 1.5](#) applies exactly here — know precisely where the assumption is wrong.

§5.1. Moments

— Slides p. 167 —

Definition 5.1.1. The **variance** of a random variable X is defined as

$$\text{Var}[X] \triangleq E[(X - E[X])^2]. \quad (5.1)$$

The **covariance** between random variables X and Y is

$$\text{Cov}[X, Y] \triangleq E[(X - \mu_X)(Y - \mu_Y)], \quad (5.2)$$

where μ_X and μ_Y are the means of X and Y , respectively. Random variables X and Y are **uncorrelated** if $\text{Cov}[X, Y] = 0$.

Expanding the products gives the computational forms used in practice,

$$\text{Var}[X] = E[X^2] - (E[X])^2, \quad \text{Cov}[X, Y] = E[XY] - E[X]E[Y],$$

and taking $Y = X$ in (5.2) recovers $\text{Cov}[X, X] = \text{Var}[X]$: variance is covariance with oneself.

Remark 5.1.1. 不相關 (uncorrelated) $\neq$ 獨立 (independent)，這是本章最容易出錯的地方。

獨立是說「 X 的任何資訊都不告訴你 Y 的任何事」；不相關只是說線性關係為零。獨立必然不相關，反之不然。

標準反例：取 $X \sim N(0, 1)$ ， $Y = X^2$ 。此時 $\text{Cov}[X, Y] = E[X^3] - E[X]E[X^2] = 0 - 0 = 0$ ，兩者不相關；但 Y 完全由 X 決定，談不上獨立。

為什麼在金融裡特別要緊？因為風險管理常以「相關係數低」來論證分散效果，而危機時真正咬人的往往是非線性的相依——平時各走各的，一起崩盤時卻同時跌。相關係數對這種結構是盲的。

§5.2. Correlation

— Slides p. 168 —

Definition 5.2.1. The **standard deviation** of X is the square root of the variance,

$$\sigma_X \triangleq \sqrt{\text{Var}[X]}.$$

The **correlation** (or **correlation coefficient**) between X and Y is

$$\rho_{X,Y} \triangleq \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y}, \quad (5.3)$$

provided both have nonzero standard deviations.¹

Note 5.2.1. Dividing by $\sigma_X \sigma_Y$ makes ρ dimensionless and confines it to $[-1, 1]$, by the Cauchy–Schwarz inequality $|\text{Cov}[X, Y]| \leq \sigma_X \sigma_Y$, with equality exactly when Y is an affine function of X . Covariance says how two variables co-move *and* how big they are; correlation strips out the size.

Remark 5.2.1. 那個腳註不是客套話，值得展開。

(一) 相關係數不穩定。 ρ 是用歷史樣本估出來的，而它本身會隨時間漂移——尤其在市場承壓時往往整體跳向 1。用平時估出來的 ρ 去計算危機時的投資組合風險，會系統性地低估損失；2008 年的結構型商品定價正是栽在這裡。

(二) 高頻下會崩解。把觀測間隔縮短到毫秒等級，兩個本來高度相關的資產（例如 S&P 500 期貨與其 ETF）算出來的相關係數會急遽下降——這就是所謂的 **Epps 效應** (Epps effect)。原因是兩邊的成交並非同時發生，取樣間隔一短，非同步性造成的雜訊就蓋過真正的共同波動。Budish、Cramton 與 Shim (2015) 指出，這個在極短時間尺度上反覆出現的價差，正是高頻交易「軍備競賽」的獵物。

結論： ρ 是一個取樣間隔與樣本期間的函數，報告時必須說清楚是用什麼頻率、什麼期間算的。

¹Wilmott (2009): “the correlations between financial quantities are notoriously unstable.” It may even break down “at high-frequency time intervals” (Budish, Cramton, & Shim, 2015).

§5.3. Variance of a Sum

— Slides p. 169 —

Proposition 5.3.1

The variance of a weighted sum of random variables equals

$$\text{Var} \left[\sum_{i=1}^n a_i X_i \right] = \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}[X_i, X_j]. \quad (5.4)$$

It becomes

$$\sum_{i=1}^n a_i^2 \text{Var}[X_i] \quad (5.5)$$

when the X_i are uncorrelated.^a

^aBienaymé (1853).

Proof. Write $Z = \sum_i a_i (X_i - \mu_i)$, so that $E[Z] = 0$ and $\text{Var}[\sum_i a_i X_i] = E[Z^2]$. Then

$$E[Z^2] = E \left[\sum_i \sum_j a_i a_j (X_i - \mu_i)(X_j - \mu_j) \right] = \sum_i \sum_j a_i a_j \text{Cov}[X_i, X_j]$$

by linearity of expectation. If the X_i are uncorrelated, every off-diagonal term vanishes and only the $i = j$ terms, $a_i^2 \text{Var}[X_i]$, survive. ■

Intuition . (5.5) is the arithmetic of diversification. Hold n uncorrelated assets in equal weights $a_i = 1/n$, each with variance σ^2 ; the portfolio variance is $n \cdot (1/n)^2 \sigma^2 = \sigma^2/n$, which vanishes as n grows. Correlation is what stops this: in (5.4) the $n^2 - n$ off-diagonal terms outnumber the n diagonal ones, so for large n the portfolio variance is governed almost entirely by the *average covariance*, not by the individual variances.

分散投資 (diversification) 之所以有效，是因為 (5.5) 的 σ^2/n 會隨 n 趨近於零。真正的限制在 (5.4)：非對角項有 $n^2 - n$ 個，對角項只有 n 個， n 一大，組合的變異數幾乎完全由平均共變異數決定——分散得掉個別風險，分散不掉共同風險。

§5.4. Conditional Expectation

— Slides p. 170 —

Definition 5.4.1. “ $X \mid I$ ” denotes X conditional on the information set I ; the information set can be another random variable’s value, or the past values of X , say. The **conditional expectation**

$$E[X \mid I]$$

is the expected value of X conditional on I . It is itself a *random variable*.

Theorem 5.4.2 (Law of iterated conditional expectations, or the tower law)

$$E[X] = E[E[X | I]]. \quad (5.6)$$

— Slides p. 171 —

Theorem 5.4.3

If I_2 contains at least as much information as I_1 , then

$$E[X | I_1] = E[E[X | I_2] | I_1]. \quad (5.7)$$

Here I_1 contains price information up to time t_1 , and I_2 contains price information up to a later time $t_2 > t_1$. In general,

$$I_1 \subseteq I_2 \subseteq \dots$$

means the players never forget past data, so the information sets are increasing over time.²

Remark 5.4.1. 這一節看起來抽象，但它是後面所有定價模型的語法，值得把三件事講清楚。

(一) 為什麼 $E[X | I]$ 是隨機變數？因為 I 本身是隨機的。明天的股價期望值取決於今天觀察到什麼；今天的觀察結果尚未實現時，這個「期望值」就還是個變數。一旦 I 實現了，它才塌縮成一個數字。

(二) (5.7) 在講什麼？用白話說：今天對「明天的預測」所做的預測，就是今天的預測。你不可能今天就知道明天會把預測往上修——若知道，今天就該先修了。這正是鞅 (martingale) 的核心性質，也是無套利定價的技術支柱。

(三) $I_1 \subseteq I_2 \subseteq \dots$ 為何要特別寫出來？它把「資訊只增不減」這個看似當然的假設明白化。數學上這叫過濾 (filtration)，是隨機過程定義「時間」的方式——沒有它，「今天知道什麼」根本無從談起。

§5.5. The Normal Distribution

— Slides p. 172 —

Definition 5.5.1. A random variable X has the **normal distribution** with mean μ and variance σ^2 if its probability density function is

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}. \quad (5.8)$$

This is expressed by $X \sim N(\mu, \sigma^2)$. The **standard normal distribution** has zero mean, unit variance, and the following distribution function:

$$\text{Prob}[X \leq z] \triangleq N(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-x^2/2} dx. \quad (5.9)$$

Note 5.5.1. The integral in (5.9) has no closed form in elementary functions, so $N(\cdot)$ is always evaluated numerically. Every option pricing formula in this course will end up

²Hirsa & Neftci (2014). This idea is formalised by sigma fields and filtrations in probability theory.

calling it, which makes a fast and accurate $N(\cdot)$ a genuine implementation concern rather than a detail.

§5.6. Moment Generating Functions

— Slides p. 173 —

Definition 5.6.1. The **moment generating function** of a random variable X is defined as

$$\theta_X(t) \triangleq E[e^{tX}].$$

Theorem 5.6.2

The moment generating function of $X \sim N(\mu, \sigma^2)$ is

$$\theta_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right). \quad (5.10)$$

Remark 5.6.1. 為什麼叫動差生成函數 (moment generating function) ? 把 e^{tX} 展開成 $\sum_{n \geq 0} t^n X^n / n!$ 再取期望值, 得

$$\theta_X(t) = \sum_{n \geq 0} \frac{E[X^n]}{n!} t^n,$$

所以 $\theta_X^{(n)}(0) = E[X^n]$ ——所有動差都藏在這一個函數裡, 微分幾次就掉出第幾階動差。
為什麼現在就要提? 因為 (5.10) 在 $t = 1$ 的值

$$\theta_X(1) = E[e^X] = e^{\mu + \sigma^2/2}$$

正是下一節對數常態分佈的平均數 (5.17)。更一般地取 $t = n$ 就得到 $E[e^{nX}] = e^{n\mu + n^2\sigma^2/2}$, 也就是 $E[Y^n]$ ——本章最後那一串對數常態公式, 全部是 (5.10) 換個寫法而已。

§5.7. The Multivariate Normal Distribution

— Slides p. 174 —

If $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then

$$\sum_i X_i \sim N\left(\sum_i \mu_i, \sum_i \sigma_i^2\right).$$

Dropping independence requires a definition rather than a theorem. Let $X_i \sim N(\mu_i, \sigma_i^2)$, which may not be independent, and suppose

$$\sum_{i=1}^n t_i X_i \sim N\left(\sum_{i=1}^n t_i \mu_i, \sum_{i=1}^n \sum_{j=1}^n t_i t_j \text{Cov}[X_i, X_j]\right) \quad (5.11)$$

for every linear combination $\sum_{i=1}^n t_i X_i$ with $\sum_{i=1}^n \sum_{j=1}^n t_i t_j \text{Cov}[X_i, X_j] \neq 0$.

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Then the X_i are said to have a **multivariate normal distribution**.³ With $M \equiv C^{-1}$ and the (i, j) th entry of the matrix M being M_{ij} , the probability density function for the X_i is

$$\frac{1}{\sqrt{(2\pi)^n \det(C)}} \exp \left[-\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (X_i - \mu_i) M_{ij} (X_j - \mu_j) \right], \quad (5.12)$$

with a positive definite covariance matrix

$$C \triangleq [\text{Cov}[X_i, X_j]]_{1 \leq i, j \leq n}.$$

Remark 5.7.1. (5.11) 的定義方式初看奇怪——為什麼不直接寫聯合密度函數，卻繞道去要求「每一個線性組合都是常態」？

理由是這個定義更廣也更好用。密度函數 (5.12) 需要 C 可逆；若某些 X_i 之間有完全線性相依（例如 $X_3 = X_1 + X_2$ ）， C 就退化、密度不存在，但 (5.11) 依然成立。投影片上那個 $\neq 0$ 的但書正是在排除退化的線性組合。

更實際的理由是：金融裡我們幾乎只關心線性組合——投資組合就是資產的加權和。(5.11) 直接告訴你任何組合的分佈，不必先算聯合密度再積分。

注意每個**邊際分佈** (marginal distribution) 都是常態並不足以保證聯合是多元常態；必須是所有線性組合都常態才行。

§5.8. Generation of Univariate Normal Distributions

— Slides p. 176 —

Monte Carlo pricing needs normal variates by the million, and a computer supplies only uniform ones. The following is the **polar method**.

- Let X be uniformly distributed over $(0, 1]$ so that

$$\text{Prob}[X \leq x] = x, \quad 0 < x \leq 1.$$

- Repeatedly draw two samples x_1 and x_2 from X until

$$\omega \triangleq (2x_1 - 1)^2 + (2x_2 - 1)^2 < 1.$$

- Then $c(2x_1 - 1)$ and $c(2x_2 - 1)$ are independent standard normal variables, where⁴

$$c \triangleq \sqrt{-2(\ln \omega)/\omega}.$$

³Corrected by Mr. Huang, Guo-Hua (R98922107) on March 10, 2010.

⁴As they are normally distributed, to prove independence it suffices to prove that they are uncorrelated, which is easy. Thanks to a lively class discussion on March 5, 2014.

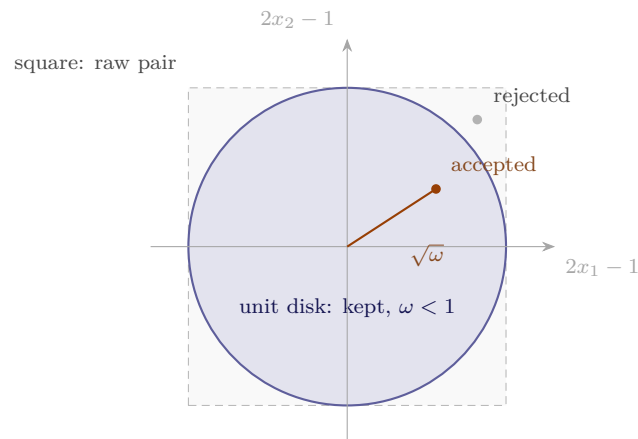


Figure 5.1.: The polar method. A raw pair lands uniformly in the square; the draw is retried unless it falls inside the unit disk. Accepted points are uniform on the disk, which makes the squared radius ω uniform on $(0, 1)$ and independent of the angle.

Remark 5.8.1. 這個看似神祕的做法其實是 **Box–Muller 變換** (Box–Muller transform) 的無三角函數版本，拆成三步就清楚了。

(一) **拒絕取樣** (rejection sampling)。 $(2x_1 - 1, 2x_2 - 1)$ 均勻分佈在邊長 2 的正方形上；只保留落入單位圓內的點，得到圓內的均勻分佈。接受率是 $\pi/4 \approx 78.5\%$ ，代價不高。

(二) 分離半徑與角度。圓內均勻分佈有一個漂亮性質：平方半徑 ω 服從 $(0, 1)$ 上的均勻分佈，且與角度 θ 獨立，而 θ 均勻分佈於 $[0, 2\pi)$ 。

(三) 套用 **Box–Muller**。該變換說：若 $\omega \sim U(0, 1)$ 與 $\theta \sim U[0, 2\pi)$ 獨立，則 $\sqrt{-2 \ln \omega} \cos \theta$ 與 $\sqrt{-2 \ln \omega} \sin \theta$ 是獨立的標準常態。而由 $\cos \theta = (2x_1 - 1)/\sqrt{\omega}$ 、 $\sin \theta = (2x_2 - 1)/\sqrt{\omega}$ ，

$$\sqrt{-2 \ln \omega} \cos \theta = \sqrt{\frac{-2 \ln \omega}{\omega}} (2x_1 - 1) = c (2x_1 - 1),$$

正是投影片上的式子。

好處：整個過程用不到 $\sin$ 與 $\cos$ ——三角函數在早期硬體上比開根號與取對數昂貴得多，而且一次產生兩個常態變量。

§5.8.i. A Dirty Trick and a Right Attitude

— Slides p. 177 —

Let ξ_i be independent and uniformly distributed over $(0, 1)$. A simple method to generate the standard normal variable is to calculate⁵

$$\left(\sum_{i=1}^{12} \xi_i \right) - 6. \quad (5.13)$$

But why use 12? Recall the mean and variance of ξ_i are $1/2$ and $1/12$, respectively.

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The general formula is

$$\frac{(\sum_{i=1}^n \xi_i) - (n/2)}{\sqrt{n/12}}, \quad (5.14)$$

⁵Jäckel (2002): “this is not a highly accurate approximation and should only be used to establish ballpark estimates.”

which is the central limit theorem applied to n uniforms, standardised. Choosing $n = 12$ makes the denominator $\sqrt{12/12} = 1$ and the numerator's correction 6, yielding a formula without the need of division and square-root operations.⁶

Always blame your random number generator last.^a Instead, check your programs first.

^a“The fault, dear Brutus, lies not in the stars but in ourselves that we are underlings.” William Shakespeare (1564–1616), *Julius Caesar*.

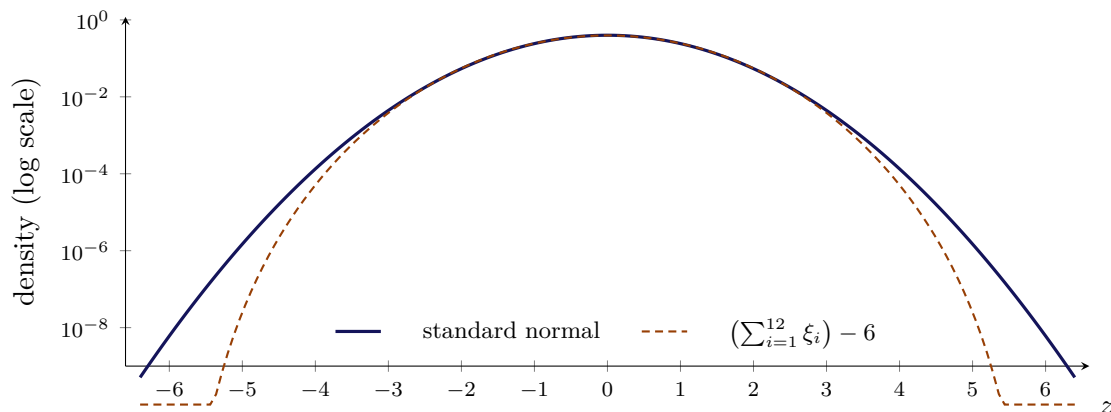


Figure 5.2.: Why (5.13) is only a ballpark approximation. On a logarithmic scale the two densities agree to the eye out to about $|z| = 2$ and then part company: at $z = 4$ the trick delivers 38% of the correct density, at $z = 5$ only 1.7%, and beyond $|z| = 6$ exactly zero, since a sum of twelve uniforms cannot leave $[-6, 6]$.

Remark 5.8.2. 這個近似的毛病正好落在金融最在意的地方。

Figure 5.2 用對數尺度畫，是因為線性尺度上兩條曲線看起來幾乎重合——中央部分確實吻合得很好。問題在尾端： $\sum_{i=1}^{12} \xi_i - 6$ 的支撐集 (support) 是 $[-6, 6]$ ，超出範圍的機率恰好為零，而真正的常態分佈在那裡仍有 2×10^{-9} 的機率。

回想 Figure 1.4：真實報酬的尾巴比常態更厚。若連常態的尾巴都模擬不出來，拿它去算風險值 (value at risk, VaR) 或深度價外 (deep out-of-the-money) 選擇權的價格，誤差方向是系統性的——一律低估極端事件。

所以 Jäckel 那句「只能用來抓個大概」要認真看待：中央矩（平均、變異數）算得很準，尾端機率則完全不能信。

§5.9. Generation of Bivariate Normal Distributions

— Slides p. 179 —

Pairs of normally distributed variables with correlation ρ can be generated as follows. Let X_1 and X_2 be independent standard normal variables, and set

$$\begin{aligned} U &\triangleq aX_1, \\ V &\triangleq a\rho X_1 + a\sqrt{1-\rho^2} X_2. \end{aligned} \tag{5.15}$$

⁶Contributed by Mr. Chen, Shih-Hang (R02723031) on March 5, 2014.

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Proposition 5.9.1

U and V are the desired random variables, with

$$\text{Var}[U] = \text{Var}[V] = a^2, \quad \text{Cov}[U, V] = \rho a^2.$$

Proof. $\text{Var}[U] = a^2 \text{Var}[X_1] = a^2$. Since X_1 and X_2 are uncorrelated, (5.5) gives

$$\text{Var}[V] = a^2 \rho^2 \text{Var}[X_1] + a^2(1 - \rho^2) \text{Var}[X_2] = a^2 \rho^2 + a^2(1 - \rho^2) = a^2.$$

Finally, using $\text{Cov}[X_1, X_2] = 0$ and $\text{Var}[X_1] = 1$,

$$\text{Cov}[U, V] = a^2 \rho \text{Var}[X_1] + a^2 \sqrt{1 - \rho^2} \text{Cov}[X_1, X_2] = \rho a^2. \quad \blacksquare$$

Note that the mapping from (X_1, X_2) to (U, V) is a one-to-one correspondence for $a \neq 0$. By (5.3), the correlation of U and V is $\rho a^2 / (a \cdot a) = \rho$, as intended.

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The mapping in matrix form is

$$\begin{bmatrix} U \\ V \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}. \quad (5.16)$$

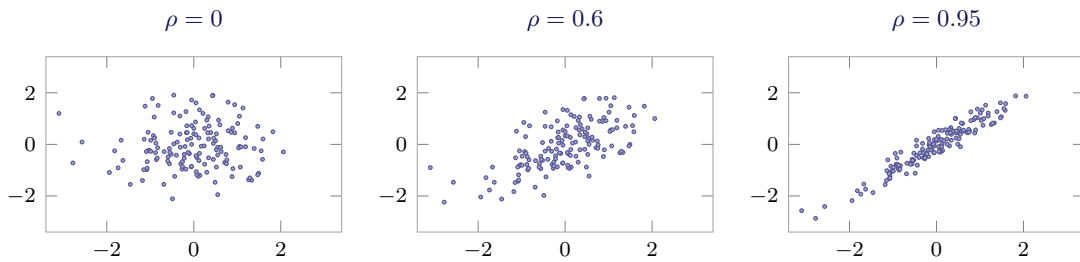


Figure 5.3.: Bivariate normal samples generated by (5.16) with $a = 1$, from the same underlying (X_1, X_2) draws. Raising ρ shears the cloud towards the diagonal without changing either marginal.

Remark 5.9.1. (5.16) 不是特例，而是一般作法的二維版本。

那個下三角矩陣正是相關係數矩陣 $\begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$ 的 **Cholesky 分解** (Cholesky decomposition) LL^T :

$$L = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{bmatrix}, \quad LL^T = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

一般地，要產生共變異數矩陣為 C 的 n 維常態向量，就把 C 做 Cholesky 分解 $C = LL^T$ ，取獨立標準常態向量 $\mathbf{X}$ ，令 $\mathbf{U} = L\mathbf{X}$ ；則 $\text{Cov}[\mathbf{U}] = L \text{Cov}[\mathbf{X}] L^T = LL^T = C$ 。

這也解釋了 (5.12) 為什麼要求 C **正定** (positive definite) ——正定才有 Cholesky 分解，才產得出來。實務上由市場資料估出的相關係數矩陣常因雜訊而不正定，必須先做修正，否則模擬根本無法開始。

§5.10. The Lognormal Distribution

— Slides p. 182 —

Definition 5.10.1. A random variable Y is said to have a **lognormal distribution** if $\ln Y$ has a normal distribution.

Let $X \sim N(\mu, \sigma^2)$ and $Y \triangleq e^X$. The mean and variance of Y are

$$\begin{aligned}\mu_Y &= e^{\mu+\sigma^2/2}, \\ \sigma_Y^2 &= e^{2\mu+2\sigma^2}(e^{\sigma^2} - 1),\end{aligned}\tag{5.17}$$

respectively. They follow from $E[Y^n] = e^{n\mu+n^2\sigma^2/2}$ — which is just (5.10) evaluated at $t = n$.

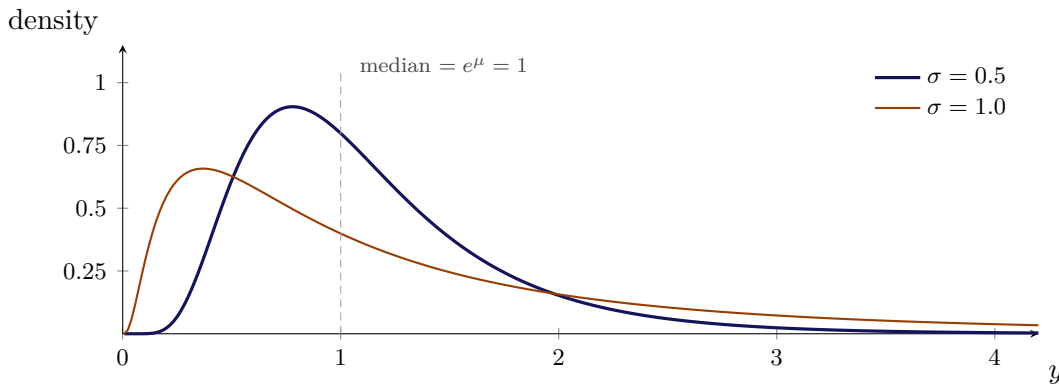


Figure 5.4.: Lognormal densities with $\mu = 0$. The support is $(0, \infty)$, the shape is right-skewed, and the right tail thickens rapidly with σ . The median sits at e^μ regardless of σ , while the mean $e^{\mu+\sigma^2/2}$ drifts to the right of it.

Remark 5.10.1. 為什麼金融上用對數常態分佈 (lognormal distribution) 描述價格，用常態描述對數報酬？三個理由，都看得見。

(一) 價格不能為負。常態分佈的支撐集是整條實線，但股價不會是負的。 $Y = e^X$ 自動把值域限制在 $(0, \infty)$ 。

(二) 報酬可加，價格可乘。回顧 Section 1.6.iii 的對數報酬 $r_t = \ln(S_t/S_{t-1})$ ：跨期報酬是逐期報酬之和。若每期 r_t 為獨立常態，其和仍為常態，於是 $S_T = S_0 \exp(\sum r_t)$ 必然是對數常態。這個封閉性正是整個模型的骨架。

(三) 平均數與中位數不同。(5.17) 裡的 $\mu_Y = e^{\mu+\sigma^2/2}$ 嚴格大於中位數 (median) e^μ ，差額 $\sigma^2/2$ 純粹來自波動本身。這就是日後反覆出現的「 $-\sigma^2/2$ 修正項」的來源：想讓價格的期望成長率等於 r ，指數裡就得寫 $r - \sigma^2/2$ 。這又是 Jensen 不等式（見 Section 4.15 的「壞」預期理論）的另一個面貌。

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Conversely, suppose Y is lognormally distributed with mean μ and variance σ^2 . Then $\ln Y$ has a normal distribution with

$$\begin{aligned}E[\ln Y] &= \ln \left[\mu / \sqrt{1 + (\sigma/\mu)^2} \right], \\ \text{Var}[\ln Y] &= \ln \left[1 + (\sigma/\mu)^2 \right].\end{aligned}\tag{5.18}$$

If X and Y are joint-lognormally distributed, then

$$\begin{aligned}E[XY] &= E[X]E[Y]e^{\text{Cov}[\ln X, \ln Y]}, \\ \text{Cov}[X, Y] &= E[X]E[Y] \left(e^{\text{Cov}[\ln X, \ln Y]} - 1 \right).\end{aligned}\tag{5.19}$$

Note 5.10.1. Beware the notation on this page: μ and σ^2 now denote the mean and variance of Y itself, not of $\ln Y$ as on the previous page. (5.18) is the inverse of (5.17), and the two must not be mixed. A quick check: substituting $\mu = e^{m+s^2/2}$ and $\sigma^2 = e^{2m+s^2}(e^{s^2} - 1)$ into (5.18) returns $E[\ln Y] = m$ and $\text{Var}[\ln Y] = s^2$, as it should.

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Theorem 5.10.2

Let Y be lognormally distributed such that $\ln Y \sim N(\mu, \sigma^2)$, with density f . Then

$$\int_a^\infty y f(y) dy = e^{\mu+\sigma^2/2} N\left(\frac{\mu - \ln a}{\sigma} + \sigma\right). \quad (5.20)$$

Proof. Substitute $y = e^x$, so that $\ln Y = X \sim N(\mu, \sigma^2)$ and the integral becomes

$$\int_{\ln a}^\infty e^x \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} dx.$$

Completing the square in the exponent,

$$x - \frac{(x-\mu)^2}{2\sigma^2} = \mu + \frac{\sigma^2}{2} - \frac{(x-\mu-\sigma^2)^2}{2\sigma^2},$$

so the integral equals

$$e^{\mu+\sigma^2/2} \int_{\ln a}^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu-\sigma^2)^2/(2\sigma^2)} dx = e^{\mu+\sigma^2/2} \text{Prob}\left[Z > \frac{\ln a - \mu - \sigma^2}{\sigma}\right],$$

with Z standard normal. By the symmetry $1 - N(z) = N(-z)$ this is $e^{\mu+\sigma^2/2} N((\mu + \sigma^2 - \ln a)/\sigma)$, which is (5.20). ■

(5.20) is the **partial expectation** of a lognormal variable — the expected value of Y restricted to the event $Y > a$.

It is not a technical curiosity: it is the entire content of the first term of the Black–Scholes formula. A call option pays $Y - a$ exactly when $Y > a$, so its expected payoff splits into $\int_a^\infty y f(y) dy - a \text{Prob}[Y > a]$ — the first piece supplying $S_0 N(d_1)$ and the second $a N(d_2)$. Every remaining chapter of this course is, in one way or another, an elaboration of that split.

(5.20) 就是 Black–Scholes 公式第一項的全部內容；買權在 $Y > a$ 時付 $Y - a$ ，期望報酬拆成這兩塊，前者給出 $S_0 N(d_1)$ ，後者給出 $a N(d_2)$ 。